Index

Setup----------1
Header & Nav Semantic----------15
Footer, Aside, Address, Section----------16
Myntra.html----------17
flipkart footer----------25
Amazon_Footer----------28
Main & CSS Colors [Netflix Design ----------32
Dialog, Article, Body----------38
Paragraphs & Blockquotes----------47
Text Formatting----------57
Data List with Terms & Definitions----------60
Containers and Images Fieldset & Legend ----------73
Card Examples----------97
Links Intra and Inter ----------141
Youtube----------154
Tables----------172
Forms----------178
Buttons Meter Progress ----------200-207
Multimedia----------208
screen_and_print----------209
Minification & Selectors----------211
CSS Selectors----------216
breadcrumbs_Pseudo_Classes----------228
Positions, Z-index, Backgrounds----------236
Display, text Styles----------242
Media Queries----------247
Transforms----------259
Transitions and Animations----------269
Setup Bootstrap----------277
Bootstrap Classes----------289
Bootstrap Components: Alert Modal Offcanvas----------309
Carousel, Collapse, Accordion----------322
Nav, Tabs, Navbar----------336
Tailwind----------343

Setup Development Environment on your PC:

1. Make sure that your PC is installed with Windows 8+ version or any Mac, Linux.

2. Download and Install “Node JS” on your PC

https://nodejs.org/

- Node JS is an open source and cross platform JavaScript runtime used by developers

to build web applications, server, scripts and command line tools.

- After installing check the version from your command prompt or terminal

windows => programs => cmd

Mac => terminal

C:\> node -v // make sure that version is above 20+

C:\> npm -v // make sure that version of npm is 10+

- NPM is a package manager. [Node Package Manager]

- Package Manager is a software tool used by developers to install, update and uninstall

various libraries required for project.

- There are various other package managers like

a) Yarn b) Bower c) Composer d) NuGet e) Ruby Gems f) GRUNT etc.

3. Download and Install “Visual Studio Code Editor”.

- Editor is a software tool that provides an IDE for developer.

- IDE is Integrated Development Environment.

- It is used by developers to build, debug, test and deploy applications.

https://code.visualstudio.com/download

Setup Web Application

Install Node JS on your PC

- Node JS - NPM

Install Visual Studio Code

3. Install Following extensions in VS Code

- Live Server [Ritwick Dey]

    * Server is required to host and run application.

    * You can setup application on local servers like

        a) IIS b) Tomcat c) WAMP d) XAMP etc.

    * You can setup application on cloud servers like

        a) Firebase b) Vercel c) Netlify d) AWS e) Azure f) GIT Hub Page etc.

- VSCode Icons [vscode icons team]

- Live Preview [microsoft]

- IntelliSense for CSS class names in HTML

4. Enable “emmet” in Visual Studio Code

- Go to “Settings” of VS Code

- Search for “emmet”

- Select the check box for “Emmet Abbreviations…”

Setup a new Web Application for Development Environment:

1. Go to any location on your PC and create a new folder for project.

    “food-delivery-app”

    D:\food-delivery-app

2. Open project folder in VS Code

3. Open terminal in folder location Ctrl + `

4. Run the following command

    > npm init -y

- It will generate a new file by name “package.json”

- It comprises of project meta data, which includes

a) App Name b) App version c) Dependencies and their version d) Commands etc.

5. Add following files into project

a) README.md

b) .gitignore : - It is a required to publish our application on to cloud.

6. Add following folders into project

a) public

- It is used to keep all static resources like html, images, documents, videos etc.

b) src

- It is used to keep all dynamic resources like css, js, ts, sass, jsx, tsx etc.

Note: You can setup entire project environment by using bundling tools like

a) Webpack b) Vite c) Parcel etc.

```
public
  |
  |_images
  |_pages
  |_docs
  |_videos

src
  |_scripts
  |_styles
  |_sass
```

For Resources : https://www.freepik.com/ https://www.canva.com/

7. Run your project by clicking “Go Live” in status of VS Code

You can request from browser:

http://127.0.0.1:5500 or http://localhost:5500

==========

HTML

- Hyper Text Markup Language.

- Hyper text navigates beyond the context.

- Marking Up is the process of preparing your content to present exactly as expected.

- HTML is a presentation language used to present content on browser.

- The latest HTML is **HTML5.**

[WHATWG started with 4 & now it is 5]

==========

Ex: Page Is refreshing every 5 second.via Meta

```
<!DOCTYPE html>

<html lang="en-in">

  <head>

      <title>Food App</title>

      <link rel="shortcut icon" href="./public/images/favicon.ico" >

      <meta name="viewport" content="width=device-width, initial-scale=1.5">

      <meta http-equiv="refresh" content="5">

      <meta charset="utf-8">

  </head>
  <body> Lorem ipsum, </body>  </html>
```

Body Element

- Body defines the content to render as a result of round-trip.

Attributes:

1. bgcolor : It sets a background color.
2. text : It sets color for body text.

HTML Colors

a) Color name

```
<body  bgcolor="yellow">
```

b) Color shade name

```
<body  bgcolor="lightyellow">
```

c) Hexadecimal color code

i) 3 Chars Hexadecimal code with "#"

#RGB

R,G,B values : 0,1,2,3,4,5,6,7,8,9,a,b,c,d,e,f

```
<body  bgcolor="#f00">
```

ii) 6 Chars Hexadecimal code with "#"

#RRGGBB

#ff0000	red
#ffffff	white
#000000	black
#00ff00	green

<body bgcolor="#ff0000" text="#fff">

3. background : It sets a background image for page. HTML can't control the background position, size etc. You have to use styles.

Styles are configure using <style> container.

CSS Background Style Attributes

background-size : cover, contain, specific width and height.

background-repeat : no-repeat, repeat, repeat-x, repeat-y

background-position : top, center, bottom, left, right, specific location with pixels.

background-attachment : scroll, fixed

background-clip : border-box, content-box

Syntax:

```
<head>
    <style>
        body {
            background-size: cover;
            background-repeat: no-repeat;
        }
    </style>
</head>
```

Ex:

```
<!DOCTYPE html>
<html lang="en-in">
```

```
  <head>
      <title>Food App</title>
      <link rel="shortcut icon" href="./public/images/favicon.ico" >
      <meta name="viewport" content="width=device-width, initial-scale=1.5">
      <meta charset="utf-8">
      <style>
        body {
          background-size: cover;
          background-repeat: no-repeat;
          background-position: center center;
          background-attachment: fixed;
        }
      </style>
  </head>
  <body background="./public/images/food-background.jpg">
    Lorem ipsum,

  </body>
</html>
```

4. align : It aligns the body text left, center, right or justify.

Syntax:

```
<body  align="justify">
```

5. alink : It sets color for active links in page.
6. vlink : It sets color for visited links in page.

Syntax:

```
<body  alink="red"  vlink="gray">
   <a href="https://www.amazon.in"> Amazon </a>
</body>
```

Ex:

```
<!DOCTYPE html>

<html lang="en-in">

  <head>

      <title>Food App</title>

      <link rel="shortcut icon" href="./public/images/favicon.ico" >

      <meta name="viewport" content="width=device-width, initial-scale=1.5">

      <meta charset="utf-8">

      <style>



      </style>

  </head>

  <body vlink="gray" alink="red">

       <a href="https://www.amazon.in"&gt; Amazon India </a>

       |

       <a href="https://www.microsoft.com"&gt; Microsoft </a>

       |

       <a href="https://www.twitter.com">Twitter</a&gt;

  </body>

</html>
```

7. leftmargin]

8. rightmargin] space between content and page border

9. topmargin]

10. bottommargin]

Syntax:

```
<body leftmargin="200" rightmargin="200" topmargin="50" bottommargin="50">

</body>
```

Body Layout Semantics

1. Header 2. Footer 3. Section 4. Main 5. Nav 6. Aside 7. Article 8. Address

Non Semantics

9. Div 10. Span

- refers to a division used to separate topic from the current context.

- refers to topic in the same context, which spans along with context but separated to highlight.

CSS Borders

border : short hand for size, style and color.

border-left : left short hand

border-right

border-top

border-bottom

a) size : thickness in pixels.

b) style : solid, double, dotted, dashed, groove etc.

c) color : name, shade, hexadecimal etc.

Syntax:

```
div {
    border: 2px solid black;        // no order dependency
  }
```

CSS Border Radius

border-radius : short hand for all directions.

border-top-left-radius

border-top-right-radius

border-bottom-left-radius

border-bottom-right-radius

Syntax:

```
div {  border-radius : 20px;        }
```

CSS Box Shadow

box-shadow : horizontal, vertical, blur and color.

Syntax:

```
div { box-shadow: 5px 10px  2px black; }
```

5px : horizontal shadow

10px : vertical shadow

2px : shadow blur [more pixels will blur more]

black : shadow color.

CSS Margin & Padding

margin : short hand for all directions [top, right, bottom, left]

margin-left, margin-right, margin-top, margin-bottom

padding : short hand for all directions [top, right, bottom, left]

padding-left, padding-right, padding-top, padding-bottom

Syntax:

div { margin: 10px 20px 10px 20px; [top, right, bottom, left] }

div { padding: 20px 10px 20px 10px; }

CSS Text Styles

font-size : text size

font-family : sets a font family [Arial, Times New Roman etc..]

font-style : italics

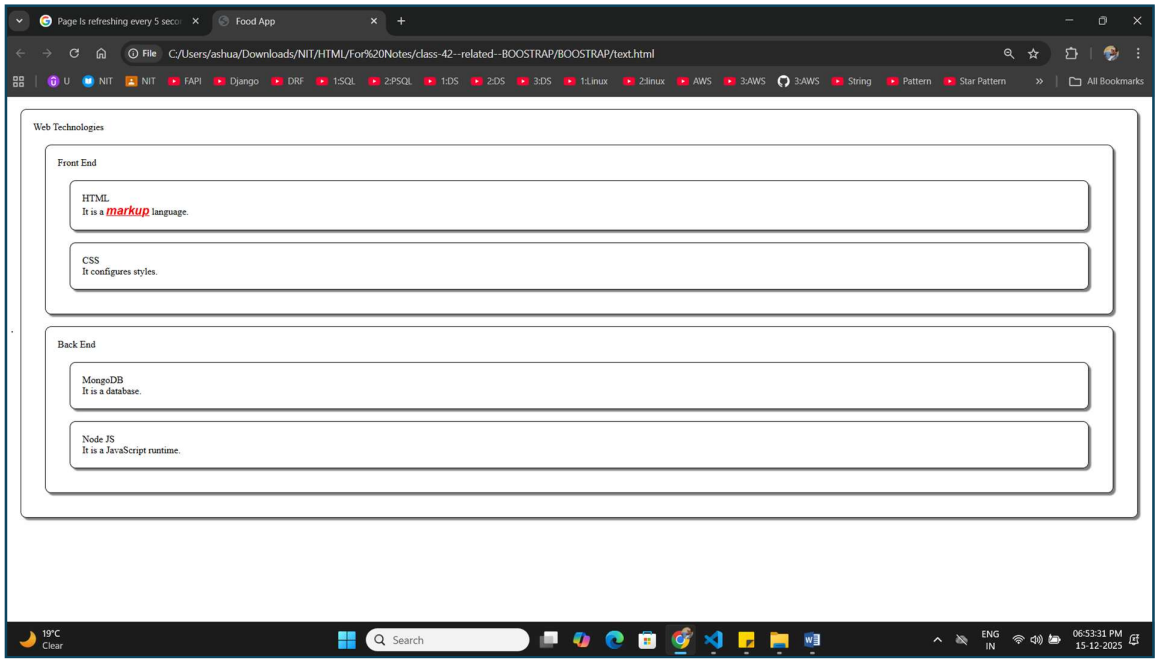
font-weight : bold

color : text color

text-align : left, center, right, justify

text-decoration : underline, overline, line-through with size, style & color

Example_1:



Ex:

```
<!DOCTYPE html>
<html lang="en-in">
  <head>
      <title>Food App</title>
      <link rel="shortcut icon" href="./public/images/favicon.ico" >
      <meta name="viewport" content="width=device-width, initial-scale=1.5">
      <meta charset="utf-8">
      <style>
         div {
            border: 1px solid black;
            margin: 20px;
            padding: 20px;
            box-shadow: 4px 4px 2px gray;
            border-radius: 10px;
         }
         span {
            color:red;
            text-decoration: underline 1px solid red;
            font-weight: bold;
            font-style: italic;
            font-size: 20px;
            font-family: Arial;
```

```
        }
    </style>
</head>
<body>
   <div>
      Web Technologies
      <div>
         Front End
         <div>
            HTML <br>
            It is a <span>markup</span> language.
         </div>
         <div>
            CSS <br>
            It configures styles.
         </div>
      </div>
      <div>
         Back End
         <div>
            MongoDB <br>
            It is a database.
         </div>
```

```
            <div>

              Node JS <br>

              It is a JavaScript runtime.

            </div>

          </div>

      </div>

   </body>

</html>
```

Setup Bootstrap Icons for Project

1. Open Terminal in your Project [VS Code]

2. Run the command

```
> npm   install   bootstrap-icons    --save
```

3. Icons are installed into “node_modules” folder, they are kept is a “.css” file.

```
node_modules

|

|_ bootstrap-icons

    |_font

        |_bootstrap-icons.css
```

4. Link “bootstrap-icons.css” to your HTML page

```
 <head>
   <link  rel=“stylesheet”  href=“./node_modules/bootstrap-icons/font/bootstrap-
icons.css”>
</head>
```

5. <div class=“bi bi-facebook”> <span class=“bi bi-cart”>
   Visit: icons.getbootstrap.com => to know icon classes

Header & Nav Semantic

1. <header>

- It is used to display content at top margin of page.

- Typically header comprises of brand name, logo, search bar, navigation etc.

https://colorlib.com/ => for templates

./ for invoking files / folders of current level

../ for invoking files / folders one level back to current folder

../../ two levels back

2. <nav>

- It is used to configure a navigation for application resources.

- It can be within the page or across pages.

CSS Flex Display

- CSS display-flex is used to keep the elements in a container flexible.

```
{ display: flex; }
```

- Flex have 2 directions : row, column

```
{
 display: flex:
 flex-direction: row | column;
}
```

- You can align flex container elements by using “justify-content”

```
{
 display:flex;
 flex-direction:row | column;
 justify-content: baseline, end, center, space-between, space-evenly,space-around;}
```

baseline [left], end [right]

- You can align items vertically using “align-items” attribute. [if direction is row]

{ align-items: baseline, end, center; }

Footer, Aside, Address, Section

3. <footer>

- It defines content to display at bottom margin of page.
- Typically footer comprises of services, copyright, address, network services etc.

4. <aside>

- It is a container that comprises of information which navigates user outside the current context.

- Typically social networking links, advertisements are kept in aside area.

5. <address>

- It is a container for contact details.

- Address can make the contact details SEO friendly.

6. <section>

- It defines the content between header and footer.

- Page body content is kept inside section.

- It is considered as workspace for user.

CSS Grid

- It allows to split the content into rows and columns.

- The content of one column will not span to next column.

Syntax:

```
{   display: grid;   }
```

- Grid can divide into columns by using “grid-template-columns”.

- A page can be divided into 12 fractions [fr], which is from 1200 pixels.

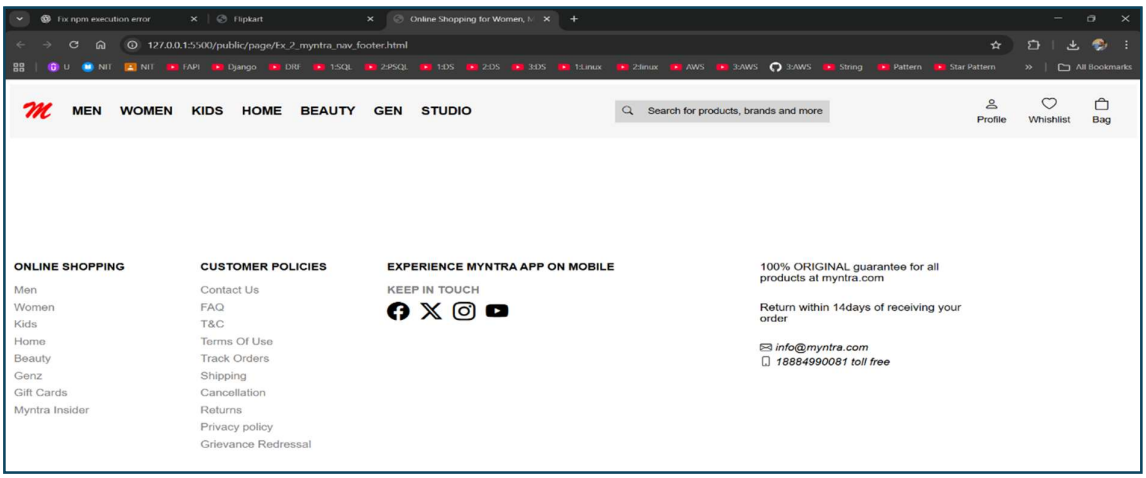
Syntax:

```
{
  display: grid;
  grid-template-columns: 6fr 6fr;          // 2 columns
}
```

CSS Relational Selectors:

nav span { }	all span inside nav. <nav> </nav>
nav + span { }	the first span after nav. [adjacent] <nav> </nav>
nav ~ span { }	all span elements after nav but inside same block.

Ex_2: Myntra.html




```
<!DOCTYPE html>

<html lang="en-in">

<head>

   <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Online Shopping for Women, Men, Kids </title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <style>

        header {

            font-size: 18px;

            font-family: Arial;

            display: flex;

            flex-direction: row;

            justify-content: space-between;

            padding: 20px;

            align-items: center;

            background-color: #f6f6f6;

        }

        .short-cuts {

            display: flex;

            flex-direction: row;

            justify-content: space-between;

            align-items: center;

        }

        .brand-nav {

            display: flex;
```

```
    flex-direction: row;
    justify-content: space-between;
    align-items: center;
}
.brand-title {
    font-size: 40px;
    font-weight: bold;
    font-family: Brush Script MT;
    color:red;
}
nav {
    padding-left: 20px;
}
nav span {
    padding-left: 10px;
    padding-right: 10px;
    font-weight: bold;
}
.search-bar {
    background-color: #d6d5d5;
    padding: 10px;
    font-size: 14px;
    width: 270px;
```

```
    display: flex;
    flex-direction: row;
    justify-content: space-between;
}
.short-cuts div {
    padding-left: 15px;
    padding-right: 15px;
    text-align: center;
    font-size: 20px;
}
.short-cut-text {
    font-size: 14px;
}
section {
    height: 200px;
}
footer {
    display: grid;
    grid-template-columns: 2fr 2fr 4fr 4fr;
    font-family: Arial;
    padding: 5px;
}
.footer-title {
```

```
            font-weight: bold;
            display: block;
            margin-bottom: 20px;
        }
        .footer-title~span {
            display: block;
            margin-bottom: 10px;
            color:gray;
        }
        aside span {
            font-size: 30px;
            padding-right: 10px;
        }
        .footer-content {
            width: 300px;
            margin-bottom: 30px;
        }
        address span {
            display: block;
            margin-bottom: 5px;
        }
    </style>
</head>
```

```
<body>
    <header>
        <div class="brand-nav">
            <span class="brand-title">M</span>
            <nav>
                <span>MEN</span>
                <span>WOMEN</span>
                <span>KIDS</span>
                <span>HOME</span>
                <span>BEAUTY</span>
                <span>GEN</span>
                <span>STUDIO</span>
            </nav>
        </div>
        <div>
            <div class="search-bar">
                <span class="bi bi-search"></span>
                <span> Search for products, brands and more </span>
            </div>
        </div>
        <div class="short-cuts">
            <div>
                <span class="bi bi-person"></span> <br> <span class="short-cut-
text">Profile</span>
```

```
            </div>

            <div>

                <span class="bi bi-heart"></span> <br> <span class="short-cut-
text">Whishlist</span>

            </div>

            <div>

                <span class="bi bi-bag"></span> <br> <span class="short-cut-
text">Bag</span>

            </div>

        </div>

    </header>

    <section>

    </section>

    <footer>

        <div>

            <span class="footer-title">ONLINE SHOPPING</span>

            <span>Men</span>

            <span>Women</span>

            <span>Kids</span>

            <span>Home</span>

            <span>Beauty</span>

            <span>Genz</span>

            <span>Gift Cards</span>

            <span>Myntra Insider</span>
```

```
</div>
<div>
    <span class="footer-title">CUSTOMER POLICIES</span>
    <span>Contact Us</span>
    <span>FAQ</span>
    <span>T&C</span>
    <span>Terms Of Use</span>
    <span>Track Orders</span>
    <span>Shipping</span>
    <span>Cancellation</span>
    <span>Returns</span>
    <span>Privacy policy</span>
    <span>Grievance Redressal</span>
</div>
<div>
    <span class="footer-title">EXPERIENCE MYNTRA APP ON MOBILE</span>
    <span class="footer-title">KEEP IN TOUCH</span>
    <aside>
        <span class="bi bi-facebook"></span>
        <span class="bi bi-twitter-x"></span>
        <span class="bi bi-instagram"></span>
        <span class="bi bi-youtube"></span>
    </aside>
```

```
        </div>
        <div>
            <div class="footer-content">
                <span>100% ORIGINAL guarantee for all products at myntra.com</span>
            </div>
            <div class="footer-content">
                <span>Return within 14days of receiving your order</span>
            </div>
            <div class="footer-content">
                <address>
                    <span class="bi bi-envelope"> info@myntra.com</span>
                    <span class="bi bi-phone"> 18884990081 toll free</span>
                </address>
            </div>
        </div>
    </footer>
</body>
</html>
```

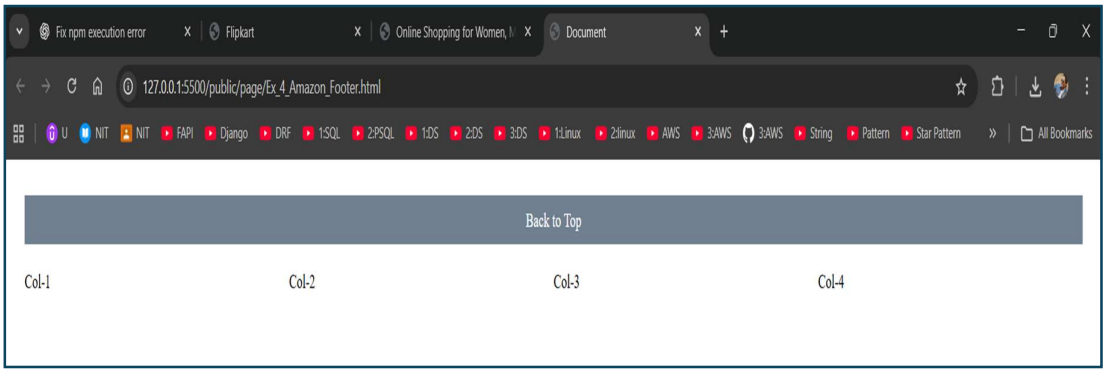
Ex_3: flipkart footer

```
!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

        footer {

            background-color: black;

            color:white;

            font-size: 20px;

            padding: 20px;

            display: grid;

            grid-template-columns: 6fr 6fr;

            height: 300px;

        }

        .col-left {

            display: grid;

            grid-template-columns: 3fr 3fr 3fr 3fr;

        }

        .col-right {

            display: grid;

            grid-template-columns: 6fr 6fr;
```

```
            border-left: 1px solid white;
            padding-left: 20px;
        }
    </style>
</head>
<body>
    <footer>
        <div class="col-left">
            <div>
                Col-1
            </div>
            <div>
                Col-2
            </div>
            <div>
                Col-3
            </div>
            <div>
                Col-4
            </div>
        </div>
        <div class="col-right">
            <div>
```

```
            Col-1
        </div>
        <div>
            Col-2
        </div>
    </div>
  </footer>
</body>
</html>
```

Ex_4_Amazon_Footer.html




```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    footer {
```

```
            padding: 20px;
        }
        .row-1 {
            background-color: slategray;
            padding: 10px;
            text-align: center;
            color:white;
        }
        .row-2 {
            display: grid;
            grid-template-columns: 3fr 3fr 3fr 3fr;
            padding-top: 20px;
        }
    </style>
</head>
<body>
    <footer>
        <div class="row-1">
            <span>Back to Top</span>
        </div>
        <div class="row-2">
            <div>
                Col-1
```

```
            </div>
            <div>
                Col-2
            </div>
            <div>
                Col-3
            </div>
            <div>
                Col-4
            </div>
        </div>
    </footer>
</body>
</html>
```

Main & CSS Colors [Netflix Design Example]

7. <main>

- Technically main is referred as “Entry Point”.

- Every computer program will have a main as entry point, from where the execution

  starts.

- Web page contains main from where a new user can start browsing or using the

application.

- A page can have multiple main containers.

CSS Colors:

1. Color Name: { color : red; }

2. Shade Name: { color: lightgreen; }

3. Hexadecimal : { color: #ff0000; (or) #f00; }

4. rgb()

   - It is a method with Red, Green & Blue values.

   - Every color value can be from 0 to 255.

   { color: rgb(redValue, greenValue, blueValue); }

   { color: rgb(255,0,0); }

5. rgba()

   - It is a method with Red, Green, Blue & Alpha value.

   - Alpha is used to set color as transparent.

   - Alpha value can be 0 to 1. [0 is complete transparent]

   { color: rgba(redValue, greenValue, blueValue, alphaValue); }

   { color: rgba(0,0,0, 0.5); }

6. hsl()

   - It is a method with hue, saturation and lightness as values.

   - Hue is a value between 0deg to 360deg.

   0 to 120deg : red
   121 to 240deg : green
   241 to 360deg : blue

- Saturation is grey shade in color between 1 to 100%

- Lightness is brightness of color between 1 to 100%

{ color: hsl(hueValue, satValue, lightValue); }

{ color: hsl (10deg, 50%, 50%); }

7. hsla()

- It is a method with hue, saturation, lightness and alpha as value.

- Alpha can be from 0 to 1.

{ color: hsla(10deg, 50%, 50%, 0.5); }

8. Gradient Colors

- You can apply multiple colors as gradient with different orientations.

- Gradient is allowed only as background image.

- You can set

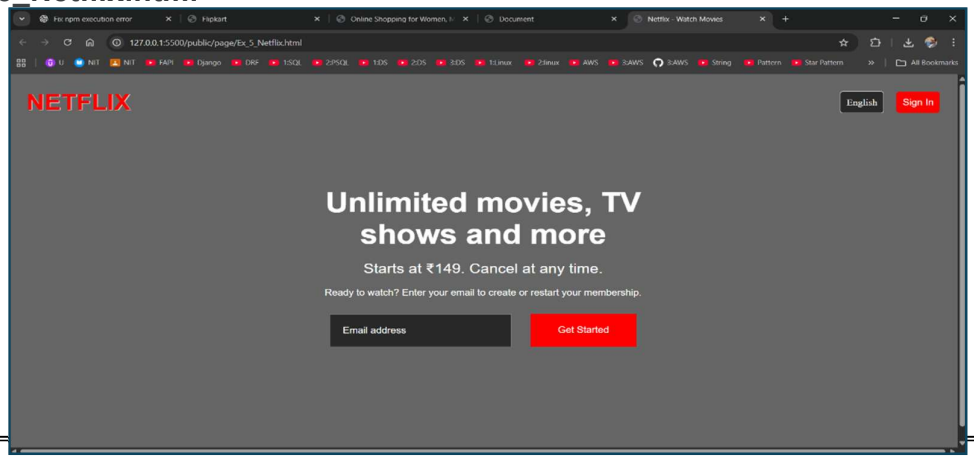
  a) Linear Gradient

  b) Radial Gradient

{ background-image : linear-gradient(direction|angle, color1, color2..); }

{ background-image : radial-gradient(color1, color2 10%); }

Ex_5_Netflix.html



```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Netflix - Watch Movies</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <style>

        body {

            background-size: cover;



        }

        .shade-cover {

            height: 100vh;

            background-color: rgba(0,0,0,0.6);

            margin-top: -10px;

            margin-left: -10px;

            width: 101.30%;

        }

        header {

            color:white;

            padding: 30px;
```

```
    display: flex;
    flex-direction: row;
    justify-content: space-between;
}
.brand-title {
    font-size: 40px;
    font-weight: bold;
    font-family: Arial;
    color:red;
    text-shadow: 1px 1px 1px white;
}
.btn-signin {
    background-color: red;
    color:white;
    border-radius: 5px;
    padding: 10px;
    font-family: Arial;
    text-align: center;
}
.language {
    border:1px solid white;
    padding: 10px;
    border-radius: 5px;
```

```
    margin-right: 20px;
    background-color: rgba(0,0,0,0.6);
}
.signin-container {
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    align-items: center;
}
section {
    color:white;
    text-align: center;
    padding-top: 100px;
    font-family: Arial;
}
.section-title {
    font-size:50px;
    font-weight: bold;
    padding-bottom: 20px;
}
.section-subtitle {
    font-size: 25px;
    padding-bottom: 20px;
```

```
        }
        .main-container {
            padding-top: 30px;
            display: flex;
            flex-direction: row;
            justify-content: center;
        }
        .email {
            background-color: rgba(0,0,0,0.6);
            padding: 20px;
            width: 250px;
            text-align: left;
            border:1px solid gray;
        }
        .btn-start {
            background-color: red;
            color:white;
            padding: 20px;
            width: 130px;
            margin-left: 30px;
        }
    </style>
</head>
```

```
<body background="./images/netflix-banner.jpg">
  <div class="shade-cover">
    <header>
      <div>
        <span class="brand-title">NETFLIX</span>
      </div>
      <div class="signin-container">
        <div class="language">
            <span class="bi bi-translate"></span>
            <span>English</span>
            <span class="bi bi-chevron-down"></span>
        </div>
        <div class="btn-signin">
            <span>Sign In</span>
        </div>
      </div>
    </header>
    <section>
        <div class="section-title">
            Unlimited movies, TV<br> shows and more
        </div>
        <div class="section-subtitle">
            Starts at ₹149. Cancel at any time.
```

```
        </div>
        <main>
            <div>
                Ready to watch? Enter your email to create or restart your membership.
            </div>
            <div class="main-container">
                <div class="email">
                    <span>Email address</span>
                </div>
                <div class="btn-start">
                    Get Started <span class="bi bi-chevron-right"></span>
                </div>
            </div>
        </main>
    </section>
  </div>
</body>
</html>
```

Dialog, Article, Body Semantics

8. Article <article>

- It is a container for displaying latest trends, notifications etc.
- It is used to publish highlights about the current context.

Syntax:

```
<article>

    … highlights …

</article>
```

- You can set animation for content in page by using <marquee> element.

- However <marquee> is depreciated and no longer recommended for modern web.

Syntax:

```
<marquee>  … your content …   </marquee>
```

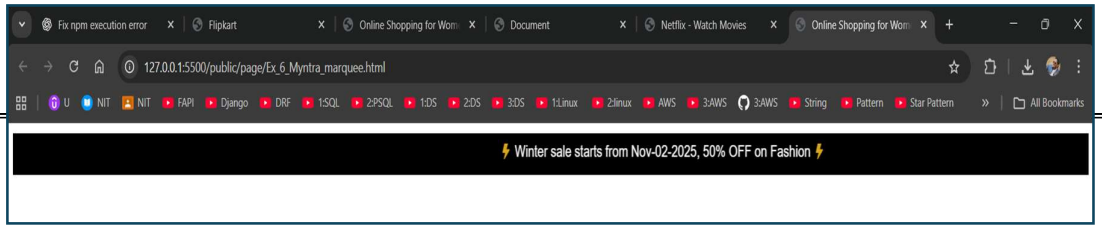
Attributes:

scrollamount	It sets marquee speed from 1 to 100.
direction	It sets scrolling direction left, right, up, down.
behavior	It changes from scroll to slide. [alternate behavior]
loop	It specifies the number of time to display after load.
bgcolor	It sets background color.
width & height	It sets width & height for marquee content.

Syntax:

```
<marquee scrollamount="20" direction="right"  loop="2"> </marquee>

<marquee behavior="alternate"> </marquee>
```

Ex_6_Myntra_marquee.html



```
<!DOCTYPE html>
<html lang="en-in">
<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Online Shopping for Women, Men, Kids </title>

    <link rel="stylesheet" href="../../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <style>

        article {

            margin-top: 10px;

            background-color: black;

            padding: 10px;

            text-align: center;

            font-family: Arial;

            color:white;

        }
```

```
        .bi-lightning-fill {

            color:goldenrod;

        }

    </style>

</head>

<body>

    <section>

        <article>

            <marquee scrollamount="20" behavior="alternate">

                <span class="bi bi-lightning-fill"></span> Winter sale starts from Nov-02-
2025, 50% OFF on Fashion <span class="bi bi-lightning-fill"></span>

            </marquee>         </article> </section>
</body> </html>
```

9. <dialog>

- It is a popup window inside page. - It is a component of page that can open and close dynamically. - It is used for push notifications, alerts, modals inside page. - It opens with content in backdrop. - Dialog is by default hidden, you have to display using “open” attribute.

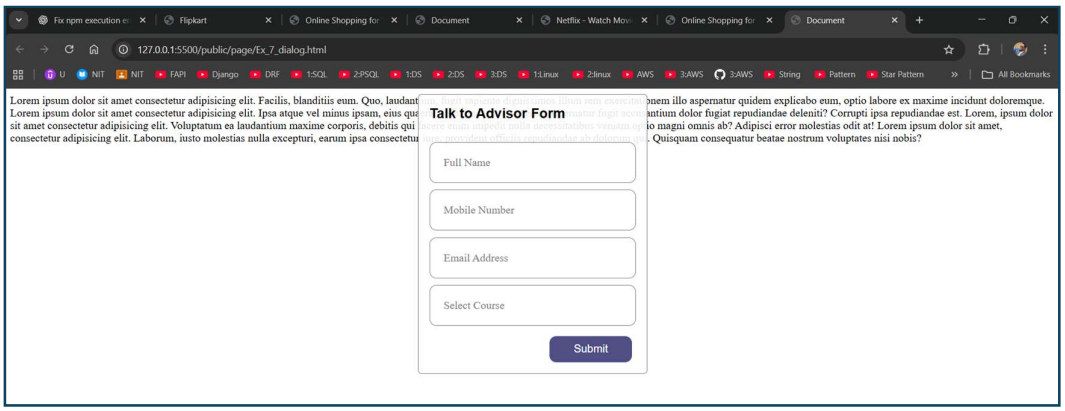
Syntax:

```
<dialog  open>… your content …</dialog>
```

- “open” is a boolean attribute with “true” as default value when defined.

- If open is not defined then by default it is “false”.

Ex_7_dialog.html




```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <style>

        dialog {

            width: 300px;

            border:1px solid gray;

            border-radius: 5px;

            background-color: rgba(255,255,255,0.9);

        }

        .dialog-header {
```

```
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    margin-bottom: 30px;
    font-family: Arial;
    font-size: 20px;
    font-weight: bold;
}
.control {
    padding: 20px;
    margin-bottom: 10px;
    border:1px solid gray;
    border-radius: 10px;
    color:gray;
}
.dialog-footer {
    padding: 5px;
}
.btn-submit {
    width: 100px;
    background-color: rgb(79, 79, 133);
    color:white;
    font-family: Arial;
```

```
            font-size: 16px;
            padding: 10px;
            text-align: center;
            border-radius: 10px;
            float: right;
        }
    </style>
</head>
<body>
    <dialog open>
        <div class="dialog-header">
            <span>Talk to Advisor Form</span>
            <span class="bi bi-x-lg"></span>
        </div>
        <div class="dialog-body">
            <div class="control">Full Name</div>
            <div class="control">Mobile Number</div>
            <div class="control">Email Address</div>
            <div class="control">Select Course</div>
        </div>
        <div class="dialog-footer">
            <div class="btn-submit">Submit</div>
        </div>
```

</dialog>

Lorem ipsum

</body>

</html>

Summary

<header> <footer> <section> <nav> <aside> <address> <article> <main> <dialog>

Body Entities and Semantics

1. Line Break

2. Blank Space   [non-breakable space]

3. Pre formatted <pre> It keeps the formats exactly as defined in editor. It is about line break & spaces you defined in editor.

4. Code Block <code> It is used to enclose any code block to keep is SEO friendly.

5. Variables <var> It is used for variables that store data in programming.

Ex_8_br_space_code_variavle.html:

```
var createHelloWorld = function(){

  return function(...args){

  }

}
```

<!DOCTYPE html>

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    var {
      color:yellow;
    }
    pre {
      background-color: rgb(14, 13, 13);
      color:white;
      width: 400px;
    }
  </style>
</head>
<body>
    <pre>
     <code>
        var <var>createHelloWorld</var> = function(){


          return function(...args){
```

```
        }


      }

    </code>

   </pre>

</body>

</html>
```

Paragraphs & Blockquotes

6. <samp> for sample text. It is used for note or output to display as result etc.

7. <small> for smaller text.

8. <big> for large text.

https://leetcode.com/problems/create-hello-world-function/description/?envType=study-plan-v2&envId=30-days-of-javascript
Ex_9_leetcode.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

  <style>

    samp {
```

```
        background-color: rgb(232, 229, 229);

        padding: 2px;

        border-radius: 2px;

      }

   </style>

</head>

<body>

     <div>

        Write a function <samp><small>createHelloWorld</small></samp>. It must always return <samp><big>"Hello World"</big></samp>.

     </div>

</body>

</html>
```

9. <p> for paragraph.

10. <blockquote> for block quote, it is used to summarize the context.

FAQ's:

1. How to set first line indent in a paragraph or blockquote?

A. By using CSS "text-indent" attribute.

```
p, blockquote { text-indent : 200px;   }
```

2. How to set space between lines, words & chars?

A. By using CSS attributes

line-height : sets space between line

word-spacing : sets space between words

letter-spacing : sets space between letters

Syntax:

```
p, blockquote {
    line-height : 30px;
    word-spacing: 5px;
    letter-spacing: 2px;
}
```

3. How to set a drop cap?

A. By using CSS "::first-letter" selector you can select the first letter of paragraph.

Then you can set "float" as left with line-height and font size.

```
p :: first-letter {
    font-size : 50px;
    font-weight: bold;
    font-family: Arial;
    float: left;
    line-height:20px;
}
```

4. How to create scrollable paragraph content in page?

A. By using CSS "overflow" set to scroll or auto.

Syntax:

```
div, p, blockquote {
```

```
                border : 1px solid gray;
                width: 400px;
                height: 300px;
                padding: 20px;
                text-align: justify;
                overflow: auto;
            }
```

Ex_10_scrollable.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    blockquote {
      padding-left: 300px;
      padding-right: 300px;
      font-style: italic;
      text-indent: 100px;
      line-height: 35px;
      word-spacing: 5px;
      letter-spacing: 2px;
```

```
        }
        .para-1::first-letter {
            font-size: 60px;
            font-weight: bold;
            font-family: Arial;
            float: left;
            line-height: 25px;
        }
        .term-header {
           display: block;
           font-size: 25px;
           font-weight: bold;
        }
        .terms {
           border: 1px solid gray;
           padding: 20px;
           text-align: justify;
           height: 300px;
           width: 500px;
           overflow: auto;
        }
    </style>
</head>
```

```
<body>
  <blockquote align="justify">
     Lorem ipsum
  </blockquote>
  <p align="justify" class="para-1">
     Lorem ipsum
  Lorem ipsum
  </p>
  <div>
    <span class="term-header">Terms of Service</span>
    <div class="terms">
      <p>Lorem.</p> <p>Lorem.</p> <p>Lorem.</p> <p>Lorem.</p>
    </div>
  </div>
</body>
</html>
```

5. How to set “ellipsis” is a paragraph to display more content?

A. By using following CSS attributes

white-space : It removes the text wrap with “nowrap” so that paragraph is displayed in single line.

overflow : It can hide the overflowing text.

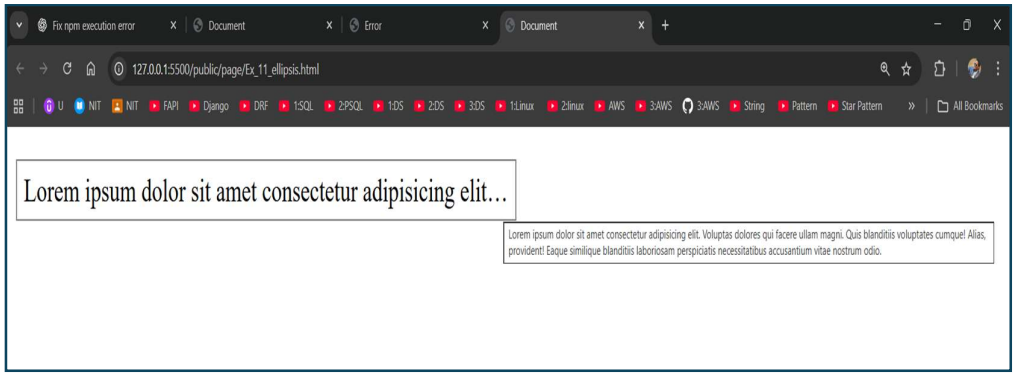
text-overflow : It sets “ellipsis” for paragraph.

Note: You can display screen tip in HTML using “title” attribute.

Syntax:

```
p {
    width: 300px;
    border:1px solid gray;
    padding: 5px;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
 }
<p title="your text">    your text </p>
```

Ex_11_ ellipsis.html:




```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
```

```
      p {
        border: 1px solid gray;
        width: 370px;
        padding: 5px;
        white-space: nowrap;
        overflow: hidden;
        text-overflow: ellipsis;
      }
   </style>
</head>
<body>
    <p title="Lorem ipsum.</p>
</body>
</html>
```

6. How to display paragraphs in multiple columns?

A. By using CSS columns attribute

columns	: It splits into specified number of columns.
column-gap	: It sets space between columns.
column-width	: It sets with for column.
column-rule	: It sets line between columns with size, style & color.

Syntax:

```
        div {
            columns : 4;
            column-gap: 20px;
            column-rule: 1px dotted gray;
        }
```

Ex_12_ paragraphs_in_multiple_columns.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>
    .news {
      columns: 4;
      column-gap: 20px;
      column-rule: 1px dotted gray;
```

```
        width: 820px;
        text-align: justify;
        font-size: 12px;
        font-family: Arial;
        padding: 20px;
      }
      .title {
        font-size: 14px;
        font-weight: bold;
      }
      .news-title {
        font-size: 73px;
      }
      .update {
        background-color: black;
        color:white;
        padding: 10px;
      }
    </style>
  </head>
  <body>
    <div class="news-title">Army hiring up in TG, south</div>
    <div class="news">
```

```
    <span class="title">OUR DC CORRESPONDENT<br>MONDAY NOV 27</span>

    <p>Lorem </p> <p>Lorem </p>

    <div class="update">Latest Updates</div>

    <p>Lorem </p> <p>Lorem ipsum </p><p>Lorem ipsum </p>
    <p>Lorem </p> <p>Lorem ipsum </p><p>Lorem ipsum </p>

  </div>

</body>

</html>
```

Text Formatting

11. Headings

- Headings are defined using <hn> element. - “n” refers to level number from 1 to 6.

Syntax: <h1> Heading Level 1 </h1> <h6> Heading Level 6 </h6>

- Headings are required to make the topics in page SEO friendly. So that crawlers can summarize the topics in page.

-What are the pre-defined styles applied to heading?

A. Heading comprises of

a) font-size b) font-weight c) display:block

-How to remove any pre-defined style?

A. By using CSS inheritance value “unset”.

Syntax:

```
h1 {   font-weight: unset;   }
h1 {    all : unset;  }
```

12. Text Formatting Elements

Note: <font> element is deprecated and no longer recommend in modern web design.

<b>	Bold
	Strong similar to bold
<i>	Italic
	Emphasized similar to italic
<u>	Underline
<ins>	Inserted, similar to underline.
<strike>	Strikeout
<del>	Deleted, similar to strikeout.
<sup>	Super Script
<sub>	Sub Script

FAQ: Why we have different elements for same purpose?

**<b> **

<u> <ins> etc.

Ans: Document have 2 modes

a) Design b) Review

-Design mode is for production. Review mode is for testing.

The review elements are

, , <ins>, <del>

Ex_13_text_formating.html:

Bold is similar to **Strong.**
Italic is similar to *Emphasized*
Underline is similar to Inserted.
~~Strikeout~~ is similar to ~~Deleted.~~
5th Anniversary
H_2O

```
<body>

    <div>

      <b>Bold</b> is similar to <strong>Strong.</strong>

    </div>

    <div>

      <i>Italic</i> is similar to <em>Emphasized</em>

    </div>

    <div>

      <u>Underline</u> is similar to <ins>Inserted</ins>.

    </div>

    <div>

      <strike>Strikeout</strike> is similar to <del>Deleted.</del>

    </div>

    <div>

      5<sup>th</sup> Anniversary

    </div>

    <div>

      H<sub>2</sub>O
```

```
</div>
```

==========

Lists in HTML

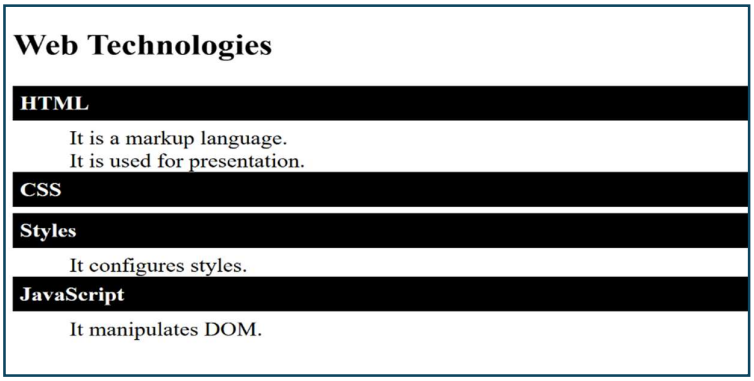
1. Data List with Terms & Definitions

```
<dl>        data list
<dt>        data term
<dd>        data definition
```

Syntax:

```
<dl>
    <dt> Term-1 </dt>
    <dd> Definition </dd>
    <dt> Term-2 </dt>
    <dd> Definition </dd>
</dl>
```

Ex_14_List_Terms_Definitions.html:



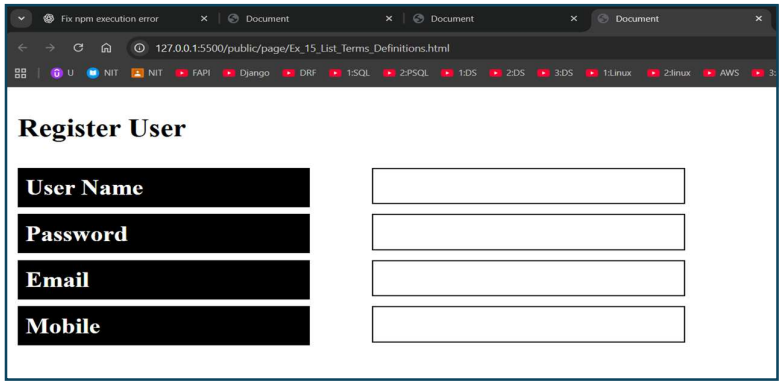

```
<style>
    dt {
```

```
            font-weight: bold;
            background-color: black;
            color:white;
            padding: 5px;
            margin-bottom: 5px;
        }
    </style>
</head>
<body>
    <h2>Web Technologies</h2>
    <dl>
        <dt>HTML</dt>
        <dd>It is a markup language. </dd>
        <dd>It is used for presentation.</dd>
        <dt>CSS </dt>
        <dt>Styles</dt>
        <dd>It configures styles.</dd>
        <dt>JavaScript</dt>
        <dd>It manipulates DOM. </dd>
    </dl>
```

Note: You can split the data list into grid with 2 columns, if list comprises of single term with relative single definition.

Syntax: `dl { display: grid; grid-template-columns: 6fr 6fr; }`

Ex_15_List_Terms_Definitions.html:




```
<style>
    dt {
        font-weight: bold;
        background-color: black;
        color:white;
        padding: 5px;
        margin-bottom: 5px;
    }
    .control {
        border:1px solid black;
        width: 200px;
        height: 25px;
    }
    dd {
        margin-bottom: 5px;
    }
    dl {
```

```
            display: grid;
            grid-template-columns: 3fr 9fr;
        }
    </style>
</head>
<body>
    <h3>Register User</h3>
    <dl>
        <dt>User Name</dt>
        <dd><div class="control"></div></dd>
        <dt>Password</dt>
        <dd><div class="control"></div></dd>
        <dt>Email</dt>
        <dd><div class="control"></div></dd>
        <dt>Mobile</dt>
        <dd><div class="control"></div></dd>
    </dl> </body> </html>
```

CSS Sticky Position

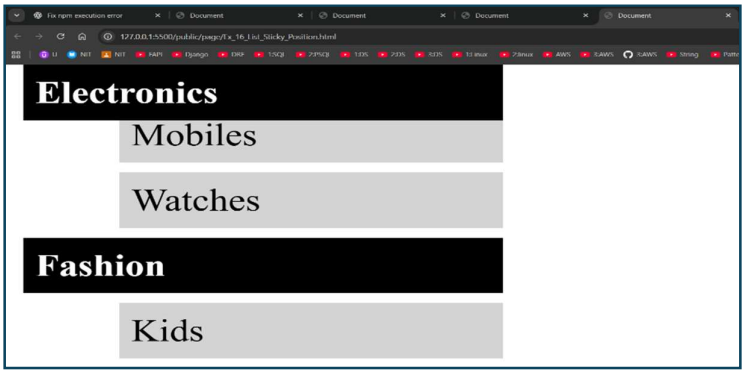
- CSS provides various positions for element.

  a) static b) fixed c) absolute d) relative e) sticky

- Sticky keeps the element according to normal of document.
- It sticks element at specified position. [top, left, right, bottom]
- It allows to scroll upto the defined position.

Syntax: dt { position: sticky; top: 0px; }

Ex_16_List_ Sticky_Position.html:




```
<style>
    dt {
        font-weight: bold;
        background-color: black;
        color:white;
        padding: 5px;
        margin-bottom: 5px;
        position: sticky;
        top: 0px;
    }
    dd {
        margin-bottom: 5px;
        background-color: lightgray;
        padding: 5px;
    }
    dl {
        width: 200px;
```

```
        }
    </style>
</head>
<body>
    <nav>
        <h3>Amazon</h3>
        <dl>
            <dt>Electronics</dt>
            <dd>Televisions</dd>
            <dd>Mobiles</dd>
            <dt>Fashion</dt>
            <dd>Kids</dd>
            <dd>Women</dd>
        </dl>     </nav> </body> </html>
```

2. Ordered List: <ol> ordered list, <li> list item

Syntax:

```
<ol>  <li> Item-1 </li>   <li> Item-2 </li>  </ol>
```

- You can change the numbering type by using “type” attribute with values: 1, A, a, i, I

Syntax: <ol type=“A”>

-You can “start” numbering from specific level by using “start” attribute. It always refers to level number.

Syntax:

```
<ol  type=“1”  start=“101”>  <ol  type=“A”  start=“5”>
```

- You can set reverse numbering by using “reversed” attribute.

Syntax: <ol type="1" reversed>

- You can configure nested numbering, but make sure that child list is inside <li> element.

WHATWG, PMD, ESLINT

Syntax:

```
<ol>
    <li>
        Parent
        <ol>
            <li> Child </li>
        </ol>
    </li>
</ol>
```

Ex_17_orderd_list.html:

1. HTML
    a. Normal Elements
        i. Bold
    b. Void Elements
2. JavaScript

```
<body>
    <ol type="1">
        <li>HTML
```

```
        <ol type="a">
        <li>Normal Elements
            <ol type="i">
                <li>Bold</li>
            </ol>
        </li>
        <li>Void Elements</li>
       </ol>
      </li>


      <li>JavaScript</li>
   </ol> </body> </html>
```

- You can display list in multiple columns by using CSS

  a) display : grid b) columns

- You can display side by side by using CSS display inline or display flex.

Ex_18_orderd_list.html:

1. _____ element will not have end token.

a. Normal b. Void c. RC Data d. Raw Text

```
   <style>
      .child-list {
        display: flex;
```

```
            justify-content: space-around;
            margin-top: 20px;
            font-size: 12px;
        }
    </style>
</head>
<body>
    <ol>
        <li>_____ element will not have end token.
            <ol type="a" class="child-list">
                <li>Normal</li>
                <li>Void</li>
                <li>RC Data</li>
                <li>Raw Text</li>
            </ol>
        </li>
    </ol> </body></html>
```

Ex_19_orderd_list.html:

1. _____ element will not have end token.

a. Normal
b. Void
c. RC Data
d. Raw Text

```
    <style>
```

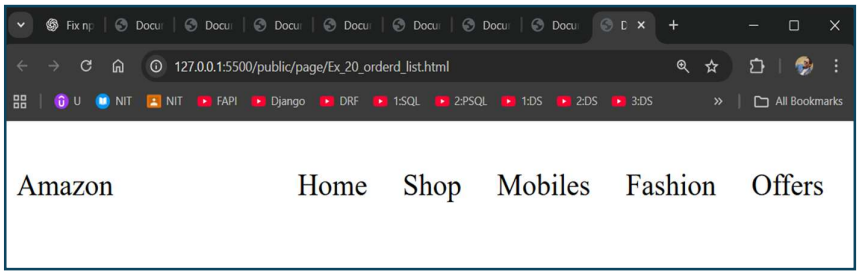
```
        .child-list {
            columns: 2;
            margin-top: 20px;
            font-size: 12px;
        }
        li {
            margin-bottom: 5px;
        }
    </style>
</head>
<body>
    <ol>
        <li>_____ element will not have end token.
            <ol type="a" class="child-list">
                <li>Normal</li>
                <li>Void</li>
                <li>RC Data</li>
                <li>Raw Text</li>
            </ol>
        </li> </ol></body></html>
```

- You can remove numbering for list by using CSS “list-style” attribute set to “none”.

Syntax:

```
ol { list-style : none;  }
```

Ex_20_nav_bar_by_order_list.html:




```
    <style>
        ol {
            list-style: none;
            display: flex;
            justify-content: space-between;
        }
        nav {
            display: flex;
            justify-content: space-between;
            align-items: center;
        }
        li {
            margin-left: 10px;
            margin-right: 10px;
        }
    </style>
</head>
<body>
    <nav>
```

```
        <div>Amazon</div>
        <ol>
            <li>Home</li>
            <li>Shop</li>
            <li>Mobiles</li>
            <li>Fashion</li>
            <li>Offers</li>
        </ol>
    </nav>
</body>
</html>
```

3. **Unordered List**

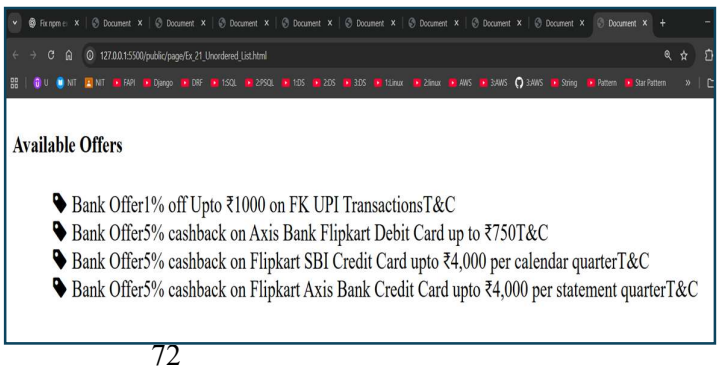
- It sets bulleted list with symbols.
- The type comprises of “disc, square & circle”.

```
<ul>        Unordered List
<li>        List Item
```

Syntax:

```
<ul type=“circle | disc | square”>
```

Ex_21_Unordered_List.html :



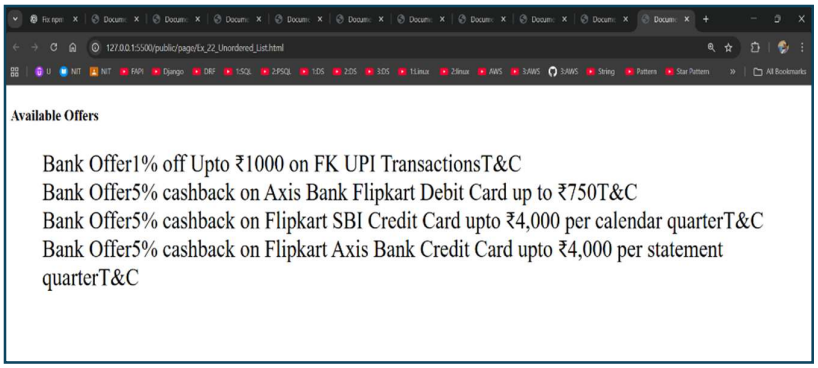
```
    <style>
        ul {
            list-style: none;
        }
    </style>
</head>
<body>
    <h4>Available Offers</h4>
    <ul>
        <li class="bi bi-tag-fill"> Bank Offer1% off Upto ₹1000 on FK UPI TransactionsT&C</li>
        <li class="bi bi-tag-fill"> Bank Offer5% cashback on Axis Bank Flipkart Debit Card up to ₹750T&C</li>
        <li class="bi bi-tag-fill"> Bank Offer5% cashback on Flipkart SBI Credit Card upto ₹4,000 per calendar quarterT&C</li>
        <li class="bi bi-tag-fill"> Bank Offer5% cashback on Flipkart Axis Bank Credit Card upto ₹4,000 per statement quarterT&C</li>
    </ul> </body> </html>
```

- You can set your custom image as bullet by using CSS “list-style-image” attribute.

Syntax:

```
ul { list-style: none; list-style-image: url(“path”);  }
```

Ex_22_Unordered_List.html:



```
    <style>
        ul {
            list-style: none;
            list-style-image: url("./images/bullet.jpg");
            font-size: 25px;
        }
    </style>
</head>
<body>
    <h4>Available Offers</h4>
    <ul>
        <li > Bank Offer1% off Upto ₹1000 on FK UPI TransactionsT&C</li>
        <li> Bank Offer5% cashback on Axis Bank Flipkart Debit Card up to ₹750T&C</li>
        <li> Bank Offer5% cashback on Flipkart SBI Credit Card upto ₹4,000 per
calendar quarterT&C</li>
        <li> Bank Offer5% cashback on Flipkart Axis Bank Credit Card upto ₹4,000 per
statement quarterT&C</li>
    </ul> </body> </html>
```

Containers & Images

Semantic Containers for a group:

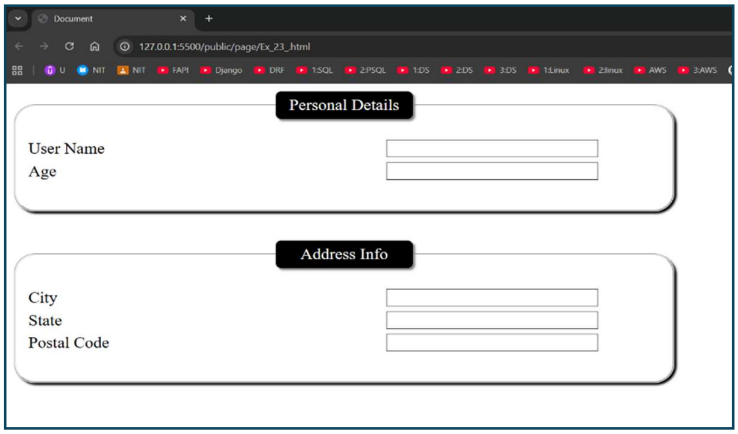
1. Fieldset & Legend

<fieldset>	It is a frame for group of topics or content.
<legend>	It is a caption for group.

Syntax:

```
<fieldset>

    <legend> Title </legend>

    … your content…

</fieldset>
```

Ex_23_Fieldset_and_Legend.html:




```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <style>

        dl {

            display: grid;
```

```
    grid-template-columns: 6fr 6fr;
}
.control {
    width: 200px;
    border:1px solid black;
}
dt, dd {
    margin-bottom: 5px;
}
fieldset {
    width: 600px;
    margin-bottom: 30px;
    box-shadow: 2px 2px 2px black;
    border-radius: 20px;
}
legend {
    background-color: black;
    color:white;
    width: 120px;
    padding: 5px;
    text-align: center;
    border-radius: 5px;
    box-shadow: 2px 2px 2px gray;
```

```
        }
    </style>
</head>
<body>
    <fieldset>
        <legend>Personal Details</legend>
        <dl>
        <dt>User Name</dt>
        <dd class="control"></dd>
        <dt>Age</dt>
        <dd class="control"></dd>
        </dl>
    </fieldset>
    <fieldset>
        <legend>Address Info</legend>
        <dl>
        <dt>City</dt>
        <dd class="control"></dd>
        <dt>State</dt>
        <dd class="control"></dd>
        <dt>Postal Code</dt>
        <dd class="control"></dd>
    </dl>
```

```
    </fieldset>
</body>
</html>
```

Ex_24_Fieldset_and_Legend.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        fieldset {
            border-left: none;
            border-right: none;
            border-bottom: none;
            width: 200px;
            border-top: 2px solid gray;
        }
        legend {
```

```
            font-size: 25px;
            color:gray;
            font-weight: bold;
            padding-left: 10px;
            padding-right: 10px;
            text-align: center;
        }
    </style>
</head>
<body>
    <fieldset>
        <legend> OR </legend>
    </fieldset>
</body>
</html>
```

2. Details & Summary

<details> It is used to configure expandable & collapsable container.

<summary> It sets caption for details.

Syntax:

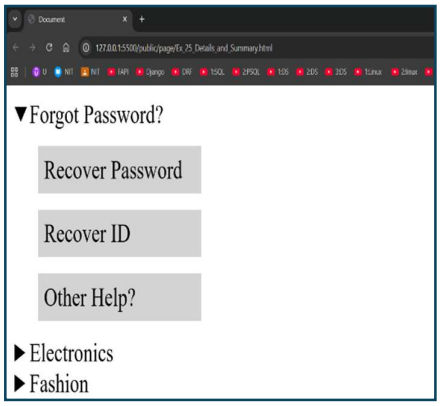
```
<details>
    <summary> Title </summary>  … your content …
</details>
```

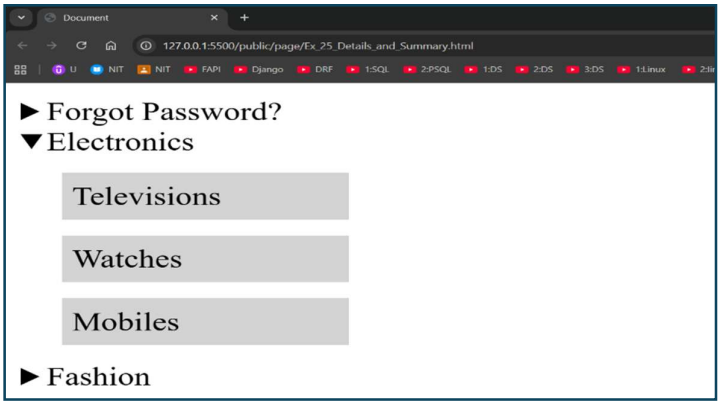
- You can keep the details always open by using “open” attribute.

Syntax:

```
<details open>  …  </details>
```

Ex_25_ Details_and_Summary.html:

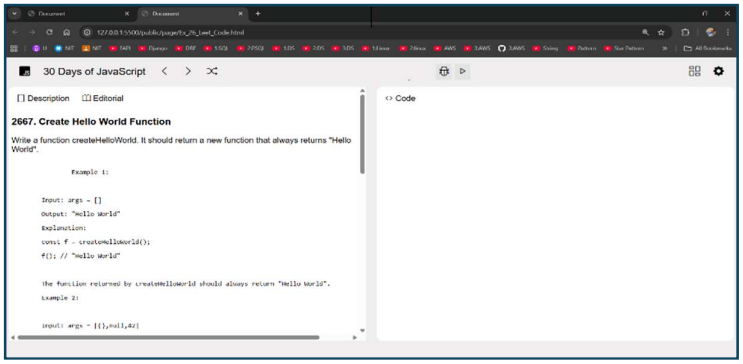





```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

        details div {

            margin-left: 20px;

            margin-bottom: 10px;

            margin-top: 10px;

            width: 130px;

            background-color: lightgray;
```

```
            padding: 5px;
        }
    </style>
</head>
<body>
    <details open>
        <summary> Forgot Password? </summary>
        <div>Recover Password</div> <div>Recover ID</div><div>Other Help?</div>
    </details>
    <nav>
        <details>
            <summary>Electronics</summary>
            <div>Televisions</div><div>Watches</div><div>Mobiles</div>
        </details>
        <details>
            <summary>Fashion</summary>
            <div>Men</div> <div>Women</div> <div>Kids</div>
        </details>
    </nav> </body> </html>
```

Ex_26_Leet_Code.html:



```
<style>
    body {
        background-color: rgb(239, 237, 237);
        font-family: Arial;
    }
    header {
        padding: 10px;
        font-size: 20px;
        display: flex;
        justify-content: space-between;
    }
    header span {
        padding-left: 10px;
        padding-right: 10px;
    }
    .bi-bug, .bi-play {
        background-color: rgb(228, 227, 227);
        padding: 5px;
    }
    section {
        display: grid;
        grid-template-columns: 6fr 6fr;
        grid-gap: 20px;
```

```
            margin-top: 10px;
        }
        .col-1, .col-2 {
            height: 520px;
            background-color: white;
            border-radius: 10px;
            overflow: auto;
            padding: 5px;
        }
        .col-1-header {
            padding: 10px;
        }
        .col-1-header span {
            padding-right: 20px;
        }
        .col-2-header span {
            padding-right: 20px;
        }
        .col-2-header {
            padding: 10px;
        }
    </style>
</head>
```

```
<body>
    <header>
        <div>
            <span class="bi bi-javascript"></span>
            <span>30 Days of JavaScript</span>
            <span class="bi bi-chevron-left"></span>
            <span class="bi bi-chevron-right"></span>
            <span class="bi bi-shuffle"></span>
        </div>
        <div>
            <span class="bi bi-bug"></span>
            <span class="bi bi-play"></span>
        </div>
        <div>
            <span class="bi bi-columns-gap"></span>
            <span class="bi bi-gear-fill"></span>
        </div>
    </header>
    <section>
        <div class="col-1">
            <div role="header" class="col-1-header">
                <span class="bi bi-file"> Description</span>
                <span class="bi bi-book"> Editorial </span>
```

```
</div>

<h3>2667. Create Hello World Function</h3>

<p>Write a function createHelloWorld. It should return a new function that always returns "Hello World".</p>

<pre>

Example 1:



Input: args = []

Output: "Hello World"

Explanation:

const f = createHelloWorld();

f(); // "Hello World"



The function returned by createHelloWorld should always return "Hello World".

Example 2:



Input: args = [{},null,42]

Output: "Hello World"

Explanation:

const f = createHelloWorld();

f({}, null, 42); // "Hello World"



Any arguments could be passed to the function but it should still always return "Hello World".
```

Constraints:

0 <= args.length <= 10

Input: args = []

Output: "Hello World"

Explanation:

```
const f = createHelloWorld();

f(); // "Hello World"
```

The function returned by createHelloWorld should always return "Hello World".

Example 2:

Input: args = [{},null,42]

Output: "Hello World"

Explanation:

```
const f = createHelloWorld();

f({}, null, 42); // "Hello World"
```

Any arguments could be passed to the function but it should still always return "Hello World".

```
        Constraints:



        0 <= args.length <= 10



            </pre>

        </div>

        <div class="col-2">

            <div role="header" class="col-2-header">

                <span class="bi bi-code"> Code</span>

            </div>

        </div>

    </section> </body> </html>
```

Images in HTML

Image Element:

- Image is embedded into a page by using “<img>” element. - If you want image SEO friendly then you can keep in <figure> container with <figcaption>

Syntax:

```
        <figure>

                <img>

                <figcaption> Keyword </figcaption>

        </figure>
```

- Image element have following attributes

1. src : It defines name and location of image [virtual path]

2. alt : It defines the alternative text to display when image fails to load.

3. title : It defines the title to display when mouse is over the image.

4. width & height : Sets width and height for image.

5. border : Sets border for image.

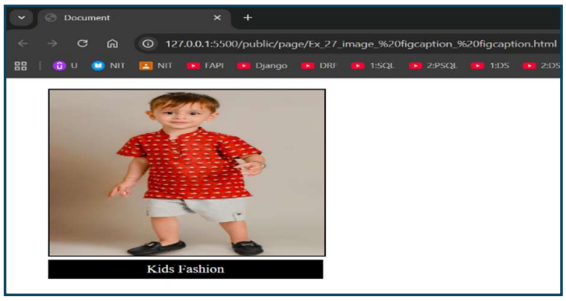
6. align : It aligns the image left or right. It can wrap text around image.

7. hspace & vspace: Sets horizontal and vertical padding.

Syntax:

```
<img src="./images/poster.jpg"  width="200" height="200" alt="alternative text"
  title="screen tip"  border="2">
```

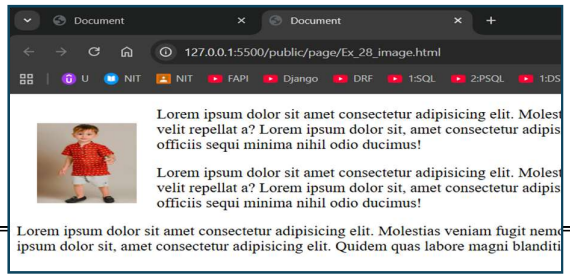
Ex_27_image_ figcaption_ figcaption.html:




```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Document</title>
   <style>
```

```
        figure {
            width: 300px;
            height: 300px;
        }
        figcaption {
            background-color: black;
            color:white;
            text-align: center;
            padding: 5px;
        }
    </style>
</head>
<body>
   <figure>
       <img border="2" alt="Kids Fashion" title="50% OFF - Winter Sale Starts Nov-10-2025" src="./images/kids-fashion.jpg" width="300" height="250">
       <figcaption>Kids Fashion</figcaption>
   </figure>
</body>
</html>
```

Ex_28_image.html:



```
<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>  </style>

</head>

<body>

   <img align="left" hspace="20" vspace="20" src="./images/kids-fashion.jpg"
width="100" height="100">

   <p>Lorem </p><p>Lorem </p><p>Lorem </p>


</body></html>
```

8. crossorign : It refers to CORS [Cross Origin Resource Sharing].

You can set restrictions for images from other origins.

It uses the values:

a) anonymous

b) use-credentials

Syntax: <img src="server.com/path/pic.jpg" crossorigin="use-credentials">

9. decoding : It defines how image has to load along with other contents in page. It can use a) async b) sync c) auto

A:Asynchronous is unblocking techniques. It loads along with others.
B: Synchronous is a blocking technique. It blocks others while loading.

Syntax: <img src="some_path" decoding="async | sync | auto">

10. importance : It sets priority for image, which can be low, high or auto.

Syntax: <img src="some_path" importance="low | high | auto">

11. loading : It can set loading style for image, which can be

a) lazy b) eager

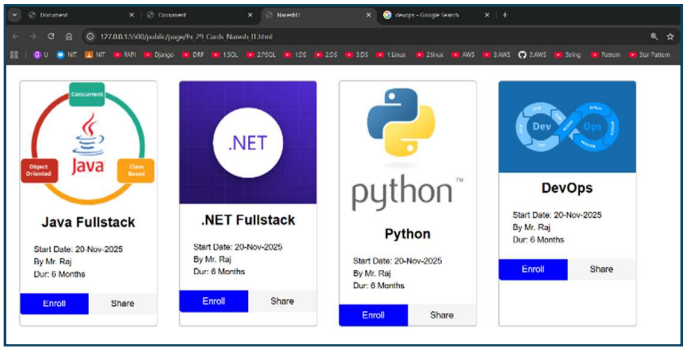
Syntax: <img src="some_path" loading="lazy | eager">

12. srcset : It configures a set of images for single image element.

Images can change dynamically.

Syntax: <img srcset="collection_of_images">

Ex_29_Cards_Naresh_IT.html:




```
<style>
    .card {
        width: 250px;
        border:1px solid gray;
        border-radius: 5px;
        font-family: Arial;
        margin: 20px;
    }
    .card-img {
        width: 100%;
        border-top-left-radius: 5px;
```

```
    border-top-right-radius: 5px;
}
.card-body {
    padding: 10px;
}
.course-title {
    font-size: 25px;
    font-weight: bold;
    padding-bottom: 10px;
    text-align: center;
}
.ratings {
    text-align: center;
    color:goldenrod;
}
.course-details {
    list-style: none;
    margin-left: -25px;
}
li {
    margin-bottom: 5px;
    font-size: 15px;
}
```

```
        .card-footer {
            display: grid;
            grid-template-columns: 6fr 6fr;
            text-align: center;
        }
        .btn-enroll {
            background-color: blue;
            color:white;
            padding: 10px;
            border-bottom-left-radius: 5px;
        }
        .btn-share {
            background-color: whitesmoke;
            padding: 10px;
            border-bottom-right-radius: 5px;
        }
        main {
            display: flex;
        }
    </style>
</head>
<body>
   <main>
```

```
<div class="card">
  <div class="card-header">
      <img src="./images/Java.png" class="card-img">
  </div>
  <div class="card-body">
      <div class="course-title">Java Fullstack</div>
      <div class="ratings">
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
      </div>
      <ul class="course-details">
         <li class="bi bi-calendar-date"> Start Date: 20-Nov-2025 </li>
         <li class="bi bi-person"> By Mr. Raj </li>
         <li class="bi bi-clock"> Dur: 6 Months </li>
      </ul>
  </div>
  <div class="card-footer">
      <div class="btn-enroll">Enroll</div>
      <div class="btn-share"> <span class="bi bi-share"></span> Share</div>
  </div>
```

```
</div>
<div class="card">
  <div class="card-header">
      <img src="./images/dot-net.png" class="card-img">
  </div>
  <div class="card-body">
      <div class="course-title">.NET Fullstack</div>
      <div class="ratings">
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
      </div>
      <ul class="course-details">
         <li class="bi bi-calendar-date"> Start Date: 20-Nov-2025 </li>
         <li class="bi bi-person"> By Mr. Raj </li>
         <li class="bi bi-clock"> Dur: 6 Months </li>
      </ul>
  </div>
  <div class="card-footer">
      <div class="btn-enroll">Enroll</div>
      <div class="btn-share"> <span class="bi bi-share"></span> Share</div>
```

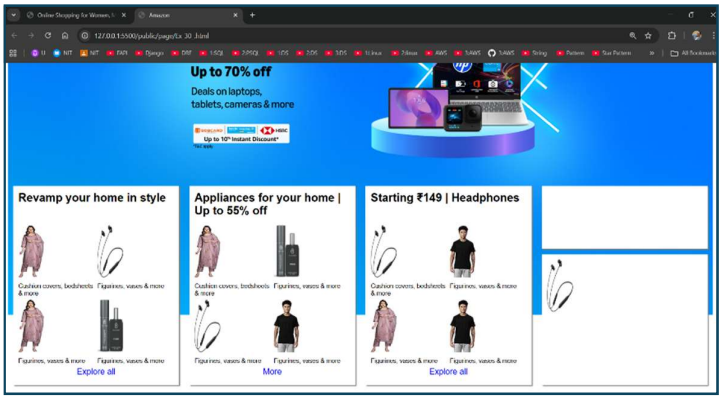
```
  </div>
</div>
<div class="card">
  <div class="card-header">
      <img src="./images/python.png" class="card-img">
  </div>
  <div class="card-body">
      <div class="course-title">Python</div>
      <div class="ratings">
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
      </div>
      <ul class="course-details">
         <li class="bi bi-calendar-date"> Start Date: 20-Nov-2025 </li>
         <li class="bi bi-person"> By Mr. Raj </li>
         <li class="bi bi-clock"> Dur: 6 Months </li>
      </ul>
  </div>
  <div class="card-footer">
      <div class="btn-enroll">Enroll</div>
```

```
      <div class="btn-share"> <span class="bi bi-share"></span> Share</div>
  </div>
</div>
<div class="card">
  <div class="card-header">
      <img src="./images/devOps.jpg" class="card-img">
  </div>
  <div class="card-body">
      <div class="course-title">DevOps</div>
      <div class="ratings">
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
          <span class="bi bi-star-fill"></span>
      </div>
      <ul class="course-details">
         <li class="bi bi-calendar-date"> Start Date: 20-Nov-2025 </li>
         <li class="bi bi-person"> By Mr. Raj </li>
         <li class="bi bi-clock"> Dur: 6 Months </li>
      </ul>
  </div>
  <div class="card-footer">
```

```
            <div class="btn-enroll">Enroll</div>

            <div class="btn-share"> <span class="bi bi-share"></span> Share</div>

        </div>

      </div>

    </main>

</body>

</html>
```

Card Examples

Ex_30_Amazon_Cards.html




```
    <style>

        section {

            display: grid;

            grid-template-columns: 3fr 3fr 3fr 3fr;

            margin-top: -260px;

        }

        .card {

            background-color: white;
```

```
    margin: 10px;
    height: 360px;
    box-shadow: 2px 2px 2px gray;
    padding: 10px;
    font-family: Arial;
}
.card-img {
  width: 50px;
  height: 100px;
}
.row {
    display: grid;
    grid-template-columns: 6fr 6fr;
    grid-gap: 5px;
}
.card-header {
    height: 65px;
}
.card-title {
    font-weight: bold;
    font-size: 22px;
}
.card-img+div {
    font-size: 12px;
```

```
    margin-top: 5px;
    margin-bottom: 5px;
}
.card-footer {
    padding: 10px;
    text-align: center;
    color:blue;
}
.card-body {
    height: 260px;
}
.last-card-header {
    height: 150px;
}
.last-card {
    background-color: white;
    padding: 10px;
    box-shadow: 2px 2px 2px gray;
    margin: 10px;
}
.last-card-row1 {
    height: 100px;
}
```

```
        .last-card-row2 {
            height: 230px;
        }
    </style>
</head>
<body>
    <header>
        <img src="./images/amazon-banner.jpg" width="100%">
    </header>
    <section>
        <div class="card">
            <div class="card-header">
                <div class="card-title">Revamp your home in style</div>
            </div>
            <div class="card-body">
               <div class="row">
                    <div>
                        <img src="./images/a1.jpg" class="card-img">
                        <div>Cushion covers, bedsheets & more</div>
                    </div>
                    <div>
                        <img src="./images/a2.jpg" class="card-img">
                        <div>Figurines, vases & more</div>
```

```
                </div>
            </div>
            <div class="row">
                <div>
                    <img src="./images/a3.jpg" class="card-img">
                    <div>Figurines, vases & more</div>
                </div>
                <div>
                    <img src="./images/a4.jpg" class="card-img">
                    <div>Figurines, vases & more</div>
                </div>
            </div>
        </div>
        <div class="card-footer">
            <div>Explore all</div>
        </div>
    </div>
    <div class="card">
<div class="card-header">
            <div class="card-title">Appliances for your home | Up to 55% off</div>
        </div>
        <div class="card-body">
            <div class="row">
```

```
        <div>
            <img src="./images/a5.jpg" class="card-img">
            <div>Cushion covers, bedsheets & more</div>
        </div>
        <div>
            <img src="./images/a6.jpg" class="card-img">
            <div>Figurines, vases & more</div>
        </div>
    </div>
    <div class="row">
        <div>
            <img src="./images/a7.jpg" class="card-img">
            <div>Figurines, vases & more</div>
        </div>
        <div>
            <img src="./images/a8.jpg" class="card-img">
            <div>Figurines, vases & more</div>
        </div>
    </div>
</div>
<div class="card-footer">
    <div>More</div>
</div>
```

```
        </div>
        <div class="card">
<div class="card-header">
            <div class="card-title">Starting ₹149 | Headphones</div>
          </div>
          <div class="card-body">
            <div class="row">
                <div>
                    <img src="./images/a9.jpg" class="card-img">
                    <div>Cushion covers, bedsheets & more</div>
                </div>
                <div>
                    <img src="./images/a10.jpg" class="card-img">
                    <div>Figurines, vases & more</div>
                </div>
            </div>
            <div class="row">
                <div>
                    <img src="./images/a11.jpg" class="card-img">
                    <div>Figurines, vases & more</div>
                </div>
                <div>
                    <img src="./images/a12.jpg" class="card-img">
```

```
                <div>Figurines, vases & more</div>
            </div>
        </div>
    </div>
    <div class="card-footer">
        <div>Explore all</div>
    </div>
</div>
<div>
    <div class="last-card last-card-row1">
    </div>
    <div class="last-card last-card-row2">
        <img src="./images/a13.png" class="card-img" height="160">
    </div>
</div>
</section>
</body>
</html>
```

Ex_31_Myntra.html

```
<!DOCTYPE html>

<html lang="en-in">
```

```
<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Online Shopping for Women, Men, Kids </title>

    <link rel="stylesheet" href="../../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <style>
        body{
            margin: 0px;
            width: 100hv;
        }
        main{
            margin: 50px;
        }
        header {

            font-size: 18px;

            font-family: Arial;

            display: flex;

            flex-direction: row;

            justify-content: space-between;
```

```
    padding: 20px;

    align-items: center;

    background-color: #f6f6f6;


    width: 97%;


}

.short-cuts {

    display: flex;

    flex-direction: row;

    justify-content: space-between;

    align-items: center;

}

.brand-nav {

    display: flex;

    flex-direction: row;

    justify-content: space-between;
```

```
    align-items: center;
}
.brand-title {
    font-size: 40px;
    font-weight: bold;
    font-family: Brush Script MT;
    color: red;
}
nav {
    padding-left: 20px;
}
nav span {
    padding-left: 10px;
    padding-right: 10px;
    font-weight: bold;
```

```
}

.search-bar {

    background-color: #d6d5d5;

    padding: 10px;

    font-size: 14px;

    width: 270px;

    display: flex;

    flex-direction: row;

    justify-content: space-between;

}

.short-cuts div {

    padding-left: 15px;

    padding-right: 15px;

    text-align: center;

    font-size: 20px;

}
```

```
.short-cut-text {
    font-size: 14px;
}

footer {
    display: grid;
    grid-template-columns: 2fr 2fr 2fr 2fr;
    font-family: Arial;
    padding: 5px;
    margin-left: 150px;

}
.footer-title {
    font-weight: bold;
    display: block;
    margin-bottom: 20px;
```

```
}

.footer-title~span {

    display: block;

    margin-bottom: 10px;

    color: gray;

}

aside span {

    font-size: 30px;

    padding-right: 10px;

}

.footer-content {

    width: 300px;

    margin-bottom: 30px;

}

address span {

    display: block;
```

```
    margin-bottom: 5px;
}
article {
    margin-top: 10px;
    background-color: black;
    padding: 10px;
    text-align: center;
    font-family: Arial;
    color: white;
}
.bi-lightning-fill {
    color: goldenrod;
}
.banner-data {
    box-shadow: 2px 2px 2px black;
```

```
    border-radius: 10px;

    background-color: white;

    margin-top: 10px;
    margin-bottom: 10px;

    height: 80px;

    border: 1px solid gainsboro;

    display: flex;

    justify-content: space-around;

    align-items: center;

}

.banner-item {
    width: 100%;
    display: flex;
    justify-content: center;
    align-items: center;
    gap: 30px;
    /* space between text and image */
}

.banner-text {
    font-size: 20px;
    font-weight: bold;
```

```
}

/* Common styling for all banner images */
.banner-img {
    object-fit: contain;
    margin-top: 20px;
    margin-bottom: 20px;
}

/*  Individual sizes for each logo */
.banner-img1 {
    width: 180px;
    height: 60px;
}

.banner-img2 {
    width: 120px;
    height: 70px;
}

.divider {
    width: auto;
    font-size: 40px;
    font-weight: bold;
    color: rgb(3, 2, 2);
}
.fashion-cards-container {
    display: flex;

    flex-direction: row;
```

```
}

.card {

    width: 360px;

    height: 400px;


    border-radius: 20px;


}

.women-fashion {


    background-image: url("../image/myntra/women-fashion.jpg");

    background-size: cover;

    display: flex;

    flex-direction: column;

    justify-content: end;

    padding: 10px;

}
```

```
.men-fashion {
    background-image: url("../image/myntra/men-fashion.jpg");
    background-size: cover;
    display: flex;
    flex-direction: column;
    justify-content: end;
    padding: 10px;
}
.kids-fashion {
    background-image: url("../image/myntra/kids-fashion.jpg");
    background-size: cover;
    display: flex;
    flex-direction: column;
    justify-content: end;
    padding: 10px;
```

```
}

.women-footwear-fashion {

    background-image: url("../image/myntra/women-footwear-fashion.jpg");

    background-size: cover;

    display: flex;

    flex-direction: column;

    justify-content: end;

    padding: 10px;

}

.home-cup-kitchen {

    background-image: url("../image/myntra/home-cup-kitchen.jpg");

    background-size: cover;

    display: flex;

    flex-direction: column;

    justify-content: end;

    padding: 10px;
```

```
}

.card-caption {

    background-color: black;

    color: white;

    font-size: 20px;

    padding: 10px;

    text-align: center;

    border-radius: 20px;

}

.card-title {

    font-weight: bold;

}

.card-subtitle {

    font-size: 15px;

}
```

```
.Brand-cards-container {
    display: flex;
    flex-direction: row;
    margin-top: 10px;
    width: 100%;

}
  .Brand-card {
    /* border: 1px solid blue; */
    margin-top: 20px;
    width: 292px;
    height: 400px;
    border-radius: 35px;
    gap: 10px;
    margin-right: 2px;

}
.Brand-cards-border {
    display: flex;
    flex-direction: column;
    justify-content: end;
    align-items: center;
    border: 2px solid #f9f6f6;
    width: 240px;
    height: 380px;
    border-radius: 35px;
    gap: 10px;

}

.Brand-hrx-men {
```

```
    margin-top: 5px;
    background-image: url("../image/myntra/hrx-discount.png");
    background-size: cover;
    justify-content: center;
    display: flex;
    align-items: center;
    padding: 1px;
    gap: 5px;

}

.Brand-h-and-m-women {
    margin-top: 5px;
    background-image: url("../image/myntra/h&m-discount.png");
    background-size: cover;
    justify-content: center;
    display: flex;
    align-items: center;
    padding: 1px;



}

.Brand-libas-women {
    margin-top: 5px;
    /* border: 1px solid gold; */
    background-image: url("../image/myntra/libas-1-discount.png");
    background-size: cover;
    justify-content: center;
    display: flex;
    align-items: center;
```

```
    padding: 1px;



}

.Brand-allen-solly-women {
    margin-top: 5px;
    /* border: 1px solid gold; */
    background-image: url("../image/myntra/Van-Heusen-discount.png");
    background-size: cover;
    justify-content: center;
    display: flex;
    align-items: center;
    padding: 1px;



}

.Brand-bershka-men {
    margin-top: 5px;
    /* border: 1px solid gold; */
    background-image: url("../image/myntra/bershka-discount.png");
    background-size: cover;
    justify-content: center;
    display: flex;
    align-items: center;
    padding: 1px;



}
```

```
.Brand-card-caption {
    width: 235px;
    height: 50px;
    display: flex;
    justify-content: center;
    background-color: rgb(252, 248, 248);
    gap: 5px;

    /* padding: 5px; */

    text-align: center;

    border-radius: 20px;

}

.Brand-card-caption span {
    display: flex;
    align-items: center;
    font-size: 30px;
    color: rgb(159, 156, 156);
}

.Brand-card-caption img {
    width: auto;
    height: 40px;
    padding: 5px;
```

```
}

.Brand-discount {
    display: flex;
    flex-direction: column;
    align-items: center;
    font-weight: bold;
    color: white;
    margin-top: 10px;
    margin-bottom: 10px;
}

sup i {
    font-size: 10px;
    margin-right: -3px;
    font-weight: bold;
    /* align-items: baseline;   */
}

.Category-cards-container {
    /* border: 1px solid black; */
    display: flex;
    flex-wrap: wrap;
    /* allows next cards to go to next row */
    flex-direction: row;
    margin: 15px;
    gap: 30px;
    justify-content: center;
    /* optional: centers the cards */

}
```

```
.Category-cards {
    background-color: rgb(251, 237, 239);
    display: flex;
    flex-direction: column;
    width: 220px;
    height: 350px;
    overflow: hidden;
    /* keeps image inside corners */
    padding: 5px;
    margin: 10px;

}

.category-cart-header {
    flex: 1;
    /* takes available space */
    height: 50%;
    overflow: hidden;

}

.category-cart-header img {
    width: 100%;
    height: 100%;
    object-fit: cover;
    /* keeps aspect ratio and fills area */
    display: block;



}
```

```
        .category-cart-footer {
            color: black;
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            display: flex;
            flex-direction: column;
            align-items: center;
            margin: 5px;
            gap: 1px;

        }

        .category-cart-footer .category-title {
            font-size: 18px;
            font-family: sans-serif;

        }

        .category-cart-footer .category-offer {
            font-size: 30px;
            font-weight: bold;

        }
    </style>

</head>

<body>

    <header>
```

```
<div class="brand-nav">
    <span class="brand-title">M</span>
    <nav>
        <span>MEN</span>
        <span>WOMEN</span>
        <span>KIDS</span>
        <span>HOME</span>
        <span>BEAUTY</span>
        <span>GEN</span>
        <span>STUDIO</span>
    </nav>
</div>
<div>
    <div class="search-bar">
        <span class="bi bi-search"></span>
        <span> Search for products, brands and more </span>
```

```
            </div>

        </div>

        <div class="short-cuts">

            <div>

                <span class="bi bi-person"></span> <br> <span class="short-cut-
text">Profile</span>

            </div>

            <div>

                <span class="bi bi-heart"></span> <br> <span class="short-cut-
text">Whishlist</span>

            </div>

            <div>

                <span class="bi bi-bag"></span> <br> <span class="short-cut-
text">Bag</span>

            </div>

        </div>

    </header>

    <Main>
```

```
<section>

    <article>

        <marquee scrollamount="20" behavior="alternate">

            <span class="bi bi-lightning-fill"></span> Winter sale starts from Nov-
02-2025, 50% OFF on Fashion <span
                class="bi bi-lightning-fill"></span>

        </marquee>

    </article>

    <img src="../image/myntra/myntra-banner.png" width="100%">

    <div class="banner-data">
        <div class="banner-item">
            <span class="banner-text">FASHION Partner</span>
            <img src="../image/myntra/logo1.png" class="banner-img banner-
img1" height="50px" width="2050px">
        </div>

        <div class="divider">|</div>

        <div class="banner-item">
            <span class="banner-text">Ethnic Partner</span>
            <img src="../image/myntra/logo2.png" class="banner-img banner-
img2">
        </div>
    </div>
```

```
<div>
    <img src="../image/myntra/myntra-banner-2.png" width="100%">
</div>

<div class="fashion-cards-container">

    <div class="card women-fashion">

        <div class="card-caption">

            <span class="card-title">DENIM Dress</span><br>

            <span class="card-subtitle">STARTING <sup>&#8377;</sup>399
</span>

        </div>

    </div>

    <div class="card men-fashion">

        <div class="card-caption">

            <span class="card-title">DENIM Dress</span><br>

            <span class="card-subtitle">STARTING <sup>&#8377;</sup>399
</span>

        </div>

    </div>
```

```
                <div class="card kids-fashion">

                    <div class="card-caption">

                        <span class="card-title">DENIM Dress</span><br>

                        <span class="card-subtitle">STARTING <sup>&#8377;</sup>399
</span>

                    </div>

                </div>
                <div class="card women-footwear-fashion">

                    <div class="card-caption">

                        <span class="card-title">DENIM Dress</span><br>

                        <span class="card-subtitle">STARTING <sup>&#8377;</sup>399
</span>

                    </div>

                </div>
                <div class="card home-cup-kitchen">

                    <div class="card-caption">

                        <span class="card-title">DENIM Dress</span><br>

                        <span class="card-subtitle">STARTING <sup>&#8377;</sup>399
</span>
```

```
        </div>

    </div>

</div>

<div>
    <img src="../image/myntra/myntra-banner-3.png" width="100%">
</div>

<div class="Brand-cards-container">

    <div class="Brand-card Brand-hrx-men">
        <div class="Brand-cards-border">

            <div class="Brand-card-caption">

                <span><img src="../image/myntra/logo-hrx.png" alt=""></span>

            </div>
            <div class="Brand-discount">
                <span style="font-size: 20px;">MIN. 50 % OFF </span>
                <span>Training Style </span>
            </div>

        </div>
    </div>
    <div class="Brand-card Brand-h-and-m-women">
        <div class="Brand-cards-border">
```

```
                    <div class="Brand-card-caption">
                        <span><img src="../image/myntra/logo-H&M.png"
alt=""></span>
                        <span>|</span>
                        <span><img src="../image/myntra/logo-victorias-secret.png"
alt=""></span>

                    </div>

                    <div class="Brand-discount">
                        <span style="font-size: 20px;">MIN. 50 % OFF </span>
                        <span>Training Style </span>
                    </div>

                </div>
            </div>
            <div class="Brand-card Brand-libas-women">
                <div class="Brand-cards-border">

                    <div class="Brand-card-caption">
                        <span><img src="../image/myntra/logo-Libas.png"
alt=""></span>
                        <span>|</span>
                        <span><img src="../image/myntra/logo-indoera.png"
alt=""></span>

                    </div>

                    <div class="Brand-discount">
                        <span style="font-size: 20px;">MIN. 75% OFF </span>
                        <span>Ethnic Edge </span>
                    </div>
```

```
                </div>
            </div>
            <div class="Brand-card Brand-allen-solly-women">
                <div class="Brand-cards-border">

                    <div class="Brand-card-caption">
                        <span><img src="../image/myntra/logo-allen-solly.png"
alt=""></span>
                        <span>|</span>
                        <span><img src="../image/myntra/logo-Van-Heusen.png"
alt=""></span>

                    </div>

                    <div class="Brand-discount">
                        <span style="font-size: 20px;">MIN. 50% OFF </span>
                        <span>Style Statement </span>
                    </div>

                </div>
            </div>
            <div class="Brand-card Brand-bershka-men">
                <div class="Brand-cards-border">

                    <div class="Brand-card-caption">
                        <span><img src="../image/myntra/logo-bershka.png"
alt=""></span>
                    </div>

                    <div class="Brand-discount">
```

```
                    <span style="font-size: 20px; align-items: baseline;">Starting
<sup><i
                            class="bi bi-currency-rupee"></i></sup>669 </span>
                    <span>Bold Looks </span>
                </div>

            </div>
        </div>
    </div>

    <div class="banner-img">
        <img src="../image/myntra/category-banner.png" width="100%">
    </div>

    <div class="Category-cards-container">
        <div class="Category-cards">
            <div class="category-cart-header">
                <img src="../image/myntra/c1.png" alt="Ethnic-Wear">
            </div>
            <div class="category-cart-footer">
                <span class="category-title">Ethnic Wear</span>
                <span class="category-offer">50-80% OFF</span>
                <span class="category-title">Shop Now</span>
            </div>
        </div>
        <div class="Category-cards">
            <div class="category-cart-header">
                <img src="../image/myntra/c2.png" alt="Ethnic-Wear">
            </div>
            <div class="category-cart-footer">
                <span class="category-title">Ethnic Wear</span>
```

```
            <span class="category-offer">50-80% OFF</span>
            <span class="category-title">Shop Now</span>
        </div>
    </div>
    <div class="Category-cards">
        <div class="category-cart-header">
            <img src="../image/myntra/c3.png" alt="Ethnic-Wear">
        </div>
        <div class="category-cart-footer">
            <span class="category-title">Ethnic Wear</span>
            <span class="category-offer">50-80% OFF</span>
            <span class="category-title">Shop Now</span>
        </div>
    </div>
    <div class="Category-cards">
        <div class="category-cart-header">
            <img src="../image/myntra/c4.png" alt="Ethnic-Wear">
        </div>
        <div class="category-cart-footer">
            <span class="category-title">Ethnic Wear</span>
            <span class="category-offer">50-80% OFF</span>
            <span class="category-title">Shop Now</span>
        </div>
    </div>
    <div class="Category-cards">
        <div class="category-cart-header">
            <img src="../image/myntra/c5.png" alt="Ethnic-Wear">
        </div>
        <div class="category-cart-footer">
            <span class="category-title">Ethnic Wear</span>
            <span class="category-offer">50-80% OFF</span>
```

```
        <span class="category-title">Shop Now</span>
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c6.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c7.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c8.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
```

```
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c9.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c10.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
    </div>
</div>
<div class="Category-cards">
    <div class="category-cart-header">
        <img src="../image/myntra/c11.png" alt="Ethnic-Wear">
    </div>
    <div class="category-cart-footer">
        <span class="category-title">Ethnic Wear</span>
        <span class="category-offer">50-80% OFF</span>
        <span class="category-title">Shop Now</span>
    </div>
```

```
            </div>
            <div class="Category-cards">
                <div class="category-cart-header">
                    <img src="../image/myntra/c12.png" alt="Ethnic-Wear">
                </div>
                <div class="category-cart-footer">
                    <span class="category-title">Ethnic Wear</span>
                    <span class="category-offer">50-80% OFF</span>
                    <span class="category-title">Shop Now</span>
                </div>
            </div>

        </div>

    </section>
</Main>
    <footer>

        <div>

            <span class="footer-title">ONLINE SHOPPING</span>

            <span>Men</span>

            <span>Women</span>

            <span>Kids</span>

            <span>Home</span>

            <span>Beauty</span>
```

```
    <span>Genz</span>

    <span>Gift Cards</span>

    <span>Myntra Insider</span>

</div>

<div>

    <span class="footer-title">CUSTOMER POLICIES</span>

    <span>Contact Us</span>

    <span>FAQ</span>

    <span>T&C</span>

    <span>Terms Of Use</span>

    <span>Track Orders</span>

    <span>Shipping</span>

    <span>Cancellation</span>

    <span>Returns</span>

    <span>Privacy policy</span>
```

```
                <span>Grievance Redressal</span>

            </div>

            <div>

                <span class="footer-title">EXPERIENCE MYNTRA APP ON
MOBILE</span>

                <span class="footer-title">KEEP IN TOUCH</span>

                <aside>

                    <span class="bi bi-facebook"></span>

                    <span class="bi bi-twitter-x"></span>

                    <span class="bi bi-instagram"></span>

                    <span class="bi bi-youtube"></span>

                </aside>

            </div>

            <div>

                <div class="footer-content">

                    <span>100% ORIGINAL guarantee for all products at
myntra.com</span>
```

```
            </div>

            <div class="footer-content">

                <span>Return within 14days of receiving your order</span>

            </div>

            <div class="footer-content">

                <address>

                    <span class="bi bi-envelope"> info@myntra.com</span>

                    <span class="bi bi-phone"> 18884990081 toll free</span>

                </address>

            </div>

        </div>

    </footer>

</body>

</html>
```

Links in HTML

Hyperlinks in HTML

- Hyperlink is a clickable text, picture or graphic that navigates the user from one location to another.

- Hyperlink is created by using **anchor** element <a>.

- Hyperlinks are classified into 2 types

    a) Intra document links

    b) Inter document links

Intra Document Links:

- An intra document link allows navigation from one location to another within the same page.

- It requires the target element defined with “ID” reference.

```
<header id=“top”> </header>

<img id=“fashion”>
```

- You can refer ID by using <a> anchor element “href” attribute.

```
<a href=“#top”>  Text | Picture | Graphic </a>

<a href=“#fashion”>
```

- You can access any ID reference directly from URL in address bar. You have to use reference id name with “#”.

http://127.0.0.1:5500/page.html#id_ref_name

FAQ: What is difference between “src” & “href” ?

Ans: “**src”** is a getter, which gets and returns a value.

“**href”** is a setter, which sets a new value into address bar by appending it into URL.

FAQ: If same ID is defined for multiple elements then how it navigates?

Ans: If same ID is defined for elements, then it navigates to the first reference.

[occurrence from top]

FAQ: How to remove underline for hyperlink?

Ans: By using CSS “text-decoration” attribute set to “none”.

Syntax:

```
a {  text-decoration: none;  }
```

FAQ: How to change color of visited and active links?

Ans: You can use HTML <body> element attributes

a) alink : active link color

b) vlink : visited link color

You can use CSS classes

a) visited

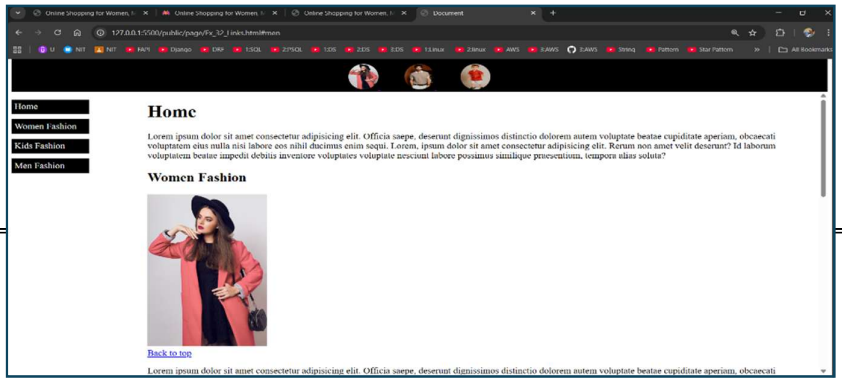
b) active

Syntax:

```
a:visited {   color: green;   }

a:active {  color :red;  }
```

Syntax: <body alink=“red” valink=“green”>

Ex_32_Intra_document_links.html:



```
<style>
    header {
        background-color: black;
        color:white;
        text-align: center;
    }
    section {
        display: grid;
        grid-template-columns: 2fr 10fr;
    }
    main {
        height: 560px;
        overflow: auto;
    }
    ul {
        list-style: none;
        margin-left: -40px;
    }
    li {
        background-color: black;
        color:white;
        padding: 5px;
```

```
            margin-top: 10px;
            margin-bottom: 10px;
            width: 120px;
        }
        nav a {
            color:white;
            text-decoration: none;
        }
        li:hover {
            background-color: blue;
        }
        header a {
            margin-left: 20px;
            margin-right: 20px;
        }
        header img {
            border-radius: 50px;
        }
    </style>
</head>
<body>
    <header>
        <h2>Amazon</h2>
```

```
        <div>

            <a href="#women"> <img src="./images/women-fashion.jpg" width="50"
height="50"> </a>

            <a href="#men"> <img src="./images/men-fashion.jpg" width="50" height="50">
</a>

            <a href="#kids"> <img src="./images/kids-fashion.jpg" width="50" height="50">
</a>

        </div>

    </header>

    <section>

        <nav>

            <ul>

                <li><a href="#home">Home</a></li>

                <li><a href="#women">Women Fashion</a></li>

                <li><a href="#kids">Kids Fashion</a></li>

                <li><a href="#men">Men Fashion</a></li>

            </ul>

        </nav>

        <main>

            <h1 id="home">Home</h1>

            <p>Lorem ipsum </p>

            <h2 id="women">Women Fashion</h2>

            <img src="./images/women-fashion.jpg" width="200" height="300">

            <div>
```

```
            <a href="#home">Back to top</a>
        </div>
        <p>Lorem </p>
        <h2 id="men">Men Fashion</h2>
        <img src="./images/men-fashion.jpg" width="200" height="300">
        <div>
            <a href="#home">Back to top</a>
        </div>
        <p>Lorem ipsum </p>
        <h2 id="kids">Kids Fashion</h2>
        <img src="./images/kids-fashion.jpg" width="200" height="300">
        <div>
            <a href="#home">Back to top</a>
        </div>
        <p>Lorem ipsum </p>
      </main>
    </section>
</body>
</html>
```

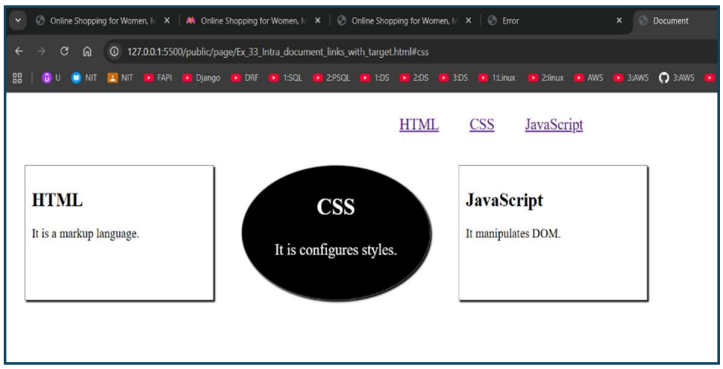
FAQ: If page can’t scroll to target, then how to display the target as active?

Ans: By using CSS “target” class you can select the link target element and apply a set of styles when it is target.

Syntax:

.name : target { }

Ex_33_Intra_document_links_with_target.html




```
<style>
    .card {
        width: 250px;
        height: 150px;
        border:1px solid gray;
        box-shadow: 2px 2px 2px black;
        margin: 20px;
        padding: 10px;
    }
    .card:target {
        background-color: black;
        color: white;
        border-radius: 50%;
        text-align: center;
        font-size: 20px;
        transform: rotateY(360deg);
```

```
            transition: 3s;
        }
        section {
            display: flex;
        }
        nav span {
            margin-left: 20px;
            margin-right: 20px;
            text-align: center;
            font-size: 20px;
        }
        header {
            display: flex;
            justify-content: center;
            padding: 20px;
        }
    </style>
</head>
<body>
    <header>
         <nav>
            <span><a href="#html">HTML</a></span>
            <span><a href="#css">CSS</a></span>
```

```
            <span><a href="#js">JavaScript</a></span>
        </nav>
    </header>
    <section>
        <div class="card" id="html">
            <h2>HTML</h2>
            <p>It is a markup language.</p>
        </div>
        <div class="card" id="css">
            <h2>CSS</h2>
            <p>It is configures styles.</p>
        </div>
        <div class="card" id="js">
            <h2>JavaScript</h2>
            <p>It manipulates DOM.</p>
        </div>
    </section>
</body>
</html>
```

Inter documents links

- Inter document links can handle various interactions

  a) Navigate to a file b) Navigate to any URL
  c) Invoke any application supported by browser
  d) Invoke any JavaScript function etc.

Navigate to File:

- You can navigate to any file in project.

- You can link any type of file with anchor <a> element. - Browser can open and display file content only when relative plugins are supported.

[plugin is an extension for browser to handle special type of content]

- If suitable plugins are not available then browser will “download” the file.

Syntax:

```
<a href=“./myntra.html”> Template </a>

<a href=“./images/pic.jpg”> Photo </a>

<a href=“./docs/content.pdf”> Tutorial </a>
```

FAQ: How to download a linked file always?

Ans: By using HTML “download” attribute with <a> anchor element.

```
Syntax:  <a href=“./images/pic.jpg”   download=“DSC0001.jpg”>   Download Image </a>
```

Ex_34_Inter_documents_links.html:

```
<body>

   <ol>

      <li> <a href="./myntra.html"> Myntra Template </a> </li>

      <li> <a href="./images/women-fashion.jpg"> Women Fashion </a> </li>

      <li> <a href="./images/kids-fashion.jpg" download="DSC0001"> Kids Fashion
</a> </li>

      <li> <a href="./docs/ui-content.pdf"> UI Content PDF </a></li>

      <li> <a href="./docs/ui-react.doc"> React with UI Document </a></li>

   </ol></body></html>
```

Navigate to URL:

- You can configure a hyperlink for any remote resource.

- If file is on another server then you can access directly by using the URL.

- URL must have a proper protocol and domain name.

Syntax:

<a href="https://www.myntra.com"> Myntra Website </a>

Ex_35_Navigate_to_URL_link.html:

```
<body>

   <ol>

      <li> <a href="https://www.myntra.com"&gt; Myntra Website </a> </li>

      <li> <a href="https://fakestoreapi.com/img/71-3HjGNDUL._AC_SY879._SX._UX._SY._UY_t.png"&gt; T-Shirt </a></li>

   </ol> </body></html>
```

Navigate to App supported by browser:

- A hyperlink can invoke various applications in browser.- Browser must support plugins for relative applications and they must be installed on your device. - Invoking application requires following prefixes

a) mailto : invokes email app
b) tel : invokes phone app
c) skype : invokes skype app

d) https://wa.me: invoke whatsApp web app

Syntax:

<a href="mailto: any@email.com">

<a href="tel: phoneNumber">

<a href="skype: skypeId@outlook.com">

<a href="https://wa.me/phoneNumber">

Ex_36_**Navigate_to_App_supported_by_browser.html**:

```
<body>

   <ol>

       <li> For more queries email us <a
href="info@nareshit.com"> info@nareshit.com </a> </li>

       <li> Call us <a href="tel:+919876543210">+919876543210</a> </li>

       <li class="bi bi-skype"> Skype us <a href="skype:nit_info@outlook.com"> Naresh
IT info </a> </li>

       <li class="bi bi-whatsapp"> Whatsapp <a
href="https://wa.me/+919876543210"&gt; +919876543210 </a> </li>

   </ol> </body></html>
```

Invoking JavaScript Functions:

- A hyperlink can execute script dynamically. - You can configure various functionalities in JavaScript, which <a> anchor can invoke. - It uses the prefix "javascript" to access and execute any JS function.

Syntax: <a href="javascript: function()"> Text | Image </a>

Ex_37_print_**Invoking_JavaScript_Functions.html**:

```
<body> <h2>Ticket</h2>
   <a href="javascript:window.print()" class="bi bi-printer-fill"> Print </a>
</body></html>
```

FAQ's:

1. How to open link target in a new tab?

A. By using <a> "target" attribute set to "_blank"

Syntax: <a href="./images/pic.jpg" target="_blank"> Image </a>

2. How to open link target in a new window?

A. By using JavaScript "window.open()" method.

Syntax: <a href="javascript:window.open('path', 'title', 'features')"> Text | Image </a>

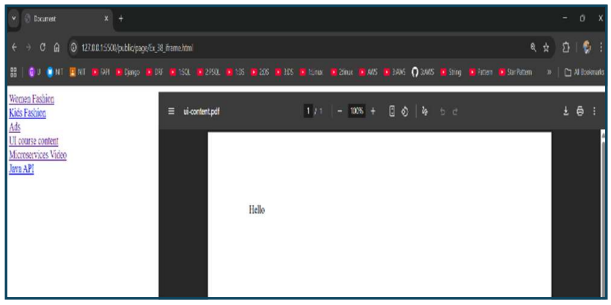
3. How to open link target in same window along with existing content in page?

A. By using HTML 5 <iframe> element to embed the target in the same page. Anchor element target must refer to <iframe> name.

Syntax: <iframe name="preview"> </iframe>

<a href="./docs/content.pdf" target="preview"> Content </a>

Ex_38_ iframe.html:




```
<link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

<style>

    body {

        display: grid;

        grid-template-columns: 3fr 9fr;

    }
```

```
        iframe {

            border:none;

        }

    </style></head><body>

   <nav>

    <a href="./images/women-fashion.jpg" target="_blank"> Women Fashion </a>

   <br>

   <a href="javascript:window.open('./images/kids-fashion.jpg', 'kids', 'width=400
height=400')"> Kids Fashion </a> <br>

   <a href="./images/a12.jpg" target="preview"> Ads </a> <br>

   <a href="./docs/ui-content.pdf" target="preview"> UI course content </a><br>

   <a href="https://www.youtube.com/embed/5mVNW8UHBIo&quot; target="preview">
Microservices Video </a><br>

   <a href="https://www.youtube.com/embed/U2PiMIS75jU&quot;
target="preview">Java API</a>

   </nav>

   <section>

        <iframe width="100%" name="preview" height="600"></iframe>

   </section></body></html>
```

Youtube Design Template

Day-21

5-11-2025

HTML Iframe

- It is used to embed any content into page.

- It can embed along with existing content in page.

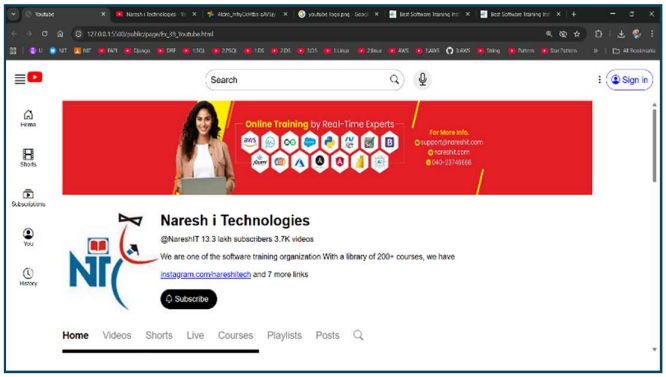
- It can embed only the content that is having relative plugin in browser.

Syntax: <iframe src="../resource_path" width="200" height="400"> </iframe>

Horizontal line:- The <hr> element can set horizontal line.- You can define size and color with noshade. a) size b) color c) noshade

Syntax: <hr size="2" noshade color="gray">

Ex_39_Youtube.html




```
<link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

<style>

    header {

        display: flex;

        flex-direction: row;

        justify-content: space-between;

        align-items: center;

        padding: 10px;

        font-family: Arial;

        font-size: 18px;

    }

    .bi-justify {

        font-size: 30px;
```

```
}
.search-container {
    display: flex;
    flex-direction: row;
    align-items: center;
}
.search-bar {
    width: 400px;
    border:1px solid gray;
    padding: 10px;
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    border-radius: 20px;
    margin-right: 20px;
}
.bi-mic {
    font-size: 25px;
    background-color: rgb(239, 238, 238);
    padding: 5px;
    border-radius: 100%;
    display: inline-block;
    width: 30px;
```

```
    height: 30px;
    text-align: center;
}
.btn-signin {
    border: 1px solid blue;
    padding: 10px;
    border-radius: 20px;
    color:blue;
}
section {
    display: grid;
    grid-template-columns: 1fr 11fr;
    margin-top: 10px;
}
.nav-menu {
    list-style: none;
    margin-left: -70px;
    font-family: Arial;
    text-align: center;
}
.nav-item span {
    display: block;
}
```

```
.nav-text {
    font-size: 12px;
}
.nav-icon {
    font-size: 22px;
}
.nav-item {
    margin-bottom: 40px;
}
article {
    display: grid;
    grid-template-columns: 2fr 10fr;
    margin-top: 20px;
    font-family: Arial;
}
.article-title {
    font-size: 30px;
    font-weight: bold;
    margin-top: 10px;
    padding-bottom: 10px;
}
.btn-subscribe {
   margin-top: 20px;
```

```
   background-color: black;
   color:white;
   padding: 10px;
   width: 100px;
   border-radius: 20px;
}
main {
    height: 500px;
    overflow: auto;
}
.nav-tabs {
    list-style: none;
    display: flex;
    margin-left: -40px;
    margin-top: 30px;
}
.nav-tabs li {
    margin-right: 30px;
    font-size: 20px;
    font-family: Arial;
    color:gray;
}
.nav-tabs li:first-child {
```

```
    font-weight: bold;
    color:black;
}
.channel-article {
    display: grid;
    grid-template-columns: 4fr 8fr;
    margin-bottom: 40px;
}
.channel-article iframe {
    height: 250px;
    border:none;
}
h3 {
    font-family: Arial;
    font-size: 22px;
}
.videos-row {
    display: grid;
    grid-template-columns: 2.4fr 2.4fr 2.4fr 2.4fr 2.4fr;
}
.video-card {
    width: 200px;
}
```

```
        .video-card iframe {
            width: 100%;
        }
        .card-body {
            padding-top: 10px;
            padding-bottom: 20px;
            font-family: Arial;
        }
        .card-title {
            font-weight: bold;
            font-size: 12px;
        }
        .card-footer {
            font-family: Arial;
            font-size: 12px;
            color:gray;
        }
    </style>
</head>
<body>
   <header>
      <div>
         <span class="bi bi-justify"></span>
```

```
        <img align="right" src="./images/youtube-logo.png">
      </div>
      <div class="search-container">
          <div class="search-bar">
              <span>Search</span>
              <span class="bi bi-search"></span>
          </div>
          <div class="bi bi-mic"></div>
      </div>
      <div>
        <span class="bi bi-three-dots-vertical"></span>
        <span class="btn-signin"> <span class="bi bi-person-circle"> Sign in</span>
</span>
      </div>
    </header>
    <section>
       <nav>
         <ul class="nav-menu">
            <li class="nav-item">
              <span class="bi bi-house-door nav-icon"></span>
              <span class="nav-text">Home</span>
            </li>
            <li class="nav-item">
```

```
        <span class="bi bi-film nav-icon"></span>
        <span class="nav-text">Shorts</span>
      </li>
      <li class="nav-item">
        <span class="bi bi-collection-play nav-icon"></span>
        <span class="nav-text">Subscriptions</span>
      </li>
      <li class="nav-item">
        <span class="bi bi-person-circle nav-icon"></span>
        <span class="nav-text">You</span>
      </li>
      <li class="nav-item">
        <span class="bi bi-clock-history nav-icon"></span>
        <span class="nav-text">History</span>
      </li>
    </ul>
  </nav>
  <main>
      <img src="./images/channel-banner.png" width="90%">
      <article>
        <div>
            <img width="100%" src="./images/nit-logo.png">
        </div>
```

```
            <div>

                <div class="article-title">Naresh i Technologies</div>

                <div class="article-subtitle"> @NareshIT 13.3 lakh subscribers 3.7K
videos</div>

                <p class="article-text"> We are one of the software training organization
With a library of 200+ courses, we have</p>

                <div>

                    <a
href="https://www.instagram.com/nareshitech">instagram.com/nareshitech</a&gt; and
7 more links

                </div>

                <div class="btn-subscribe bi bi-bell"> Subscribe</div>

            </div>

        </article>

        <ul class="nav-tabs">

            <li>Home</li>

            <li>Videos</li>

            <li>Shorts</li>

            <li>Live</li>

            <li>Courses</li>

            <li>Playlists</li>

            <li>Posts</li>

            <li class="bi bi-search"></li>

        </ul>

        <div class="channel-article">
```

<iframe width="100%" src="https://youtube.com/embed/yBw-6rr5hbA"></iframe>

<pre>

Best Software Training Institute with Placements - Naresh i Technologies

2,54,425 views

5 years ago

Register For Free Demos On New Batches :-

Java Real Time Project | 9:00 AM | Link : https://t.ly/JRTP-13

Data Analytics & Business Analytics | 7:30 PM | Link : https://t.ly/Da-17M

DevOps with AWS | 4:00 PM | Link : https://t.ly/DevA-17M

Ms Azure + Azure DevOps | 7:00 PM | Link : https://t.ly/AZ-DeO

Unix|Linux | 5:30 PM | Link : https://t.ly/UX-LX1

Python & Full Stack Python | 9:00 AM | Link : https://t.ly/fsPk-17M

Spring Boot & Microservices | 7:00 PM | Link : https://t.ly/SbmsK-20M

Join Us on Telegram: https://t.me/nareshit

</pre>

```
<hr noshade color="lightgray">

<h3>Videos</h3>

<div class="videos-row">

  <div class="video-card">

    <div class="card-header">

      <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

    </div>

    <div class="card-body">

      <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

    </div>

    <div class="card-footer">

       <span>69 views</span>

       <span>15 hrs ago</span>

    </div>

  </div>

  <div class="video-card">

    <div class="card-header">

      <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

    </div>

    <div class="card-body">

      <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>
```

```
            </div>
            <div class="card-footer">
                <span>69 views</span>
                <span>15 hrs ago</span>
            </div>
        </div>
        <div class="video-card">
            <div class="card-header">
                <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;
            </div>
            <div class="card-body">
                <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>
            </div>
            <div class="card-footer">
                <span>69 views</span>
                <span>15 hrs ago</span>
            </div>
        </div>
        <div class="video-card">
            <div class="card-header">
                <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;
```

```
                </div>

                <div class="card-body">

                    <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

                </div>

                <div class="card-footer">

                     <span>69 views</span>

                     <span>15 hrs ago</span>

                </div>

            </div>

            <div class="video-card">

                <div class="card-header">

                    <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

                </div>

                <div class="card-body">

                    <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

                </div>

                <div class="card-footer">

                     <span>69 views</span>

                     <span>15 hrs ago</span>

                </div>

            </div>

        </div>
```

```
        <h3>Past live streams</h3>

        <div class="videos-row">

            <div class="video-card">

                <div class="card-header">

                    <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

                </div>

                <div class="card-body">

                    <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

                </div>

                <div class="card-footer">

                     <span>69 views</span>

                     <span>15 hrs ago</span>

                </div>

            </div>

            <div class="video-card">

                <div class="card-header">

                    <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

                </div>

                <div class="card-body">

                    <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>
```

```
            </div>

            <div class="card-footer">

                <span>69 views</span>

                <span>15 hrs ago</span>

            </div>

        </div>

        <div class="video-card">

            <div class="card-header">

                <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

            </div>

            <div class="card-body">

                <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

            </div>

            <div class="card-footer">

                <span>69 views</span>

                <span>15 hrs ago</span>

            </div>

        </div>

        <div class="video-card">

            <div class="card-header">

                <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;
```

```
            </div>

            <div class="card-body">

                <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

            </div>

            <div class="card-footer">

                <span>69 views</span>

                <span>15 hrs ago</span>

            </div>

        </div>

        <div class="video-card">

            <div class="card-header">

                <iframe
src="https://www.youtube.com/embed/B7X5vDz3x4g"></iframe&gt;

            </div>

            <div class="card-body">

                <span class="card-title">Beginner React FAQ: What Does
Reconciliation Mean?</span>

            </div>

            <div class="card-footer">

                <span>69 views</span>

                <span>15 hrs ago</span>

            </div>

        </div>
```

```
        </div>
</main>   </section></body></html>
```

Tables in HTML

- HTML 5 provides following elements for tables

<table>	container with rows and columns.
<caption>	makes the table SEO friendly
<thead>	header section
<tbody>	body section
<tfoot>	footer section
<tr>	table row in head, body & footer
<th>	table heading cell
<td>	table data [normal table cell]
<colgroup>	groups columns for formatting

Table Attributes:

1. Frame, Rules & Border

frame : It sets a frame for table with value: box, rhs, lhs, above, below, void.

rules : It sets rules for rows, columns & groups with values: rows, cols, groups.

border : It sets border for cells.

Syntax:

```
<table frame="void | box | rhs | lhs | above | below">

<table rules="all | none | rows | cols | groups">

<table border="0 | 1">      0=false, 1=true
```

Note: You can't set border when rules attribute is defined for table.

To display border remove rules from table.

2. Cellspacing & Padding

cellspacing It sets space between cells.

cellpadding It sets space between border and content inside cell.

Syntax: <table border="1" cellspacing="20" cellpadding="10">

3. Width & Height

width : It is configured for <td> or <th>

height : It is configure for <tr>

Syntax: <td width="50"> <tr height="200">

4. Align & Valign

align : It aligns horizontally left, center, right & justify.

valign : It aligns vertically top, center, bottom.

Syntax: <tr align="center" valign="top"> <td align="right" valign="center">

Note: You can't align all contents in a column. You have to defined individually for every cell or row.

5. Background color & image

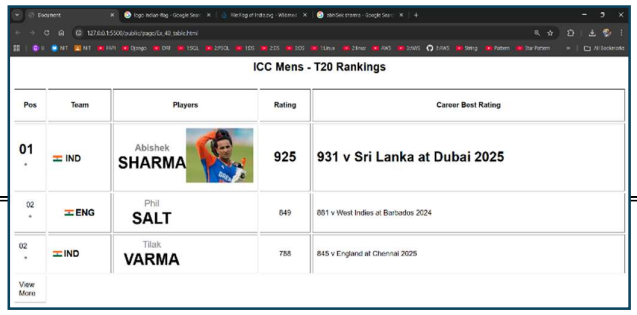
bgcolor : It sets background color for row, table, group, cell.

background : It sets background image.

Syntax:

<table bgcolor="yellow"> <table background="./images/pic.jpg">

Ex_40_table.html:



ICC Mens - T20 Rankings

Pos	Team	Players	Rating	Career Best Rating
01	IND	Abishek SHARMA	925	931 v Sri Lanka at Dubai 2025
02	ENG	Phil SALT	849	881 v West Indies at Barbados 2024
02	IND	Tilak VARMA	788	845 v England at Chennai 2025

View More

```
<style>
    table {
        font-family: Arial;
    }
    .flag {
        width: 20px;
    }
    .flag+span {
        font-size: 20px;
        font-weight: bold;
    }
    .player-img {
        width: 150px;
    }
    .player-details {
        display: grid;
        grid-template-columns: 6fr 6fr;
        background-size: cover;
    }
    .col-1 {
        display: flex;
        justify-content: center;
```

```
    align-items: center;
    text-align: center;
}
.col-1 span {
    display: block;
}
.player-subtitle {
    font-size: 35px;
    font-weight: bold;
}
.player-title {
    font-size: 20px;
    color:gray;
}
caption {
    padding-bottom: 30px;
    font-weight: bold;
    font-size: 25px;
}
tbody tr:first-child {
    font-size: 30px;
    font-weight: bold;
}
```

```
        .bi-dot {
            font-size: 30px;
            color:gray;
        }
    </style>
</head>
<body>
    <table   width="100%" border="1" cellspacing="5" cellpadding="10" frame="void">
        <caption>ICC Mens - T20 Rankings</caption>
        <colgroup span="1"></colgroup>
        <thead >
            <tr height="70">
                <th width="50">Pos</th>
                <th>Team</th>
                <th width="300">Players</th>
                <th>Rating</th>
                <th>Career Best Rating</th>
            </tr>
        </thead>
        <tbody>
            <tr  valign="center">
                <td>01 <span class="bi bi-dot"></span> </td>
                <td><img class="flag" src="./images/ind-flag.png"> <span>IND</span> </td>
```

```
            <td class="player-details" background="./images/cup-transparent.png">
                <div class="col-1">
                    <div>
                        <span class="player-title">Abishek</span>
                        <span class="player-subtitle">SHARMA</span>
                    </div>
                </div>
                <div class="col-2">
                    <img class="player-img" src="./images/abhi.png">
                </div>
            </td>
            <td align="center">925</td>
            <td>931 v Sri Lanka at Dubai 2025</td>
        </tr>
        <tr align="center">
            <td>02 <span class="bi bi-dot"></span> </td>
            <td><img class="flag" src="./images/england.png">
<span>ENG</span></td>
            <td class="player-details">
                <div class="col-1">
                    <div>
                        <span class="player-title">Phil</span>
                        <span class="player-subtitle">SALT</span>
                    </div>
```

```
            </div>
        </td>
        <td>849</td>
        <td align="left">881 v West Indies at Barbados 2024</td>
    </tr>
    <tr>
        <td>02 <span class="bi bi-dot"></span> </td>
        <td><img class="flag" src="./images/ind-flag.png"> <span>IND</span></td>
        <td class="player-details">
          <div class="col-1">
              <div>
                <span class="player-title">Tilak</span>
               <span class="player-subtitle">VARMA</span>
              </div>
          </div>
        </td>
        <td align="center">788</td>
        <td>845 v England at Chennai 2025</td>
    </tr>
</tbody>
<tfoot>
    <tr>
        <td>View More</td>
```

```
</tr></tfoot></table></body></html>
```

Forms

Day-23

7-11-2025

Table Elements

<table>,<caption>,<thead>,<tbody>,<tfoot>,<tr>,<th>,<td>,<colgroup>

Table Attributes

- frame, - rules, - border, - cellspacing, - cellpading, - align, - valign, - bgcolor,

- background, - height

6. Merging rows & columns

rowspan It merges specified number of rows in to a cell.

colspan It merges specified number of columns into one cell.

- You can define only for <th> or <td> elements.

Syntax: <th colspan="4"> <td rowspan="6">

Ex_41_table.html:

Name		Address		
First Name	Last Name	City	State	Postal Code
		DELHI		

```
<table width="100%" border="1">
    <thead>
        <tr>
            <th colspan="2">Name</th>
            <th colspan="3">Address</th>
```

```
    </tr>
    <tr>
        <th>First Name</th>
        <th>Last Name</th>
        <th>City</th>
        <th>State</th>
        <th>Postal Code</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td> </td>
        <td> </td>
        <td rowspan="6" align="center">D<br>E<br>L<br>H<br>I</td>
        <td> </td>
        <td> </td>
    </tr>
    <tr>
        <td> </td>
        <td> </td>
        <td> </td>
        <td> </td>
    </tr>
```

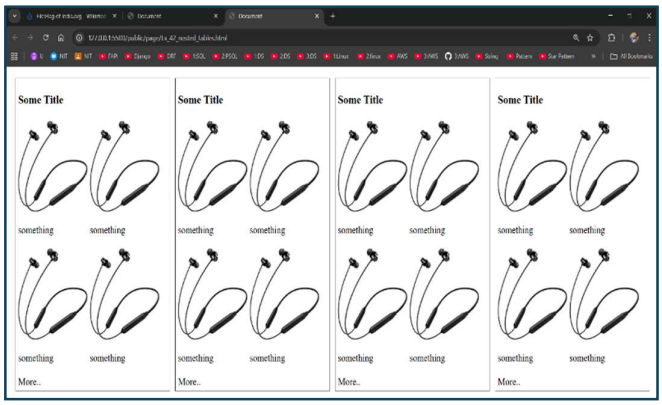
```
<tr>
    <td> </td>
    <td> </td>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
    <td> </td>
    <td> </td>
```

```
        </tr>

      </tbody></table></body></html>
```

Note: You can define nested tables, but make sure that child table is inside <td> element.

```
        <table>

            <tr>

                <td>

                    <table>

                        <tr> <td> </td> </tr>

                    </table>

                </td>

            </tr>

        </table>
```

Ex_42_nested_tables.html:




```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>
</head>
<body>
    <table width="100%" border="1" frame="void" cellspacing="10" cellpadding="2">
        <tr height="400">
            <td>
                <table>
                    <tr>
                        <td><h3>Some Title</h3></td>
                    </tr>
                    <tr>
                        <td><img src="./images/a1.jpg" width="100%">
                            <p>something</p>
                        </td>
                        <td><img src="./images/a2.jpg" width="100%">
                            <p>something</p>
                        </td>
                    </tr>
                    <tr>
                        <td><img src="./images/a1.jpg" width="100%">
                            <p>something</p>
                        </td>
                        <td><img src="./images/a2.jpg" width="100%">
```

```
                <p>something</p>
            </td>
        </tr>
        <tr>
            <td>More..</td>
        </tr>
    </table>
</td>
<td>
    <table>
        <tr>
            <td><h3>Some Title</h3></td>
        </tr>
        <tr>
            <td><img src="./images/a1.jpg" width="100%">
                <p>something</p>
            </td>
            <td><img src="./images/a2.jpg" width="100%">
                <p>something</p>
            </td>
        </tr>
        <tr>
            <td><img src="./images/a1.jpg" width="100%">
```

```
                <p>something</p>
            </td>
            <td><img src="./images/a2.jpg" width="100%">
                <p>something</p>
            </td>
        </tr>
        <tr>
            <td>More..</td>
        </tr>
    </table>
</td>
<td>
    <table>
        <tr>
            <td><h3>Some Title</h3></td>
        </tr>
        <tr>
            <td><img src="./images/a1.jpg" width="100%">
                <p>something</p>
            </td>
            <td><img src="./images/a2.jpg" width="100%">
                <p>something</p>
            </td>
```

```
        </tr>
        <tr>
            <td><img src="./images/a1.jpg" width="100%">
                <p>something</p>
            </td>
            <td><img src="./images/a2.jpg" width="100%">
                <p>something</p>
            </td>
        </tr>
        <tr>
            <td>More..</td>
        </tr>
    </table>
</td>
<td>
    <table>
        <tr>
            <td><h3>Some Title</h3></td>
        </tr>
        <tr>
            <td><img src="./images/a1.jpg" width="100%">
                <p>something</p>
            </td>
```

```
                <td><img src="./images/a2.jpg" width="100%">
                    <p>something</p>
                </td>
            </tr>
            <tr>
                <td><img src="./images/a1.jpg" width="100%">
                    <p>something</p>
                </td>
                <td><img src="./images/a2.jpg" width="100%">
                    <p>something</p>
                </td>
            </tr>
            <tr>
                <td>More..</td>
            </tr>
        </table>
    </td>
</tr></table></body></html>
```

HTML Forms

- It is a generic container with set of elements like buttons, textbox, checkbox, radio, listbox, dropdown etc.- Generic element have predefined functionality.

- HTML <form> is a generic container that handles actions like

a) submit b) reset c) validate

Form Attributes:

1. id]
2. name] used for reference in CSS & JavaScript
3. class]
4. method : It defines the actions to perform.

A form can have 2 methods

a) GET

b) POST

GET is for fetching data from server. [submits to fetch]

POST is for submitting data to server in order to insert, update, delete.

Syntax: <form method="get | post"> The default form method is "**get**".

FAQ: Can we submit & save data on server using GET method?

Ans: Yes. But not recommended.

GET Method:- It submits for fetching. - It submits the data is **query string**. - Query string is appended into URL and displayed in browser address bar. - Everyone can view the data. - It is not safe. - It is easy to hack your data. - It can be bookmarked. - You can't submit complex data, like binary type. - There is limit for data for 2048 to 65535 chars.- It can **"cache"** the data and save **round-trip.**

POST Method

- It submits for saving, updating & deleting. - It submits as **"Form Body Data".** - It is not visible to users directly. - It is safer when compared to GET. - It is not saved in browser logs. - It can't be bookmarked. - It is hard to hack your data. - You can submit complex data. - There is no limit for data. - It can't cache the data.

Form Attributes: 1. Id 2. Name 3. Class 4. Method – GET - POST

5. Action : It refers to the target location where the form data is submitted. Typically it is a server side or client side page.

Syntax: <form method="get" action="page.html | page.php | page.asp"> </form>

6. Novalidate : It is used to disable the default HTML validation. Default Validations are disabled when developers write custom validation.

Syntax: <form method="get" novalidate></form>

TextBox

- It is used to display text, which can be submitted to server. - It allows to edit text. - Text refers to a string with group chars. [alphabet, number & special chars] - <input> element with **type** as "text" can define a textbox.

Syntax: <input type="text">

Note: "text" is not textbox, it refers to "string" data type.

Attributes:

1. Id]

2. Name] used for reference

3. Class]

Note: Every form element must have "name" attribute. If name is not defined then form can't submit the data of element.

4. Value : It defines the default value to display in textbox.

5. readonly : It will not allow to modify the value. But it can submit the value.

6. disabled : It will not allow to modify and submit.

Syntax: <input type="text" name="qty" value="1" disabled | readonly>

7. placeholder : It defines the water mark text to display in textbox.

Syntax: <input type="text" name="email" placeholder="Your email">

8. autofocus : It sets focus to element automatically on page load.

Syntax: <input type="text" name="user" autofocus>

9. size : It sets the width for textbox. The default size is 20. It is not limit for value, it is just width of textbox.

Syntax: <input type="text" name="age" size="2">

10. maxlength : It limits the number of characters in a textbox.

Syntax: <input type="text" name="age" size="2" maxlength="2">

11. minlength : It validates the minimum number of chars in a textbox.

Syntax: <input type="text" name="age" size="2" minlength="2" maxlength="2">

12. required : It will not allow to submit empty. It is used to for mandatory fields.

Syntax: <input type="text" name="age" required size="2" minlength="2" maxlength="2">

13. list : It refers to a data list of form.

Data list is used to display auto complete text for textbox.

The elements used for configuring form data list are:

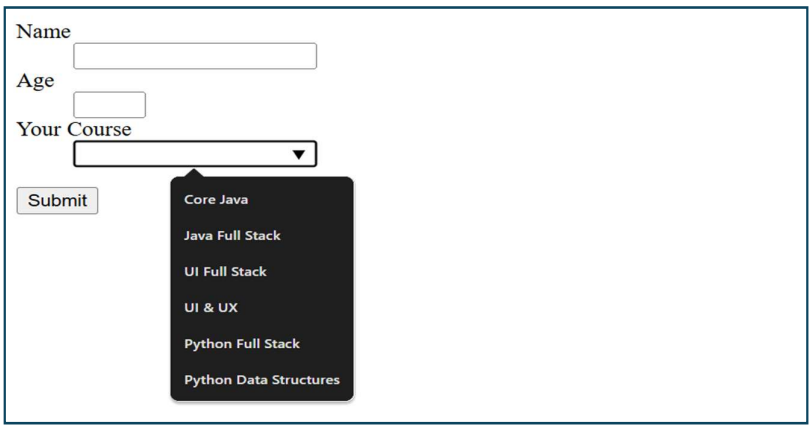
<datalist>

<option>

Syntax:

```
<input type="text"  name="course"  list="courses">

<datalist id="courses">

 <option> Java </option>

 <option> UI </option>

</datalist>
```

Ex_43_form.html:



```
<form>
    <dl>
        <dt>Name</dt>
        <dd><input type="text" name="Name" maxlength="10" autofocus></dd>
        <dt>Age</dt>
        <dd><input type="text" name="Age" required autofocus size="2" minlength="2"
maxlength="2"></dd>
        <dt>Your Course</dt>
        <dd>
            <input type="text" name="Course" list="courses">
            <datalist id="courses">
                <option>Core Java</option>
                <option>Java Full Stack</option>
            </datalist>
        </dd>
    </dl>
    <button>Submit</button>
</form></body></html>
```

14. pattern: It is used to verify the format of input value. It validates the format using a regular expression. Regular expression is built by using meta characters & quantifiers.

Syntax:pattern="\d{3}" => only 3 digits number is allowed.

pattern="[A-Z]{2}\d{2}" => AB34

{n, m} minimum 'n' and maximum 'm'

Syntax: pattern="[A-Z]{2,10}" => only uppercase letters between 2 to 10.

{n, } minimum 'n' and maximum any

Syntax: pattern="[A-Z]{2, }" => uppercase minimum 2 and max any

Task:

1. **Write a pattern for password 4 to 15 chars alpha numeric with underscore.**

   pattern = "\w{4,15}"

2. **Write a pattern to validate mobile number starting with +91 and 10 digits.**

   pattern = "\+91\d{10}"

3. **Write a pattern to validate US mobile number format "+(1)(425) 555-0100"**

   pattern ="\+\(1\)\(\d{3}\)\s\d{3}-\d{4}"

   UK +(44)(20) 1234 5678

4. **Built in expressions**

   (?=.*[A-Z]) at least one uppercase letter

   (?=.*[0-9]) at least one number

   (?=.*[!@#$%]) at least one specified special chars

Write a pattern for user name, It must start with uppercase letter and must have minimum 4 to 10 chars. Only alphabet allowed.

pattern="[A-Z][a-z,A-Z]{3,9}

Write a pattern for user name 4 to 10 chars alphabet & numbers allowed but at least one uppercase letter is mandatory.

pattern="(?=.*[A-Z])[A-Z,a-z,0-9]{4,10}"

Write a pattern for user name 4 to 10 chars alphabet & numbers allowed but at least one uppercase letter & at least one number is mandatory.

pattern="(?=.*[A-Z])(?=.*[0-9])[A-Z,a-z,0-9]{4,10}"

Write a pattern for user name 4 to 10 chars alphabet, numbers & special chars allowed but at least one uppercase letter, at least one number & at least one special

char is mandatory.

pattern=

Password Input

- It is same as textbox but can mask the chars with “*”.- All attributes same as textbox.

Syntax: <input type=“password” name=“password” required pattern=“ ” .. >

- “list” is not recommended for password input.

Number Input

- It restricts value to number type.

- You can set limits with: a) min b) max c) step

Syntax:
<input type=“number” name=“rate” required min=“10.45” max=“18.45” value=“10.45” step=“0.01”>

Range Input

- It is similar to number but provides a slider to choose value. - You can’t input a value, you can just choose.

Syntax: <input type=“range” name=“rating” min=“1” max=“5” step=“0.1” value=“1”>

Color Input

- It uses a color picker. - Color allows only 6 chars hexadecimal value.

Syntax: <input type=“color” name=“color” value=“#ff0000”>

Email Input

- It is same like textbox but validates email using “@“ character. - All attributes same as you define for textbox.

Syntax: <input type=“email” name=“email” required>

URL Input

- It validates URL. - A valid URL must have protocol and domain.

Syntax: <input type="url" name="url" required>

www.amazon.in : // invalid, https://www.amazon.in :// valid

Date & Time Input

- There are various values for date and time type.

- Input type for date & time can be

  a) date b) time c) datetime-local d) week e) month f) year

- Date format for developer is always "yy-mm-dd hrs:min:sec.milliSec"

- You can restrict date by using : a) min b) max

Syntax: <input type="date" name="date" min="2025-11-11" max="2025-11-22">

File Input

- It adds a file browser into page. - It allows to select files on client device. - You can define filters optionally. - Filter can be configured by using "accept" attribute. However filter is not permanent.

Syntax: <input type="file" name="photos" accept=".jpg, .png">

- You can enable selection of multiple files by using the attribute "multiple".

Syntax: <input type="file" name="photos" multiple>

Radio Button

- Radio buttons allow to select any option in UI. - Radio allows to select one or multiple from a group of choices. - But radio can't allow to deselect the selected option. - It requires to configure **"Mutex".** - Mutex refers to mutual exclusion. - You can configure mutex by specifying same name for all radios in a group. - Radios don't have a caption. - Captions can be defined by using "label" element.

Syntax: <input type="radio" name="gender"> <label for="gender"> Male </label>

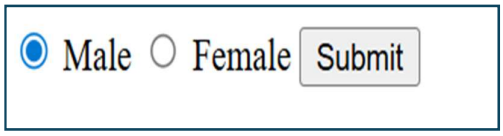
- You can use **"checked"** attribute to select by default.

  Syntax: <input type="radio" name="gender" checked>

- Radios submit "on" as value. Hence it is mandatory to define a value of every radio.

Syntax: <input type="radio" name="gender" value="Male"> <label for="gender"> Male </label>

Ex_44_form_radio.html:




```
<form>

    <input type="radio" value="Male" name="gender" checked> <label
for="gender">Male</label>

    <input type="radio" value="Female" name="gender"> <label
for="gender">Female</label>

    <button>Submit</button>

</form></body></html>
```

Checkbox

- It is similar to radio button.- But a checkbox allows to select or deselect any option dynamically. - It will not use "Mutex" even when the same name is defined for a group of options. - You can select one or multiple from a group of choices.

Syntax:

```
<input type="checkbox" name="course" value="UI" checked>

<label for="course"> UI Full Stack </label>
```

Ex_45_form_**Checkbox**.html:

```
<form>

    <input type="checkbox" name="course" value="Java"> <label
for="course">Java</label> <br><br>
```

```
        <input type="checkbox" name="course" value=".NET"> <label
for="course">.NET</label>

        <br><br>

        <button>Submit</button>

    </form></body></html>
```

Ex_46_checkbox_with_slider.html:

```
    <style>

        .user {

            border-top: none;

            border-left: none;

            border-right: none;

            outline: none;

        }

        ul {

            list-style: none;

            width: 120px;

            border: 1px solid gray;

            padding: 10px;

            height: 50px;

            overflow: auto;
```

```
        }
    </style>
</head>
<body>
    <form>
        <input placeholder="User Name" class="user" type="text">
        <br><br>
         <ul>
            <li> <input type="checkbox"> <label>HTML</label> </li>
            <li> <input type="checkbox"> <label>CSS</label> </li>
            <li> <input type="checkbox"> <label>Sass</label> </li>
            <li> <input type="checkbox"> <label>Bootstrap</label> </li>
            <li> <input type="checkbox"> <label>JavScript</label> </li>
         </ul>
        <button>Submit</button>
    </form></body></html>
```

Dropdown

- It enables user to select any one option from a group of choices.- Dropdown is designed using **<select>** element. - Items in dropdown are defined using **<option>** element.

Syntax: <select><option> Text </option></select>

- Every option is RC type, it can display only plain text content.

[alphabet, number or special chars]

- Symbols, graphics and complex formats are not allowed in <option>.

- Every option comprises of 3 attributes

  a) value It defines the value to submit.

  b) selected It keeps the option selected by default.

  c) disabled It will not allow to select.

Syntax:

```
<option  value="to_submit"  selected>  Text </option>

<option  value="to_submit"  disabled>  Text </option>
```

Ex_47_option.html:

```
<form>

  <dl>

  <dt>Departure</dt>

  <dd>

    <select name="city">

      <option>Choose Departure</option>

      <option disabled value="MUM">Chatrapati Shivaji Terminal[Mumbai]</option>

      <option selected value="RGI">Hyderabad International Airport</option>

    </select>

  </dd>

</dl>

<button>Submit</button>
```

```
</form></body></html>
```

Note: A dropdown can be transformed into ListBox by using following attributes

a) size b) multiple

Multiple allows to select more than one. [changes to list box]

Size changes to listbox.

Syntax: <select size="3" multiple></select>

Textarea

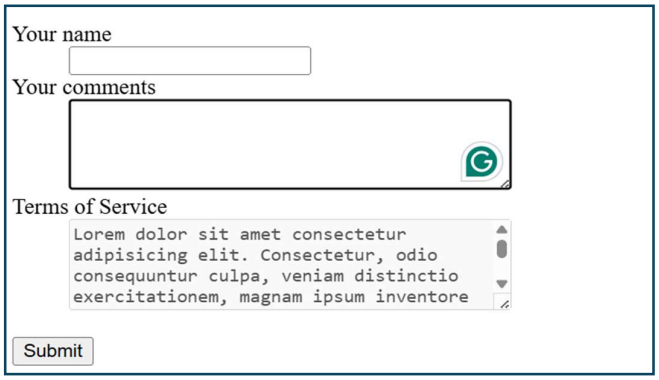
- It is used for handling multiline text.- Textbox enables only single line text. - It can have attributes

a) rows b) cols c) readonly d) disabled e) required f) maxlength g) minlength etc..

- It is RC type.

Syntax: <textarea rows="4" cols="40"> Your text optional </textarea>

Ex_48_textarea.html:




```
<form>
    <dl>
        <dt>Your name</dt>
        <dd><input type="text"></dd>
        <dt>Your comments</dt>
        <dd>
```

```
            <textarea rows="4" cols="40"></textarea>
        </dd>
        <dt>Terms of Service</dt>
        <dd>
            <textarea disabled rows="4" cols="40">Lorem dolor!</textarea>
        </dd>
    </dl>
  <button>Submit</button>
  </form></body></html>
```

Buttons & Multimedia

Meter - It is used to display grade meter. - It can show value but controlled dynamically.

Syntax: <meter min="1" max="100" value="100" low="0" high="0"> </meter>

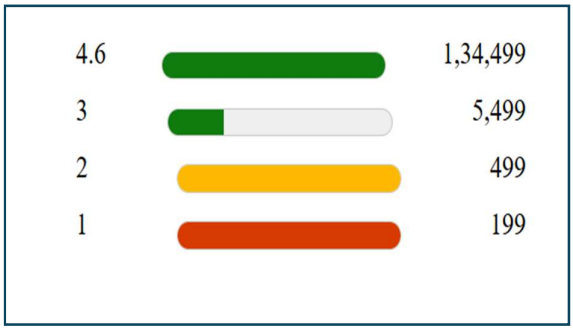
- "low" must be above min & below high.- "high" must be above low & below value.- "value" must be between min & max.

If low & high are zero success [green]

If low & high range is more warning [yellow]

If low & high range is less danger [red]

Ex_49_**Meter.html**:



```
    <style>
        meter {
            width: 150px;

            height: 30px;

        }

        li {

            display:  flex;

            justify-content: space-between;

        }

        ul {

            width: 300px;

            list-style: none;

        }

    </style>

</head>

<body>

    <form>

        <ul>

            <li> <span>4.6 <span class="bi bi-star-fill"></span> </span> <meter min="1"
max="5" value="5"></meter> <span>1,34,499</span> </li>

            <li> <span>3 <span class="bi bi-star-fill"></span> </span> <meter min="1"
max="5" value="2"></meter> <span>5,499</span> </li>

            <li> <span>2 <span class="bi bi-star-fill"></span> </span> <meter low="40"
high="80" min="1" max="100" value="100"></meter> <span>499</span> </li>

            <li> <span>1 <span class="bi bi-star-fill"></span> </span> <meter min="1"
max="100" value="100" low="60" high="80"></meter> <span>199</span> </li>
```

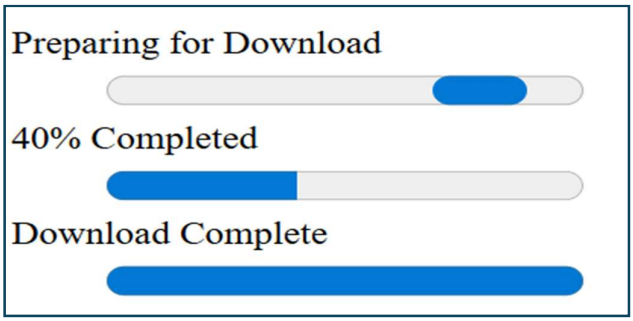
```
    </ul>
</form></body></html>
```

Progress

- It can display are progress bar. - It is used for suspense features in page. - It is for actions like downloading, uploading, copying, installing etc.

Syntax:<progress min="1" max="100" value="30"> </progress>

Ex_50_**Progress.html**:




```
  <style>
    progress {
      width: 200px;
      height: 30px;
    }
  </style>
</head>
<body>
  <dl>
    <dt>Preparing for Download</dt>
    <dd><progress></progress></dd>
    <dt>40% Completed</dt>
    <dd><progress min="1" max="100" value="40"></progress></dd>
    <dt>Download Complete</dt>
```

```
<dd><progress min="1" max="100" value="100"></progress></dd>
</dl></body></html>
```

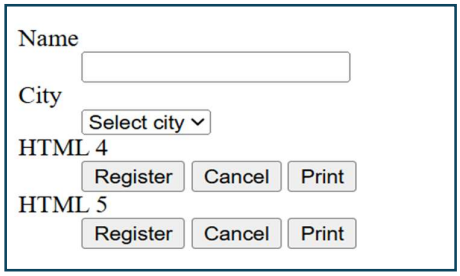
Buttons

syntax:<button type="submit"> Text | Image | Icon </button>

<button type="reset"> Text | Image | Icon </button>

<button type="button"> Text | Image | Icon </button>

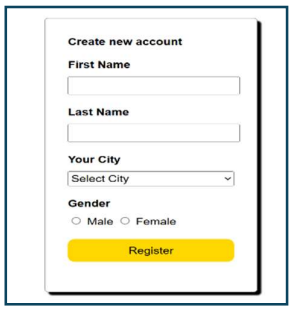
Ex_51_Button.html:




```
<form>
    <dl>
        <dt>Name</dt>
        <dd><input type="text" name="Name"></dd>
        <dt>City</dt>
        <dd><select><option>Select
city</option><option>Delhi</option></select></dd>
        <dt>HTML 4</dt>
        <dd>
            <input type="submit" value="Register">
            <input type="reset" value="Cancel">
            <input type="button" value="Print">
        </dd>
        <dt>HTML 5</dt>
```

```
        <dd>

            <button type="submit"> <span class="bi bi-person"></span> Register
</button>

            <button type="reset"> <span class="bi bi-x-circle"></span> Cancel
</button>

            <button type="button"> <span class="bi bi-printer-fill"> Print </span>
</button>

        </dd>

      </dl></form></body></html>
```

Ex_52_registration_form.html:




```
  <style>

      body {

          display: flex;

          justify-content: center;

      }

      form {

          padding: 30px;

          border:1px solid gray;

          border-radius: 5px;

          margin-top: 40px;

          font-family: Arial;
```

```
    font-size: 16px;
    width: 220px;
    box-shadow: 5px 5px 2px black;
}
.form-group {
    margin-bottom: 20px;
}
.form-title {
    font-weight: bold;
    padding-bottom: 20px;
}
.form-control {
    width: 100%;
    font-size: 16px;
    height: 25px;
}
.form-label {
    font-weight: bold;
    margin-bottom: 10px;
    display: block;
}
.btn-register {
    font-size: 16px;
```

```
            padding: 10px;

            border: none;

            background-color: gold;

            border-radius: 10px;

            width: 100%;

        }

    </style>

</head>

<body>

    <form>

        <div class="form-title"> <span class="bi bi-person-circle"></span> Create new
account</div>

        <div class="form-group">

            <label for="FirstName" class="form-label">First Name</label>

            <div>

                <input type="text" name="FirstName" class="form-control">

            </div>

        </div>

        <div class="form-group">

            <label for="LastName" class="form-label">Last Name</label>

            <div>

                <input type="text" name="LastName" class="form-control">

            </div>

        </div>
```

```
        <div class="form-group">
            <label for="City" class="form-label">Your City</label>
            <div>
                <select name="City" class="form-control">
                    <option>Select City</option>
                    <option>Delhi</option>
                </select>
            </div>
        </div>
        <div class="form-group">
            <label for="Gender" class="form-label">Gender</label>
            <div>
                <input type="radio" name="gender" value="Male"> <label class="form-
check-label"> Male </label>
                <input type="radio" name="gender" value="Female"> <label class="form-
check-label"> Female </label>
            </div>
        </div>
        <div class="form-group">
            <button class="btn-register">Register</button>
        </div>
    </form>
</body>
</html>
```

Multimedia

- Multimedia refers to animation, audi & video. - HTML provides various elements for handling multimedia

a) <marquee> b) <video> c) <audio> d) <embed> Deprecated for multimedia

Syntax:

<video src="path" controls width="100" height="100" poster="path" autoplay> </video>

<audio src="path" … > </audio>

Ex_53_Audio_video.html:

```
<html><body><video src="./videos/js-class.mp4" poster="./images/Java.png" width="400" controls height="400"></video></body></html>
```

Styles

Syntax: Your text

FAQ's:

1. Where to embed the <style> container?

A. You can embed in <head> and <body>

2. What is difference between styles in <head> & styles in <body>?

A. Styles in head can save round-trip.

Styles in body will load always as a result of round-trip.

3. Can we have multiple style containers in a page?

A. Yes.

4. If there are multiple style containers and you are configuring same style for element in different containers then which set will apply?

A. It always uses **LIFO** mechanism. [Last-In-First-Out].If different attributes are

configured then all will apply.

5. If styles are defined by using inline & embedded then which set will apply?

A. Always first priority is for inline. If different styles are configured then all will apply.

6. What is MIME type of styles?

A. MIME refers to content type. Style content type is defined as “text/css”.

Syntax: <style type=“text/css”> </style>

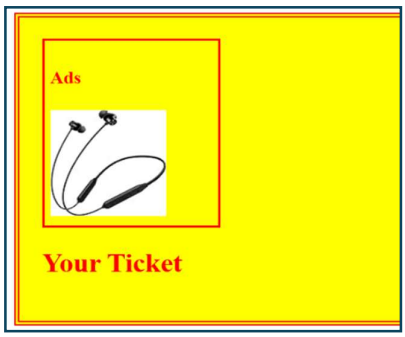
7. What is media type for styles?

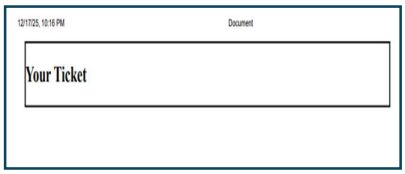
A. It allows to configure styles for various outputs, which include screen, printer & audio.You can have media type as “all, screen, print and speech”.

Syntax: <style type=“text/css” media=“print | speech | screen | all”>

EX_54_css_media_screen_and_print.html:

screen **print**






```
<style type="text/css" media="screen">

    .ticket {

        background-color: yellow;

        padding: 20px;

        border:5px double red;

        color:red;

    }
```

```
        aside {
            width: 140px;
            border:2px solid red;
            padding: 5px;
        }
    </style>
    <style type="text/css" media="print">
        .ticket {
            background-color: none;
            color:black;
            border: 2px solid black;
        }
        aside {
            display: none;
        }
    </style>
</head>
<body>
    <div class="ticket">
        <aside>
            <h4>Ads</h4>
            <img src="../public/images/a10.jpg" width="100" height="100">
        </aside>
```

```
    <h2>Your Ticket</h2>
  </div></body></html>
```

External Style Sheet

1. Add a new style sheet into “src/styles” folder by using “index.css”

2. Write styles directly inside index.css

3. Link the stylesheet to any HTML page.

```
<link rel=“stylesheet” href=“../src/styles/index.css”>
```

1st Priority : Inline

2nd Priority : Depends or which one is latest [embedded or external]

CSS Minification & Selectors

- Minification is a technique of compressing the code. - It is the process of reducing file size by removing additional blank spaces, line breaks, using short hand techniques for writing styles etc.- Uncompressed files are used in development and minified files are used in production.

Steps:

1. Copy the actual CSS code 2. Go to any CSS minifier

   https://www.toptal.com/developers/cssminifier

3. Paste the code => click Minify => copy minified code. 4. Go to project src/styles

5. Add a new file “home.min.css” 6. Paste minified code

7. Link minified file to your HTML page.

```
<link rel=“stylesheet” href=“../src/styles.home.min.css”>
```

home.css development

home.min.css production

CSS Selectors:- A selector is required to select element in page, in order to apply set of styles. - Inline styles doesn’t require a selector as styles are configured directly for element. - Embedded and external styles require selector. - There are no official categories for selectors but most commonly used category names are

1. Primary Selectors 2. Relational Selectors 3. Structural Selectors

4. Attribute Selectors 5. Dynamic Pseudo classes 6. Element State Selectors

7. Validation State Selectors 8. Universal Selector 9. Root Selector

10. Language Selector etc.

Primary Selectors

1. Type Selector 2. ID Selector 3. Class Selector

Type Selector:

- It refer to the element name.

- It sets effects to every occurrence of element in page.

- You can't skip of any specific occurrence.

Syntax: p, img, header, aside, nav {}

ID Selector: #text-style {….}

 <p id="text-style">

- Every element can refer to only one ID. - You can't configure multiple sets of styles for one element using ID.

Class Selector:- HTML element can implement multiple classes. - You can configure multiple sets of styles for one element.

Syntax: .class-1{…….} <p class="class-1 class-2 …">

Ex:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>

    <style>

        .text-style {

            color:red;

            text-align: center;

        }

        .bg-style {

            background-color: yellow;

        }

        .border-style {

            padding: 10px;

            border:4px double red;

        }

    </style>

</head>

<body>

    <h1 class="bg-style border-style text-style">Web Technologies</h1>

    <h1 class="text-style border-style">HTML</h1>

</body>

</html>
```

FAQ's:

1. If element is configured with type, id and class selectors then which set will apply?

A. If different styles are configured then all will apply. If same style attribute is defined then it uses the priority : 1st :ID Selector, 2nd :Class Selector, 3rd :Type Selector

2. If ID is defined using embedded and external techniques then which ID will apply?

A. The latest will apply. [LIFO]

Relational Selectors:

1:child selector : parent child { } It refers to all child elements at any level.

2: direct child selector : parent > child { } It refers only to direct child elements.

3: general sibling : A ~ B { } It refers to all elements after specified. But in the same scope.

4:adjacent sibling : A + B { } It refers to first occurrence element.

Ex_55_direct_child.html:

```
    <style>

        header > span {

            color: red;

        }

    </style>

</head>

<body>

    <header>

        <span>Amazon</span>

        <nav>

            <span> Home </span>

            <span> Shop </span>

        </nav>

    </header></body></html>
```

Ex_56_ general_sibling.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        h1~p {
            color: red;
        }
    </style>
</head>
<body>
    <div>
        <h1>Front End</h1>
        <p> HTML </p>
        <p> CSS </p>
        <p> Bootstrap </p>
        <p> JavaScript </p>
    </div>
        <p> React JS </p>
```

</body></html>

CSS Selectors

Attribute Selectors:

element[attribute] : It refers to elements that have given attribute.

Conditions:

element[attribute=value] : It refers to exact match

element[attribute^=value] : It refers to start with.

element[attribute$=value] : It refers to end
width. element[attribute|=value] : It must start with specified and can separate with“-“delimiter.

element[attribute~=value] : It refers to individual occurrence with space delimiter.
[It can be at any occurrence]
Ex_57_**Attribute_Selectors.html**:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

        input[type="button"] {

            background-color: yellow;

        }

        img[src] {
```

```
            display: inline;
        }
        img {
            display: none;
        }
    </style>
</head>
<body>
    <img width="100" height="100">
    <img width="100" src="../public/images/a2.jpg" height="100">
    <img width="100" height="100">
    <form>
        <dl>
            <dt>First Name</dt>
            <dd><input type="text"></dd>
            <dt>Last Name</dt>
            <dd><input type="text"></dd>
        </dl>
        <input type="button" value="Login">
        <input type="button" value="Cancel">
    </form>
</body>
</html>
```

Ex_58_**Attribute_Selectors.html**:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

       p[class~="Effect"] {

          color:red;

       }

    </style>

</head>

<body>

    <p class="Effect">Para-1</p>

    <p class="ParaEffect">Para-2</p>

    <p class="TextEffectPara">Para-3</p>

    <p class="Text Effect">Para-4</p>

    <p class="Effect Text" >Para-5</p>

    <p class="Text Effect Para">Para-6</p>

    <p class="Text-Effect">Para-7</p>

    <p class="Effect-Text">Para-8</p>
```

```
    <p class="Text-Effect-Para">Para-9</p>
</body></html>
```

Structural Selectors:

:first-child : first child element.

:last-child : last child element.

:nth-child() : specific level child element or even & odd occurrence.

:nth-of-type() : every nth occurrence from top.

:nth-last-of-type() : every nth occurrence from bottom.

Syntax:

```
tr : first-child { }

tr : nth-child(2) { }

tr : nth-child(even) { }

tr : nth-of-type(2n) { }

tr : nth-of-type(2n+1) { }

tr : nth-last-of-type(2n+1) { }
```

Ex_59_Structural_Selectors.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>
```

```
        tbody tr:nth-last-of-type(3n+2) {
            background-color: #c6e9c6;
        }
        thead tr {
            background-color: green;
            color:white;
        }
    </style>
</head>
<body>
    <table border="1" width="100%">
        <thead>
            <tr>
                <th>Name</th>
                <th>Price</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td> </td>
                <td> </td>
            </tr>
            <tr>
```

```
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
<tr>
    <td> </td>
    <td> </td>
</tr>
```

```
            <tr>
                <td> </td>
                <td> </td>
            </tr>
            <tr>
                <td> </td>
                <td> </td>
            </tr>
            <tr>
                <td> </td>
                <td> </td>
            </tr>


        </tbody>
    </table>
</body>
</html>
```

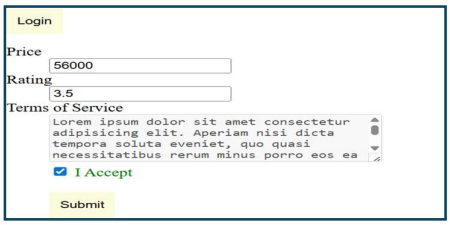
Dynamic Pseudo Classes:

- Dynamic refers to styles applied to element according to state and situation. - Pseudo refers to a false appearance for element. - Class is a program template with pre-defined data & logic, which you can implement by customizing according to the requirements.

Element State Classes

: hover, : focus, : active, : disabled, : read-only, : checked, : target, : visited

Ex_60_Element_State.html:




```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

        button {

            border:none;

            padding: 10px;

            background-color: rgb(251, 251, 222);

        }

        button:hover {

             border:none;

            background-color: rgb(246, 246, 47);

            padding: 10px;

            cursor: grab;

        }

        button:active {

            border:4px solid red;

        }

        #rating:read-only {

            cursor: not-allowed;
```

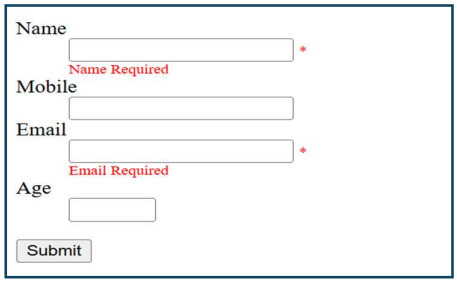
```
        background-color: rgb(235, 232, 232);
    }
    #rating+div {
        display: none;
    }
    #rating:focus+div {
        display: block;
        font-size: 12px ;
        color:red;
    }
    #terms+label {
        color:red;
    }
    #terms:checked+label {
        color:green;
    }
    #terms~button {
        display: none;
    }
    #terms:checked~button {
        display: block;
    }
</style>
```

```
</head>
<body>
    <button>Login</button>
    <dl>
        <dt>Price</dt>
        <dd><input type="text" value="56000" readonly></dd>
        <dt>Rating</dt>
        <dd>
            <input id="rating" type="text" value="3.5">
            <div>Rating 1 to 5 only</div>
        </dd>
        <dt>Terms of Service</dt>
        <dd>
            <textarea rows="4" cols="40" disabled>Lorem ipsum </textarea>
        </dd>
        <dd>
            <input type="checkbox" id="terms"> <label>I Accept</label>
            <br><br>
            <button>Submit</button>
        </dd>
    </dl></body></html>
```

Element Validation State Classes

: required, : optional, : in-range, : out-of-range, : valid, : invalid

Ex_61_Element_Validation_State.html:




```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

        #Email:optional+span, #Mobile:optional+span, #Name:optional+span {

            display: none;

        }

        #Email:required+span, #Mobile:required+span, #Name:required+span {

            display: inline;

            font-size: 12px;

            color:red;

        }

        #Name:valid+span, #Email:valid+span, #Mobile:valid+span{

            display: none;
```

```
        }
        #Name:valid~div, #Mobile:valid~div, #Email:valid~div {
            display: none;
        }
        #Name:invalid~div, #Mobile:invalid~div, #Email:invalid~div {
            display: block;
            color:red;
            font-size: 12px;
        }
        #Mobile:invalid {
            border:1px solid red;
            outline: none;
        }
        #Age:in-range+div {
            display: none;
        }
        #Age:out-of-range+div {
            display: block;
            color:red;
            font-size: 12px;
        }
    </style>
</head>
<body>
```

```
<dl>
    <dt>Name</dt>
    <dd>
        <input type="text" id="Name" required>
        <span>*</span>
        <div>Name Required</div>
    </dd>
    <dt>Mobile</dt>
    <dd>
        <input type="text" id="Mobile" pattern="\+91\d{10}">
        <span>*</span>
        <div>Invalid Mobile: +91 & 10 digits</div>
    </dd>
    <dt>Email</dt>
    <dd>
        <input type="text" id="Email" required>
        <span>*</span>
        <div>Email Required</div>
    </dd>
    <dt>Age</dt>
    <dd><input type="number" min="15" max="30" id="Age">
        <div>Age 15 to 30 only</div>
    </dd>
```

```
    </dl>

    <button>Submit</button>
</body></html>
```

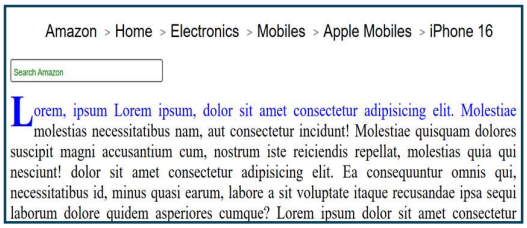
CSS selectors & CSS Units

Structural & Occurrence Pseudo Classes

::first-letter, ::first-line, ::placeholder, ::before, ::after, ::selection

Note: The entity "::" refers to implementation with set of limitations[rules].

Ex_62_breadcrumbs_Pseudo_Classes.html:




```
    <style>
        p {
            text-align: justify;
        }
        p::first-letter {
            font-size: 40px;
            font-weight: bold;
            float: left;
            line-height: 20px;
        }
        p::first-line {
            color:blue;
```

```
}
p::selection {
    background-color: yellow;
    text-decoration: underline solid red;
    font-size: 30px;
}
input::placeholder {
    color:green;
    font-size: 8px;
}
ul {
    list-style: none;
    display: flex;
}
li::after {
    content: ' > ';
    padding-left: 5px;
    padding-right: 5px;
    color:gray;
    font-size: 12px;
}
li {
    font-family: Arial;
```

```
        }

        li:last-child::after {

            content: '';

        }

    </style>

</head>

<body>

    <ul>

        <li>Amazon</li> <li>Home</li><li>Electronics</li><li>Mobiles</li><li>Apple Mobiles</li><li>iPhone 16</li>

    </ul>

    <input type="text" placeholder="Search Amazon">

    <p>Lorem, </p></body></html>
```

Universal Selector: * {… } - It selects all elements in page.

Root Selector: :root {…} - It is used for elements in shadow DOM. - Shadow DOM is configured for components. [date, time, color, file ..]

Language Selector: :lang("en-in") {…} - It is used for handling multilingual content in a page.

Summary

- CSS Integration - CSS Selectors - CSS Rules

CSS Inheritance

- Inheritance is a mechanism where the child element inherits the features of parent. - However inheritance is abnormal in behaviour as child can implement only few features. - You can control inheritance by using following values: a) inherit b) initial c) unset

- "**inherit"** gets the parent feature and assigns to child.
- "**initial"** removes the parent feature assigned to child.
- **"unset"** removes the default feature assigned to element.

Ex_63_Inheritance.html:

```
    <style>
        .container {
            color:red;
            border:4px double red;
            padding: 20px;
        }
        p {
            border: inherit;
            padding: inherit;
            color:initial;
        }
        h1 {
            font-weight: unset;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>Heading</h1>
        Container Text
        <p>Para in container</p>
    </div></body></html>
```

CSS Units:- Units are used to change position, scale and orientation of element. - Units are categorized into 2 types: **a) Absolute Units** **b) Relative Units**

Absolute Units:- These are individual units.- They are not effected by parent or adjacent elements. - The primary unit is **pixel.**

Unit	Name	Equivalent to
cm	Centimeters	1cm = 96px/2.54
mm	Millimeters	1mm = 1/10th of 1cm
Q	Quarter-millimeters	1Q = 1/40th of 1cm
in	Inches	1in = 2.54cm = 96px
pc	Picas	1pc = 1/6th of 1in
pt	Points	1pt = 1/72th of 1in
px	Pixels	1px = 1/96th of 1in

Relative Units:- These units depends on the parent elements. - They use a specific ratio of parent. - Child element position, size, orientation depends on its parent.

a) em- **Element Size** b) rem- **Root Element Size**

c) %- **Ratio of parent** d) vh- **Viewport height** e) vw- **Viewport width**

Note:**- "em" & "rem" are used for text.**

Ex_64_em_rem_vh.html:

```
<style>
    .container {
        font-size: 30px;
        border :1px solid black;
        padding: 10px;
    }
```

```
        .container p {
            font-size: 0.8em;
        }
        span {
            font-size: 1rem;
        }
        .box {
            border : 2px solid red;
            height: 50px;
            width: 50%;
        }
    </style>
</head>
<body>
    body text
    <div class="container">
        container text
        <p> para in container </p>
        <span>Span in container</span>
        <div class="box"></div>
    </div></body></html>
```

Summary: - CSS Integration, - CSS Selectors, - CSS Rules, - CSS Inheritance, - CSS Units, - CSS Colors

CSS Box Model

1. Margin 2. Padding 3. Border 4. Border Radius 5. Width 6. Height

Margin:

margin : short hand for all directions [top, right, bottom, left orientation]

margin-left, margin-right, margin-top, margin-bottom

Padding

padding : short hand for all directions [top, right, bottom, left orientation]

Border

border : It is short hand for all directions and all styles

border-width : It sets size

border-color : It sets color

border-style : It sets style double, solid, dotted, dashed, groove etc.

border-left : left short hand for style, width, color.

border-left-width

border-left-color

border-left-style

… similar for all directions …

Border Radius

border-radius : short hand for all direction.

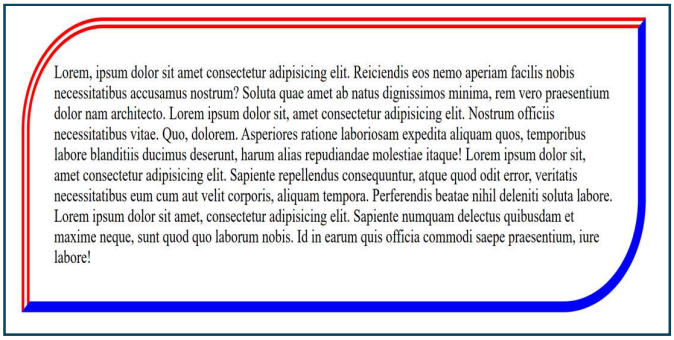
border-top-left-radius

border-top-right-radius

border-bottom-left-radius

border-bottom-right-radius

Ex_65_Box_Model.html:



```
    <style>
        p {
            border-left: 10px double red;
            border-top: 10px double red;
            border-right: 10px solid blue;
            border-bottom: 10px solid blue;
            margin: 50px;
            padding: 30px;
            border-top-left-radius: 100px;
            border-bottom-right-radius: 100px;
        }
    </style>
</head>
<body>    <p>Lorem, ipsum </p></body></html>
```

CSS Positions, Z-index, Backgrounds

border-image : It sets image as border with style and offset.

Syntax: { border-image : url(“path”) stretch | space offset% ; }

Note: To set border image you have to define border style with transparent color.

Syntax:

```
p {
    border : 30px solid transparent;
    border-image: url(“path”)  stretch 70;
}
```

Ex_67_ border_image.html:

```
    <style>
        p {
            border: 40px solid transparent;
            padding: 20px;
            border-image: url("../public/images/border-1.png") stretch 50;
        }
    </style>
</head>
<body> <p>Lorem, ipsum </p> </body> </html>
```

CSS Positions

- You can set position for element with values

  a) static b) absolute c) relative d) fixed e) sticky

- Position requires the attributes a) top

  b) left c) right d) bottom

Static:- It keeps the element according to normal flow of document. - It will not allow to change the position using left, right, top & bottom. - It is the default position.

Absolute:- It removes element from normal flow of document. - It allows to change the position using left, right, top & button.- But It keeps the element with regard to content in page.

Fixed:- It removes element from normal flow of document. - It allows to change position. - But it keeps element with regard to browser window. - It locks scrolling of element.

Relative:- It is defined for parent element to keep the child elements relative to parent. - It keeps element according to normal flow of document. - It will not allow to change the position. - It anchors the parent to keep child elements relative to context.

Note: It is mandatory to configure child element with “absolute” position, when the parent is set to relative.

Ex_68_ badge.html:

```
<link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

<style>

    .btn {

        border: none;

        padding: 10px;

        background-color: yellow;

        font-size: 30px;

        position: relative;

    }

    .badge {

        display: inline-block;

        width: 20px;

        height: 20px;

        text-align: center;

        border-radius: 20px;

        background-color: red;

        color:white;

        font-size: 14px;

        padding: 2px;

        position: absolute;
```

```
            top: 0px;

            right: -10px;

        }

    </style>

</head>

<body>

    <button class="btn bi bi-cart4"> <span class="badge">2</span> </button>

</body></html>
```

Sticky: - It keeps element according to normal flow of document. - It allows to scroll along with content. - It sticks to the window at specified position.

Syntax: h1 { position: sticky; top: 0px;   }

CSS Z-Index

- It is used to control the placement of overlapping elements. - You can send back or backward and front or forward. - Index from bottom starts with “0”.

Syntax: element1 { z-index : 0; }   element2 { z-index : 1; }

Ex_69_Z-Index.html:

```
    <style>

        .men-fashion {

            background-image: url("../public/images/men-fashion.jpg");

            background-size: cover;

            height: 300px;


            width: 500px;
            z-index: 1;

        }

         .women-fashion {
```

```
        background-image: url("../public/images/women-fashion.jpg");
        background-size: cover;
        height: 300px;
        width: 300px;
        left: 300px;
        top: 200px;
        z-index: 2;
    }
     .kids-fashion {
        background-image: url("../public/images/kids-fashion.jpg");
        background-size: contain;
        height: 200px;
        width: 400px;
        left: 450px;
        top: 50px;
        z-index: 0;
    }
    .card {
        border-radius: 30px;
        box-shadow: 2px 2px 2px black;
        position: absolute;
    }
</style>
```

```
</head>

<body>

    <div class="card men-fashion"></div>

    <div class="card women-fashion"></div>

    <div class="card kids-fashion"></div>

</body></html>
```

CSS Backgrounds

a) background-color b) background-image c) background-size d) background-position
e) background-repeat f) background-attachment g) background-clip

Note : You can set multiple background images. It requires all relative attributes separated with “,” delimiter.

Syntax : { background-image : url(“pic1”), url(“pic2”);
background-repeat: no-repeat, repeat; }

Ex_70_Backgrounds.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <style>

       body {

          background-image: url("../public/images/a5.jpg"), url("../public/images/a11.jpg");

          background-repeat: no-repeat, repeat;

          background-position: center center;
```

```
        background-size: 300px 300px, 100px 100px;

        background-attachment: fixed, scroll;

        height: 100vh;

      }

   </style>

</head>

<body>



</body>

</html>
```

CSS Display, text Styles

CSS Text Styles

font-size, font-family, font-weight, font-style, font-variant [uppercase] => small-caps,color [text color]

text-align, text-decoration [underline, overline, line-through], text-shadow, text-overflow, text-transform [uppercase, lowercase, capitalize], text-indent [first line space]

letter-spacing, word-spacing,

line-height

word-break [break-all]

white-space [nowrap]

writing-mode [sideways-lr, rl, vertical-lr, rl]

overflow

direction [rtl, ltr]

Syntax: p { writing-mode: sideways-lr ; text-transform : capitalize; }

CSS Float & Clear:

a) float : left, right b) clear : left, right, both

Syntax: h1 { float :left; } p { clear: left; }

CSS Display

- CSS **display** attribute can control the display of elements using following values

1. none : It hides element and removes allocated space.
2. Inline : It allows content beside the element in same line.
3. Block : It sets a block style that keeps adjacent elements in new line.
4. Inline-block: It keeps in same line but adds block style to change width, height etc.

FAQ: What is difference between "display:none & visibility:hidden"?

Ans: display:none removes the allocated space after hiding element. visibility:hidden keeps the allocated space after hiding element.

5. Flex : It is the style defined for container to keep its child elements flexible.

Flex container can be controlled by using following attributes

a) flex-direction : [row, column, row-reverse, column-reverse]

b) flex-wrap : [wrap, nowrap]

c) justify-content : [baseline, center, end, space-between | around | evenly]

d) align-items : [baseline, center, end]

FAQ: What is difference between inline & flex?

Ans: Inline transforms a block element into inline, so that you can keep any content beside.Flex is defined for a container to keep its child elements flexible.

FAQ: What is flex wrap?

Ans : It can set content in flex box wrapped into multiple rows or columns. It aligns content into rows or columns according to container width & height.

6. Grid : It is used to create a responsive layout that can change the contents position in rows and columns dynamically.

A responsive layout adjusts content according the device. It requires various

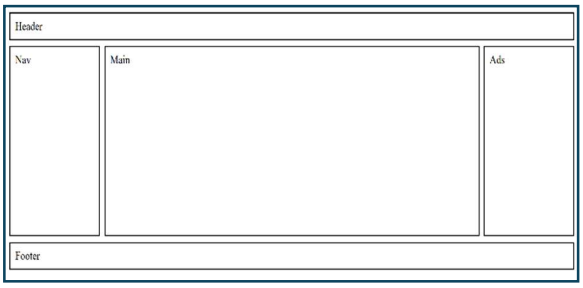
media queries. [@media]

Grid provides “template-columns” which can be adjusted with in 12 fractions.

Grid requires attributes:

a) grid-template-columns b) grid-row c) grid-column d) grid-gap

Ex_71_grid.html:

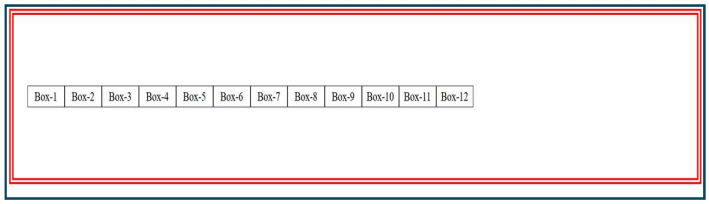



```
<style>
    header, nav, main, aside, footer {
        border:2px solid black;
        padding: 10px;
        font-size: 20px;
        margin: 5px;
    }
    main {
        height: 300px;
    }
    body {
        display: grid;
        grid-template-columns: 2fr 8fr 2fr;
```

```
        }
        header {
            grid-row: 1;
            grid-column: 1/4;
        }
        nav {
            grid-row: 2;
            grid-column: 1;
        }
        main {
            grid-row: 2;
            grid-column: 2;
        }
        aside {
            grid-row: 2;
            grid-column: 3;
        }
        footer {
            grid-row: 3;
            grid-column: 1/4;
        }
    </style>
</head>
```

```
<body>
    <header>Header</header>
    <nav>Nav</nav>
    <main>Main</main>
    <aside>Ads</aside>
    <footer>Footer</footer>
</body></html>
```

Ex_72_box.html:




```
    <style>
      .box {
        width: 70px;
        border:1px solid black;
        padding: 5px;
        text-align: center;
        font-size: 20px;

      }
      .container {
         border: 10px double red;
         padding: 30px;
         display: flex;
```

```
        flex-direction: row;

        flex-wrap: wrap  ;

        justify-content: baseline;

        align-items: center;

        height: 200px;

      }

   </style>

</head>

<body>

   <div class="container">

      <div class="box">Box-1</div>

      <div class="box">Box-2</div>

   </div></body></html>
```

Media Queries

CSS Media Queries - Media queries are required to design responsive pages. - A responsive page can adjust its content according to device width and orientation.

Syntax: @media type and (query) {……..}

```
        type            : screen, print

        query            : orientation, width, min-width, max-width

        orientation    : landscape, portrait
```

Ex_73_Media_Queries.html:

```
   <style>
      @media screen and (orientation: landscape){
```

```
        body {
            background-color: yellow;
        }
    }
    @media screen and (orientation: portrait) {
        body {
            background-color: greenyellow;
        }
    }
  </style>
</head><body>
</body></html>
```

FAQ: What is difference between “width, min-width & max-width”?

Ans:

width : It refers to exact units.

min-width : It specifies the starting point.

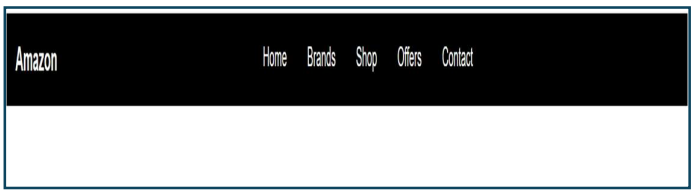
max-width : It specifies the ending point.

(width: 600px) => exactly at 600px

(min-width: 600px) => starting from 600px upto end.

(max-width: 600px) => starting from 0 to 600px.

Ex_74_media.html:

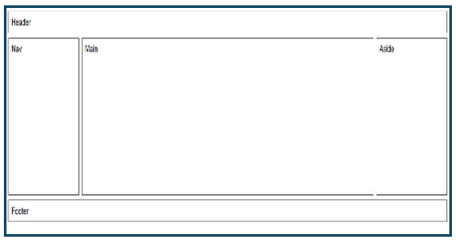


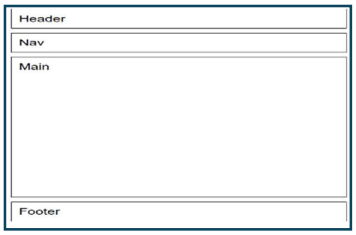
```
<style>
  header {
    padding: 20px;
    font-family: Arial;
    font-size: 20px;
    background-color: black;
    color:white;
   display:  flex;
   flex-direction: row;
   justify-content: space-between;
  }
  .brand-title {
   font-weight: bold;
   font-size: 24px;
  }
  @media screen and (orientation: landscape){
    nav span {
      padding: 0px 20px 0px 20px;
    }
    .bi-justify {
      display: none;
    }
    .bi-cart4 {
```

```
            font-size: 24px;
          }
        }
        @media screen and (orientation: portrait) {
            nav {
                display: none;
            }
            .bi-cart4 {
                display: none;
            }
            .bi-justify {
                display: inline;
            }
        }
    </style>
</head>
<body>
    <header>
        <div>
            <span class="brand-title">Amazon</span>
        </div>
        <div>
            <nav>
```

```
            <span>Home</span>
            <span>Brands</span>
            <span>Shop</span>
            <span>Offers</span>
            <span>Contact</span>
          </nav>
        </div>
        <div>
          <span class="bi bi-cart4"></span>
          <button class="bi bi-justify"></button>
        </div>
    </header></body></html>
```

Ex_75_media.html:






```
    <style>
        header, nav, main, aside, footer {
            border : 1px solid black;
            padding: 10px;
            font-family: Arial;
            font-size: 20px;
            margin: 5px;
        }
```

```
main {
    height: 300px;
}
@media screen and (orientation: landscape) {
     body {
        display: grid;
        grid-template-columns: 2fr 8fr 2fr;
     }
     header {
        grid-row: 1;
        grid-column: 1/4;
     }
     nav {
        grid-row: 2;
        grid-column: 1;
     }
     main {
        grid-row: 2;
        grid-column: 2;
     }
     aside {
        grid-row: 2;
        grid-column: 3;
```

```
    }
    footer {
      grid-row: 3;
      grid-column: 1/4;
    }
}
@media screen and (orientation: portrait) {
    body {
      display: grid;
      grid-template-columns: 12fr;
    }
    header {
      grid-row: 1;
    }
    nav {
      grid-row: 2;
    }
    main {
      grid-row: 3;
    }
    aside {
      grid-row: 4;
      display: none;
```

```
            }
            footer {
              grid-row: 5;
            }
        }
    </style>
</head>
<body>
    <header>Header</header>
    <nav>Nav</nav>
    <main>Main</main>
    <aside>Aside</aside>
    <footer>Footer</footer>
</body></html>
```

Ex: Design a page for mobile & pc screen

1. Open “Google Stitch” and login with your google account

   http://stitch.withgoogle.com

2. Enter the following prompts

   - Create 2 home pages for a shopping app

   - First home page for PC Responsive

   - Second home page for Mobile responsive [create Mobile Layout]

   - Home page must have a navigation with brand name, menu items, cart icon on right.

- A hero banner

- A section with cards displaying featured products

- First home page section with 2 x 4 cards

- Second home page section with 2x2 cards

- Footer with services, social links and contact details.

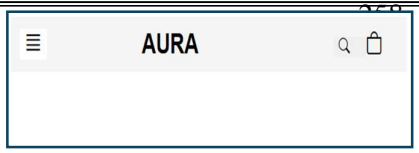
Ex : src/styles/shopping.css

```
@media screen and (orientation: landscape){
    nav span {
        font-size: 18px;
        padding: 0px 10px 0px 10px;
    }
    .bi-bag {
       margin-left: 20px;
       color:blue;
    }
    .bi-search {
       border:none;
       font-size: 18px;
    }
    .bi-justify {
       border: none;
       background-color: white;
       font-size: 22px;
       display: none;
    }
```

```
}
@media screen and (orientation: portrait) {
    nav {
      display: none;
    }
    input {
      display: none;
    }
    .bi-justify {
       border: none;
      background-color: white;
      font-size: 22px;
      display: inline;
      float: left;
    }
    .brand-title {
      display: flex;
      justify-content: center;
      margin-left: 150px;
    }
    .bi-search {
       border: none;
       padding-right: 20px;
    }
```

```
}
 .quick-links {
        display: flex;
        flex-direction: row;
        padding-right: 20px;
     }
header {
    font-family: Arial;
    font-size: 20px;
    padding: 10px;
    background-color: whitesmoke;
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    align-items: center;
}
.brand-title {
    font-size: 24px;
    font-weight: bold;
}
```

Ex_76_shopping.html



```
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../src/styles/shopping.css">

</head>

<body>

    <header>

        <div>

          <button class="bi bi-justify"></button>

          <span class="brand-title">AURA</span>

        </div>

        <div>

          <nav>

              <span>Home</span>

              <span>Women Fashion</span>

              <span>Men Fashion</span>

              <span>New Arrivals</span>

          </nav>

        </div>

        <div  class="quick-links">

          <div>
              <input type="text" placeholder="Search Aura.in">
```

```
            <button class="bi bi-search"></button>
        </div>
        <div>
            <span class="bi bi-bag"></span>
        </div>
      </div>
  </header></body></html>
```

CSS Transforms

Ex:

1. shopping.css

```
@media screen and (orientation: landscape){
    nav span {
        font-size: 18px;
        padding: 0px 10px 0px 10px;
    }
    .bi-bag {
       margin-left: 20px;
       color:blue;
    }
    .bi-search {
       border:none;
       font-size: 18px;
    }
    .bi-justify {
```

```
   border: none;
   background-color: white;
   font-size: 22px;
   display: none;
}
.hero-banner {
 background-image:  url("../../public/images/men-banner.jpg");
 background-size: cover;
 height: 450px;
 text-align: center;
 font-family: Arial;
 display: flex;
 justify-content: center;
 align-items: center;
}
.banner-title {
  font-size: 40px;
  font-weight: bold;
  text-shadow: 2px 2px 2px black;
  color:white;
}
.banner-subtitle {
   font-size: 16px;
```

```
        color:white;
    }
}
@media screen and (orientation: portrait) {
    nav {
        display: none;
    }
    input {
        display: none;
    }
    .bi-justify {
         border: none;
        background-color: white;
        font-size: 22px;
        display: inline;
        float: left;
    }
    .brand-title {
        display: flex;
        justify-content: center;
        margin-left: 150px;
    }
    .bi-search {
```

```
        border: none;
        padding-right: 20px;
    }
    .hero-banner {
    background-image:  url("../../public/images/women-banner.jpg");
    background-size: cover;
    height: 450px;
    text-align: center;
    font-family: Arial;
    display: flex;
    justify-content: center;
    align-items: center;
    }
    .banner-title {
     font-size: 40px;
     font-weight: bold;
     text-shadow: 2px 2px 2px black;
     color:white;
    }
    .banner-subtitle {
       display: none;
    }
}
```

```
 .quick-links {
        display: flex;
        flex-direction: row;
        padding-right: 20px;
     }

header {
    font-family: Arial;
    font-size: 20px;
    padding: 10px;
    background-color: whitesmoke;
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    align-items: center;
}
.brand-title {
    font-size: 24px;
    font-weight: bold;
}

.btn {
```

```
    background-color: rgb(255, 255, 254);

    border:none;

    font-size: 20px;

    margin-top: 30px;

    padding: 20px;

    width: 130px;

    text-align: center;

}
```

Ex_77_Shopping.html

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../src/styles/shopping.css">

</head>

<body>

    <header>

        <div>

            <button class="bi bi-justify"></button>
```

```
        <span class="brand-title">AURA</span>
      </div>
      <div>
        <nav>
            <span>Home</span>
            <span>Women Fashion</span>
            <span>Men Fashion</span>
            <span>New Arrivals</span>
        </nav>
      </div>
      <div  class="quick-links">
        <div>
            <input type="text" placeholder="Search Aura.in">
            <button class="bi bi-search"></button>
        </div>
        <div>
            <span class="bi bi-bag"></span>
        </div>
      </div>
  </header>
  <section>
     <div class="hero-banner">
        <div>
            <div class="banner-title">Discover our Newest Collection</div>
```

```
            <div class="banner-subtitle">Fresh styles and modern designs have arrived. Shop now for a limited time and redefine your look.</div>

            <button class="btn">Signup</button>

          </div>

        </div>

    </section></body></html>
```

CSS Transforms

- Transformation includes position, size and orientation.

- CSS provides 2 types of transformation methods

    1. 2D Transforms

    2. 3D Transforms

2D Transforms:

- 2D refers to two dimensions.

- It is along “X” and “Y” axis.

- 2D transform methods are

    a) translate() b) scale() c) skew() d) rotate()

Syntax: { transform : translate() | scale() | skew() | rotate();}

translate()

- It moves element from one location to another along X and Y axis.- It provides

    a) translate() b) translateX() c) translateY()

Syntax: { transform : translate(400px, 100px) ; // x = 400, y=100 }

scale() - It changes the width and height of element. - It can scale along X and Y axis. a) scale() b) scaleX() c) scaleY()

Syntax: { transform : scale(2, 1.5); 2 = 200% width, 1.5 = 150% height }

Skew()- It tilts the element by specified angle. - It can tilt along X and Y axis.

a) skew() b) skewX() c) skewY()

Syntax: { transform : skewX(20deg); }

rotate()- It rotates element by specified angle. - It can rotate along X and Y axis.

a) rotate() b) rotateX() c) rotateY()

Syntax: { transform : rotateX(20deg); }

Ex_78_Transforms.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        .side-1 {
            width: 200px;
            height: 200px;
            background-color: red;
            left: 1000px;
        }
        .side-2 {
            width: 200px;
            height: 200px;
```

```
    background-color: green;
    left: 550px;
    transform: skew(-40deg) scaleY(0.7);
    top: 260px;
}
.side-3 {
    width: 200px;
    height: 200px;
    background-color: blue;
    left: 520px;
    transform: skew(60deg) scaleY(0.5);
    top: 140px;
}
.side-4 {
    width: 200px;
    height: 200px;
    background-color: yellow;
    left: 370px;
    top: 160px;
    transform: skew(-40deg) scaleY(0.7);
}
.box {
    position: absolute;
```

```
        }
    </style>
</head>
<body style="display: flex; justify-content: center; align-items: center; height: 50vh;">
    <div>
        <div class="side-1 box"></div>
        <div class="side-2 box"></div>
        <div class="side-3 box"></div>
        <div class="side-4 box"></div>
    </div>
</body>
</html>
```

CSS 3D Transforms:

a) translate3D() b) scale3D() c) skew3D() d) roate3D()

Syntax: { transform : scale3d(2, 2, 3); x, y, z }

CSS Transitions and Animations

Day-37

CSS Transitions

a) transition b) transition-duration c) transition-delay d) transition-property

e) transition-timing-function

- **duration** refers to total time taken to complete transformation.
- **delay** refers to the time taken to start transformation.
- **property** refers to the properties that use transition timing.

- **timing-function** refers to pre-defined transition effects.

Syntax:

```
{
    transition-duration: 5s;
    transition-delay: 2s;
    transition-property: width, height ..;
    transition-timing-function: ease-in, steps() ..;
}
```

Ex_79_transition_duration.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        div {
            width: 50px;
            height: 30px;
            background-color: red;
            transition-duration: 1s;
            transition-timing-function: steps(4);
        }
```

```
        div:hover {
            width: 100%;
            background-color: blue;
            transition-property: width;
            transition-duration: 5s;
            transition-delay: 2s;
            transition-timing-function: steps(20);
        }
    </style>
</head>
<body>
    <div></div>
</body></html>
```

CSS Keyframes & Animations:

- Keyframe is the source of animation. - Frame is a container with elements changing position, size or orientation. - Keyframe can animate element.

- Keyframes have

    a) Initial State b) Break Point c) Final State

- Initial state is the starting point defined with

    a) 1%{ } or b) from { }

- Final state is ending point defined with

    a) 100%{ } or b) to { }

- Break point can be between 1 to 100% or between from & to.

Syntax:

```
@keyframes Name
{
    from { }
    50% { }
    to { }
}
```

- Keyframes are applied to element by using animation properties

  1. animation-name 2. animation-duration 3. animation-delay

  4. animation-iteration-count 5. animation-direction 6. animation-timing-function

Ex_80_Keyframes_Animations.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
    <style>
        @keyframes Run {
            form {
                transform: translateX(0px);
```

```
    }
    49% {
        transform: translateX(1200px);
    }
    50% {
        transform: translateX(1200px) rotateY(180deg);


    }
    51% {
        transform: translateX(1200px);
    }
    52% {
        transform: translateX(1200px) rotateY(180deg);
    }
    to {
        transform: translateX(0px) rotateY(180deg) ;
    }
}
.truck {
    animation-name: Run;
    animation-duration: 10s;
    animation-iteration-count: infinite;
    display: inline-block;
```

```
        }
        body {
            overflow: hidden;
        }
    </style>
</head>
<body>
    <span class="bi bi-truck-flatbed truck" style="font-size: 40px;"></span>
</body>
</html>
```

https://cssloaders.github.io/

Ex_81_Keyframes_Animations.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
    <style>
        @keyframes Blink{
            from {
                opacity: 0;
```

```
            transform: rotate(0deg) scale(0.1) translateX(-500px);
        }
        to {
            opacity: 1;
            transform: rotate(360deg) scale(1) translateX(0px);
        }
    }
    body {
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
        background-color: black;
    }
    .bi-star-fill {
        color:white;
        font-size: 100px;
        animation-name: Blink;
        animation-duration: 500ms;
        animation-iteration-count: infinite;
    }
  </style>
</head>
```

```
<body>

    <div class="bi bi-star-fill"></div>

</body>

</html>
```

Ex_82_Keyframes_Animations.html:

```
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <style>

        @keyframes Spin{

            from {

                transform: rotate(0deg);

            }

            to {

                transform: rotate(360deg);

            }

        }

        body {

            display: flex;

            justify-content: center;

            align-items: center;

            height: 100vh;

            background-color: black;

        }
```

```
        .box {
            width: 200px;
            height: 200px;
            border-top:20px solid white;
            border-right:20px solid white;
            border-bottom:20px solid white;
            border-left:20px solid transparent;
            border-radius: 100%;
            animation-name: Spin;
            animation-duration: 1s;
            animation-iteration-count: infinite;
            animation-timing-function: linear;


        }
    </style>
</head>
<body>
    <div class="box"></div>
</body></html>
```

==========

Bootstrap

1. Setup Bootstrap - Install following libraries

```
>npm install  bootstrap    bootstrap-icons --save
```

2. Link bootstrap resources

```
<link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
<link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">
<script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"> </script>
```

Bootstrap Classes

1. Box Model Classes

Margins:

m-{size}	size can be 1 to 5 all directions		
ms-{ }	left [start]		
me-{ }	right [end]		
mx-{ }	left and right	mt-{ }	top
mb-{ }	bottom		
my-{ }	top & bottom		

Padding:

p-{ }, ps-{ }, pe-{ }, px-{ }, pt- { }, pb-{ }, py-{ }

Border

border
border-{size}
border-{contextual} primary, secondary, danger, warning, success, info etc.

Border Radius:

rounded, rounded-{ }, rounded-pill, rounded-circle

Shadow

shadow, shadow-{ }

Width & Height

w-{ }	25, 50, 75, 100
h-{ }	25, 50, 75, 100

Text & Background Color

text-{contextual}
bg-{contextual}

Syntax:

```
<img src="" class="border border-4 border-danger rounded rounded-circle">
<p class="m-4 px-4 text-danger bg-warning">
```

Ex_83_Bootstrap.htm:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body>

    <div class="m-5 bg-warning p-3">

        <img class="border border-4 border-warning rounded rounded-circle p-2 shadow
shadow-lg" src="../public/images/women-banner.jpg" width="300" height="300">

    </div>

    <div class="border border-4 border-danger mt-3 mx-5 p-4 rounded rounded-pill bg-
warning text-light w-50">

        <h1 class="text-danger">Bootstrap</h1>

        <p>Frontend toolkit for rapidly building responsive UI.</p>

    </div>

</body>

</html>
```

2. Position Classes

.position-static, .position-absolute, .position-fixed, .position-relative, .position-sticky

.top-{ }

.bottom-{ } => 0, 25, 50, 100

.start-{ }

.end-{ }

Ex_84_Position_Classes.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

    <style>

        .women-fashion {

            background-image: url("../public/images/women-banner.jpg");

            background-size: cover;

            height: 400px;

            width: 400px;

        }
```

```
    </style>

</head>

<body class="p-5">

    <div class="women-fashion position-relative">

        <div style="width:60px; height:60px; text-align: center;" class="bg-danger text-
white p-2 position-absolute top-0 end-0 rounded rounded-circle">30% Off</div>

    </div>

    <p>Lorem ipsum </p>

     <h1 class="bg-dark position-sticky top-0 text-white p-2">Navbar</h1>

    <p>Lorem ipsum </p>

</body> </html>
```

3. Display Classes

.d-none, .d-block, .d-inline, .d-inline-block, .d-flex, .flex-wrap, .flex-row, .flex-row-reverse, .flex-column, .flex-column-reverse, .justify-content-between, .justify-content-evenly, .justify-content-around, .justify-content-center, .justify-content-baseline, .justify-content-end, .align-items-baseline, .align-items-end, .align-items-center

4. Grid Classes

.row, .col

.col-{ } 1 to 12

Syntax:

```
    <div class="row">

        <div class="col-3"> </div>  <div class="col-9"> </div>

    </div>
```

Ex_85_Grid_Classes.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

    <style>

        .women-fashion {

            background-image: url("../public/images/women-banner.jpg");

            background-size: cover;

            height: 400px;

            width: 400px;

        }

    </style>

</head>

<body class="p-5">

    <header class="bg-light border border-2 p-3 d-flex flex-row justify-content-between
align-items-center">

         <div>Shopping.</div>

         <div>

            <nav>
```

```
            <span>Home</span>
            <span class="mx-4">Shop</span>
            <span>Deals</span>
            <span class="mx-4">Contact</span>
        </nav>
    </div>
    <div>
        <span class="bi bi-person-fill"></span>
        <span class="bi mx-3 bi-cart4"></span>
    </div>
</header>
<section class="row p-3">
    <nav class="border col-2 border-2 bg-info">Nav</nav>
    <main style="height:330px" class="border p-3 col-8 border-2">
        <h3 class="bi mb-3 bi-person-circle"> Register User</h3>
        <dl class="row border border-2 p-4 w-75">
            <dt class="col-3">First Name</dt>
            <dd class="col-9"><input type="text"></dd>
            <dt class="col-3 my-3">Last Name</dt>
            <dd class="col-9 my-3"><input type="text"></dd>
            <dt class="col-3">City</dt>
            <dd class="col-9">
                <select>
```

```
                <option>Select City</option>
            </select>
        </dd>
    </dl>
  </main>
  <aside class="border col-2 border-2 bg-warning">Ads</aside>
</section>
<footer class="bg-light mt-3 row p-2">
  <div class="col">
    <ul>
      <li>Service1</li>
      <li>Service1</li>
      <li>Service1</li>
    </ul>
  </div>
  <div class="col">
    <ul>
      <li>Service1</li>
      <li>Service1</li>
      <li>Service1</li>
    </ul>
  </div>
  <div class="col">
```

```
            <ul>
                <li>Service1</li>
                <li>Service1</li>
                <li>Service1</li>
            </ul>
        </div>
        <div class="col">
            <ul>
                <li>Service1</li>
                <li>Service1</li>
                <li>Service1</li>
            </ul>
        </div>
    </footer>
</body>
</html>
```

5. Form Classes

.form-control	text, email, url, password, number, date, textarea, etc.
.form-range	range input
.form-control-color	color input
.form-control-file	file input
.form-label	label
.form-check-input	checkbox & radio

.form-check-label label for checkbox & radio

.form-switch container that have checkbox & radio

.form-select dropdown and listbox

6. Button Classes

.btn [base]

.btn-{contextual}

.btn-outline-{contextual}

.btn-sm

.btn-lg

.btn-group

.btn-group-vertical

.btn-toolbar

.btn-link

.btn-close

.btn-close-white

Ex_86_Form_Classes.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
```

```
    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>



</head>

<body class="d-flex justify-content-center">

     <form class="border w-25 border-1 shadow border-secondary rounded rounded-2
p-4 mt-2">

        <h5 class="bi bi-person-circle d-flex justify-content-between"> Register User
<button class="btn btn-close"></button> </h5>

        <div class="mb-2">

            <label class="form-label">First Name</label>

            <div>

                <input type="text" class="form-control">

            </div>

        </div>

        <div class="mb-2">

            <label class="form-label">Last Name</label>

            <div>

                <input type="text" class="form-control">

            </div>

        </div>

        <div class="mb-2">

            <label class="form-label">City</label>

            <div>
```

```
            <select class="form-select">
                <option>Select City</option>
            </select>
        </div>
    </div>
    <div class="mb-2">
        <label class="form-label">Rating</label>
        <div>
            <input  class="form-range" type="range">
        </div>
    </div>
    <div class="mb-2">
        <label class="form-label">Gender</label>
        <div>
            <input class="form-check-input" type="radio" name="Gender"> <label
class="form-check-label">Male</label>
            <input class="form-check-input" type="radio" name="Gender"> <label
class="form-check-label">Female</label>
        </div>
    </div>
    <div class="mb-2">
        <label class="form-label">Subscribe</label>
        <div class="form-switch">
            <input type="checkbox" class="form-check-input"> <label class="form-
```

```
check-label">Yes</label>
            </div>
        </div>
        <div>
            <button class="btn btn-danger w-100">Register</button>
            <button class="btn btn-link">Forgot Password</button>
        </div>
    </form>
</body>
</html>
```

==========

Bootstrap Classes

Day-39

28-11-2025

Bootstrap

- **Box Model**
- **Position**
- **Display**
- **Grid**
- **Forms**
- **Buttons**

Input Group Classes:

.input-group

.input-group-text

.input-group-lg

Syntax:

```
<div class="input-group">
    <span class="input-group-text"> text </span>
    … form elements…
</div>
```

Ex_87_Bootstrap.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">
    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>


</head>
<body class="p-4">
    <form class="input-group">
        <select class="input-group-text">
            <option>All</option>
        </select>
```

```
        <input type="text" placeholder="Search Amazon.in" class="form-control">

        <button class="btn btn-warning bi-search"></button>

    </form>

    <h3>Register</h3>

    <form class="w-25">

       <div class="input-group">

        <span class="bi bi-person-circle input-group-text"></span> <input type="text"
placeholder="User Name" class="form-control"> <span class="input-group-
text"><input type="checkbox"></span>

       </div>

       <div class="input-group mt-3">

        <span class="bi bi-key-fill input-group-text"></span> <input type="password"
placeholder="Password" class="form-control"> <span class="input-group-text"><input
type="checkbox"></span>

       </div>

       <div class="mt-3 input-group">

           <span class="bi bi-currency-rupee input-group-text"></span>

           <input type="text" class="form-control">

           <span class="input-group-text">.00</span>

       </div>

    </form>

</body>

</html>
```

Text Styles

.text-{contextual}

.text-center

.text-start

.text-end

.text-justify

.text-decoration-none

.text-decoration-underline

.text-decoration-overline

.text-decoration-line-through

.fs-{ } font size 1 to 7

.fw-bold bold

.fst-italics italic

.h{ } heading size 1 to 6

Ex:

```
<div class="text-center fs-1 fw-bold fst-italic text-decoration-
underline">Bootstrap</div>
```

Badge Class

.badge

.bg-{contextual}

.text-{contextual}

Syntax:

```
<span class="badge bg-danger text-white rounded rounded-circle"> 2 </span>
```

Ex_88_ badge.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>



</head>

<body class="p-4">

    <button class="btn btn-warning position-relative bi bi-cart4">

        <span class="badge bg-danger text-white rounded rounded-circle position-absolute">2</span>

    </button>

</body>

</html>
```

Bread Crumb Classes

.breadcrumb

.breadcrumb-item

Note: You can set bread crumb separator by using the following variable.

“--bs-breadcrumb-divider”.

Syntax:

```
<ul  style=“--bs-breadcrumb-divider: ‘ > ‘ “   class=“breadcrumb”>

      <li class=“breadcrumb-item”>  </li>

</ul>
```

Ex_89_Bread_Crumb.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>



</head>

<body class="p-4">
```

```
  <ul class="breadcrumb" style="--bs-breadcrumb-divider:' > '">
    <li class="breadcrumb-item">Flipkart</li>
    <li class="breadcrumb-item">Home</li>
    <li class="breadcrumb-item">Electronics</li>
    <li class="breadcrumb-item">Mobiles</li>
    <li class="breadcrumb-item">Apple Mobiles</li>
    <li class="breadcrumb-item">iPhone 17</li>
  </ul>
</body>
</html>
```

Pagination Classes:

.pagination

.page-item

.page-link

.pagination-sm | lg

.active [defined for link]

.disabled

Syntax:

```
<ul class="pagination">
    <li class="page-item"> <a class="page-link"> </a> </li>
</ul>
```

Ex_90_Pagination.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>



</head>

<body class="p-4">

    <ul class="pagination">

        <li class="page-item"><a class="page-link">&laquo;</a></li>

        <li class="page-item"><a class="page-link">1</a></li>

        <li class="page-item"><a class="page-link">2</a></li>

        <li class="page-item"><a class="page-link active">3</a></li>

        <li class="page-item"><a class="page-link disabled">4</a></li>

        <li class="page-item"><a class="page-link">5</a></li>

        <li class="page-item"><a class="page-link">&raquo;</a></li>

    </ul>

</body>

</html>
```

List Group & List Style:

.list-unstyled

.list-group

.list-group-horizontal

.list-group-item

.list-group-item-{contextual}

Syntax:

```
<ul class="list-unstyled list-group">
    <li class="list-group-item list-group-item-primary"> </li>
</ul>
```

Ex_91_List_Group_and_List_Style.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">
    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">
    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>


</head>
<body class="p-4">
```

```
    <ul class="list-unstyled list-group-horizontal list-group">
        <li class="list-group-item list-group-item-primary">Home</li>
        <li class="list-group-item  list-group-item-danger">Men Fashion</li>
        <li class="list-group-item list-group-item-warning">Women Fashion</li>
        <li class="list-group-item  list-group-item-info">Kids Fashion</li>
        <li class="list-group-item list-group-item-dark">Offers</li>
    </ul>
</body>
</html>
```

Table Classes:

.table

.table-hover

.table-bordered

.table-borderless

.table-striped

.table-dark

.table-light

.table-{contextual}

.caption-top

.table-responsive [bootstrap 5 sets responsive for container that contains table]

Syntax:

```
<div class="table-responsive">
    <table class="table table-hover">
```

```
        </table>

      </div>
```

Ex_92_**Table.html**:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>



</head>

<body class="p-4">

    <h2>Bootstrap Table</h2>

     <div class="w-50 table-responsive">

        <table class="table caption-top table-bordered table-striped table-dark table-
hover">

        <caption class="fs-3 fw-bold text-center bg-light mb-2">Student Details</caption>

        <thead>

            <tr>

                <th colspan="2">Name</th>

                <th colspan="3">Address</th>
```

```
</tr>
<tr>
    <th>First Name</th>
    <th>Last Name</th>
    <th>City</th>
    <th>State</th>
    <th>Postal Code</th>
    <th>First Name</th>
    <th>Last Name</th>
    <th>City</th>
    <th>State</th>
    <th>Postal Code</th>
    <th>First Name</th>
    <th>Last Name</th>
    <th>City</th>
    <th>State</th>
    <th>Postal Code</th>
    <th>First Name</th>
    <th>Last Name</th>
    <th>City</th>
    <th>State</th>
    <th>Postal Code</th>
    <th>First Name</th>
```

```
<th>Last Name</th>

<th>City</th>

<th>State</th>

<th>Postal Code</th>

<th>First Name</th>

<th>Last Name</th>

<th>City</th>

<th>State</th>

<th>Postal Code</th>

<th>First Name</th>

<th>Last Name</th>

<th>City</th>

<th>State</th>

<th>Postal Code</th>

<th>First Name</th>

<th>Last Name</th>

<th>City</th>

<th>State</th>

<th>Postal Code</th>

<th>First Name</th>

<th>Last Name</th>

<th>City</th>

<th>State</th>
```

```
        <th>Postal Code</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td> </td>
        <td> </td>
        <td rowspan="6" align="center">D<br>E<br>L<br>H<br>I</td>
        <td> </td>
        <td> </td>
    </tr>
    <tr>
        <td> </td>
        <td> </td>
        <td> </td>
        <td> </td>
    </tr>
    <tr>
        <td> </td>
        <td> </td>
        <td> </td>
        <td> </td>
    </tr>
```

```
            <tr>
                <td> </td>
                <td> </td>
                <td> </td>
                <td> </td>
            </tr>
            <tr>
                <td> </td>
                <td> </td>
                <td> </td>
                <td> </td>
            </tr>
            <tr>
                <td> </td>
                <td> </td>
                <td> </td>
                <td> </td>
            </tr>
        </tbody>
    </table>
     </div>
</body>
</html>
```

Card Classes:

.card
.card-header
.card-body
.card-footer
.card-title
.card-subtitle
.card-img-top
.card-img-bottom
.card-img-overlay

Syntax:

```
<div class="card m-2 p-2">

    <div class="card-header">

        <img class="card-img-top">

    </div>

    <div class="card-body"> </div>

    <div class="card-footer"> </div>

</div>
```

Ex_93_card.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">
```

```
    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>
</head>
<body class="p-4">
    <section class="d-flex">
        <div class="card m-2 p-2" style="width:250px">
            <img src="../public/images/dot-net.png" height="120" class="card-img-top">
            <div class="card-header text-center">
                <div class="card-title fw-bold">.NET</div>
                <div>
                    <span class="bi bi-star-fill text-warning"></span>
                    <span class="bi bi-star-fill text-warning"></span>
                    <span class="bi bi-star-fill text-warning"></span>
                    <span class="bi bi-star-fill text-warning"></span>
                    <span class="bi bi-star-fill text-warning"></span>
                </div>
            </div>
            <div class="card-body">
                <ol>
                    <li>Topic-1</li>
                    <li>Topic-2</li>
                    <li>Topic-3</li>
                    <li>Topic-4</li>
                </ol>
```

```
    </div>
    <div class="card-footer">
        <button class="btn btn-warning w-100">Join Course</button>
    </div>
</div>




<div class="card m-2 p-2" style="width:250px">
    <img src="../public/images/devOps.jpg" height="120" class="card-img-top">
    <div class="card-header text-center">
        <div class="card-title fw-bold">.NET</div>
        <div>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
        </div>
    </div>
    <div class="card-body">
        <ol>
            <li>Topic-1</li>
            <li>Topic-2</li>
```

```
            <li>Topic-3</li>
            <li>Topic-4</li>
        </ol>
    </div>
    <div class="card-footer">
        <button class="btn btn-warning w-100">Join Course</button>
    </div>
</div>


<div class="card m-2 p-2" style="width:250px">
    <img src="../public/images/python.png" height="120" class="card-img-top">
    <div class="card-header text-center">
        <div class="card-title fw-bold">.NET</div>
        <div>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
        </div>
    </div>
    <div class="card-body">
        <ol>
```

```
            <li>Topic-1</li>
            <li>Topic-2</li>
            <li>Topic-3</li>
            <li>Topic-4</li>
        </ol>
    </div>
    <div class="card-footer">
        <button class="btn btn-warning w-100">Join Course</button>
    </div>
</div>


<div class="card m-2 p-2" style="width:250px">
    <img src="../public/images/Java.png" height="120" class="card-img-top">
    <div class="card-header text-center">
        <div class="card-title fw-bold">.NET</div>
        <div>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
            <span class="bi bi-star-fill text-warning"></span>
        </div>
    </div>
```

```
            <div class="card-body">
                <ol>
                    <li>Topic-1</li>
                    <li>Topic-2</li>
                    <li>Topic-3</li>
                    <li>Topic-4</li>
                </ol>
            </div>
            <div class="card-footer">
                <button class="btn btn-warning w-100">Join Course</button>
            </div>
        </div>
    </section>
</body>
</html>
```

Bootstrap Components

Day-40

01-12-2025

- Box Model, - Display, - Positions, - Form, - Grid, - Input Group, - Badge, - Button, - Text, - Breadcrumb, - Pagination, - Card, - Table, - List Group

Progress Bar & Spinner

.progress

.progress-bar

.progress-bar-animated

.progress-bar-striped

Note: The progress value is defined using **style:width** in “%”.

Syntax:

```
<div class=“progress”>
    <div class=“progress-bar” style=“width:40%”> </div>
    … you can define multiple progress bars …
    </div>
</div>
```

.spinner-border

.spinner-border-sm

.spinner-grow

.spinner-grow-sm

Note: Spinner color is defined using text color.

Syntax:

```
<span class=“spinner-border spinner-border-sm text-success”> </span>
```

Ex_94_Progress_Bar_and_Spinner.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="p-4">

    <div class="progress">

        <div class="progress-bar bg-success progress-bar-animated progress-bar-striped" style="width: 40%;">

            40% completed

        </div>

        <div class="progress-bar bg-warning progress-bar-striped progress-bar-animated" style="width: 60%;">

            60% remaining

        </div>

    </div>

    <div class="spinner-grow mt-2 text-danger bg-warning">20%</div>

    <button class="btn btn-primary"> <span class="spinner-border spinner-border-sm"></span> Loading </button>

</body>

</html>
```

Container Classes

.container-sm	mobile
.container-md	tab

.container-lg PC

.container-xl Laptop

.container-xxl laptop wide screen

.container-fluid responsive according to device.

Note: “container” is built with pre-defined media query.

Syntax:

```
<div class=“container-sm”>
<body class=“container-fluid”>
```

Bootstrap Components

- Components are building blocks for modern web applications.
- A component comprises of

  a) Design : HTML

  b) Styles : CSS

  c) Actions : JavaScript / jQuery
- Component based design allows

  a) Easy reusability

  b) Easy extensibility

  c) Clean separation

  d) Maintainability &

  e) Testability
- Bootstrap provides built-in components for faster design, which you can customize and

  implement according to requirements.

1. alert

- It is used to display embedded message box in page.

- You can dynamically control the messages. [open, close]

classes:

.alert

.alert-{contextual}

.alert-dismissible

.alert-title

.alert-subtitle

.alert-link

attribute:

data-bs-dismiss : It is defined for clickable element to close the component.

It refers to component name. [alert]

Syntax:

```
<div class="alert alert-danger alert-dismissible">
    <button data-bs-dismiss="alert"> Text </button>
    …
    your content
    …
</div>
```

Ex_95_ alert.html:

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h2>Alerts</h2>

    <aside class="alert alert-warning alert-dismissible w-25">

        <h4>Ads.</h4>

        <button data-bs-dismiss="alert" class="btn btn-close"></button>

        <img src="../public/images/women-fashion.jpg" width="200" height="200">

    </aside>

    <div class="alert alert-danger alert-dismissible w-50">

        <button class="btn btn-close" data-bs-dismiss="alert"></button>

        <h2 class="alert-title">Delete Record</h2>

        <p class="alert-subtitle">Selected record deleted successfully..</p>

        <a class="alert-link" href="#">Help ?</a>

    </div>



    <div class="alert alert-success w-25">
```

```
        <h2 class="alert-title">Registered</h2>

        <p class="alert-subtitle">Your account created successfully..</p>

        <a class="alert-link" href="#">Help ?</a>

        <br><br>

        <button data-bs-dismiss="alert" class="btn btn-success">OK</button>

    </div>

</body>

</html>
```

2. Modal

- It is a dialog.
- It keeps content in backdrop.
- It can popup into page.
- It is hidden by default.

classes:

.modal

.modal-dialog

.modal-dialog-centered

.modal-dialog-scrollable

.modal-fullscreen

.modal-content

.modal-header

.modal-body

.modal-footer

attributes:

data-bs-dismiss : It refers the component to close.

data-bs-toggle : It refers the component to show / hide [toggle]

data-bs-target : It refers to the “id” of component.

Syntax:

```
<div class=“modal”>
    <div class=“modal-dialog”>
        <div class=“modal-content”>
            … header …
            … body …
            … footer …
        </div>
    </div>
</div>
```

Note: Modal can use “fade” as animation.

```
<div class=“modal fade”>
```

Note: Nested modals are disabled in bootstrap 5.

Ex_96_modal.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h3>Modals</h3>

    <button data-bs-toggle="modal" data-bs-target="#login" class="btn btn-
primary">Login</button>

    <button data-bs-toggle="modal" data-bs-target="#ads" class="btn btn-
warning">Show Ads</button>



    <div class="modal fade" id="login">

        <div class="modal-dialog modal-dialog-centered modal-dialog-scrollable">

            <div class="modal-content">

                <div class="modal-header">

                    <h3 class="bi bi-person-circle"> User Login</h3>

                    <button class="btn btn-close" data-bs-dismiss="modal"></button>

                </div>

                <div class="modal-body">

                    <dl>

                        <dt>User Name</dt>

                        <dd><input type="text" class="form-control"></dd>

                         <dt>Password</dt>
```

```
<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>

 <dt>Password</dt>

<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>

 <dt>Password</dt>

<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>

 <dt>Password</dt>

<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>

 <dt>Password</dt>

<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>

 <dt>Password</dt>

<dd><input type="password" class="form-control"></dd>

<dt>User Name</dt>

<dd><input type="text" class="form-control"></dd>
```

```
                <dt>Password</dt>

                <dd><input type="password" class="form-control"></dd>

            </dl>

        </div>

        <div class="modal-footer">

            <button class="btn btn-warning">Login</button>

            <button class="btn btn-danger" data-bs-
dismiss="modal">Cancel</button>

        </div>

      </div>

    </div>


    <div class="modal fade" id="ads">

        <div class="modal-dialog modal-dialog-centered">

            <div class="modal-content">

                <div class="modal-header">



                    <h3>Ads</h3>

                    <button class="btn btn-close" data-bs-dismiss="modal"></button>

                </div>

                <div class="modal-body">

                    <img src="../public/images/women-banner.jpg" width="200" height="200">
```

```
            </div>

          </div>

        </div>

    </div>

</body>

</html>
```

3. Offcanvas

- It is similar to modal with content in backdrop but allows to stick to specific direction left, right, top & bottom.

classes:

.offcanvas, .offcanvas-start, .offcanvas-end, .offcanvas-top, .offcanvas-bottom, .offcanvas-header, .offcanvas-body, .offcanvas-title

attributes

data-bs-toggle, data-bs-target, data-bs-dismiss

Ex_97_offcanvas.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>
```

```
</head>

<body class="container-fluid">

    <header class="border border-1 mt-2 p-2 d-flex justify-content-between">

        <div>

            <button data-bs-target="#menu" data-bs-toggle="offcanvas" class="btn btn-
light bi bi-justify"></button>

            <span>Amazon</span>

        </div>

        <div>

            <button data-bs-target="#cart" data-bs-toggle="offcanvas" class="btn btn-
warning bi bi-cart4"></button>

        </div>

    </header>

    <div class="offcanvas offcanvas-start" id="menu">

        <div class="offcanvas-header">

            <h3>Amazon</h3>

            <button class="btn btn-close" data-bs-dismiss="offcanvas"></button>

        </div>

        <div class="offcanvas-body">

            <ul>

                <li>Home</li>

                <li>Shop</li>

                <li>Deals</li>

                <li>Contact</li>
```

```
            </ul>
        </div>
    </div>


    <div class="offcanvas offcanvas-end" id="cart">
        <div class="offcanvas-header">
            <h3>Your Cart Items</h3>
            <button class="btn btn-close" data-bs-dismiss="offcanvas"></button>
        </div>
        <div class="offcanvas-body">
           <table class="table table-hover">
            <thead>
                <tr>
                    <th>Title</th>
                    <th>Price</th>
                </tr>
            </thead>
           </table>
        </div>
    </div>
</body>
</html>
```

Carousel, Collapse, Accordion

1. Alert 2. Modal 3. Offcanvas

4. Carousel : - It is used to for sliding and fading content in page. - It provides auto and manual control for content.

Basic Carousel Classes:

.carousel

.carousel-inner

.carousel-item

Syntax:

```
<div class="carousel">
    <div class="carousel-inner">
        <div class="carousel-item">
            … your content …
        </div>
        … other items …
    </div>
</div>
```

- Make sure that any one carousel items is **".active".**

```
<div class="carousel-item active">
```

- Make sure that carousel is set with animation to start slide or fade.

- Animation is defined by using "**data-bs-ride**" attribute.

```
<div class="carousel"  data-bs-ride="carousel">
```

- The default animation is not linear, you have to apply the classes

.slide

.carousel-fade

<div class=“carousel slide”>

<div calss=“carousel carousel-fade”>

Ex_98_carousel.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body>

    <div class="carousel slide" id="banners" data-bs-ride="carousel">

        <div class="carousel-inner">

            <div class="carousel-item active">

                <img src="../public/images/slide-1.png" class="w-100" height="250">

            </div>

            <div class="carousel-item">

                <img src="../public/images/slide-2.png" class="w-100" height="250">
```

```
            </div>
            <div class="carousel-item">
                <img src="../public/images/slide-3.png" class="w-100" height="250">
            </div>
        </div>
    </div>
</body>
</html>
```

Carousel with Controls:

- Controls allow to navigate between items manually.
- You can move to next or previous item in collection.

classes:

.carousel-control-prev

.carousel-control-prev-icon

.carousel-control-next

.carousel-control-next-icon

attributes:

data-bs-slide="prev | next"

data-bs-target : It refers to carousel ID.

Syntax:

```
<button class="carousel-control-prev"  data-bs-slide="prev"  data-bs-target="#id">
```

Ex_99_carousel_with_Controls.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body>

    <div class="carousel slide" id="banners" data-bs-ride="carousel">

        <div class="carousel-inner">

            <div class="carousel-item active">

                <img src="../public/images/slide-1.png" class="w-100" height="250">

            </div>

            <div class="carousel-item">

                <img src="../public/images/slide-2.png" class="w-100" height="250">

            </div>

            <div class="carousel-item">

                <img src="../public/images/slide-3.png" class="w-100" height="250">

            </div>

        </div>
```

```
        <button data-bs-slide="prev" data-bs-target="#banners" class="carousel-control-prev">

            <span class="carousel-control-prev-icon"></span>

        </button>

         <button data-bs-slide="next" data-bs-target="#banners" class="carousel-control-next">

            <span class="carousel-control-next-icon"></span>

        </button>

    </div>

</body>

</html>
```

Carousel with Indicators: - Indicators allow random navigation.- User can choose any item directly to view.

class

.carousel-indicators

attributes

data-bs-slide-to="index_number" [first slide = 0]

data-bs-target="carousel_id"

- Make sure that any one indicator is active.

Syntax:

```
<div class="carousel-indicators">

    … buttons …

</div>
```

Ex_100_carousel_with_indicators.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body>

    <div class="carousel slide" id="banners" data-bs-ride="carousel">

        <div class="carousel-inner">

            <div class="carousel-item active">

                <img src="../public/images/slide-1.png" class="w-100" height="250">

            </div>

            <div class="carousel-item">

                <img src="../public/images/slide-2.png" class="w-100" height="250">

            </div>

            <div class="carousel-item">

                <img src="../public/images/slide-3.png" class="w-100" height="250">

            </div>

        </div>
```

```
        <button data-bs-slide="prev" data-bs-target="#banners" class="carousel-control-prev">

            <span class="carousel-control-prev-icon"></span>

        </button>

         <button data-bs-slide="next" data-bs-target="#banners" class="carousel-control-next">

            <span class="carousel-control-next-icon"></span>

        </button>

        <div class="carousel-indicators">

            <button class="active" data-bs-slide-to="0" data-bs-target="#banners"></button>

            <button data-bs-slide-to="1" data-bs-target="#banners"></button>

            <button data-bs-slide-to="2" data-bs-target="#banners"></button>

        </div>

    </div>

</body>

</html>
```

Note: You can set timing for slides using the attribute “data-bs-interval”.

```
        <div class=“carousel-item”  data-bs-interval=“4000”>      4000ms  =4 secs
```

Ex_101_carousel_with_interval.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body>

    <div class="carousel slide" id="banners" data-bs-ride="carousel">

        <div class="carousel-inner">

            <div class="carousel-item active" data-bs-interval="4000">

                <img src="../public/images/slide-1.png" class="w-100" height="250">

            </div>

            <div class="carousel-item bg-dark" data-bs-interval="5000">

                <div class="d-flex justify-content-center align-items-center"
style="height:250px">

                    <div>

                        <div class="input-group">

                            <input type="text" class="form-control" placeholder="Your email">

                            <button class="btn btn-danger">Get Started <span class="bi bi-
chevron-right"></span> </button>

                        </div>

                    </div>

                </div>

            </div>
```

```
            <div class="carousel-item" data-bs-interval="1000">

                <img src="../public/images/slide-2.png" class="w-100" height="250">

            </div>

            <div class="carousel-item" data-bs-interval="5000">

                <img src="../public/images/slide-3.png" class="w-100" height="250">

            </div>

        </div>

        <button data-bs-slide="prev" data-bs-target="#banners" class="carousel-control-
prev">

            <span class="carousel-control-prev-icon"></span>

        </button>

         <button data-bs-slide="next" data-bs-target="#banners" class="carousel-control-
next">

            <span class="carousel-control-next-icon"></span>

        </button>

        <div class="carousel-indicators">

            <button class="active" data-bs-slide-to="0" data-bs-
target="#banners"></button>

            <button data-bs-slide-to="1" data-bs-target="#banners"></button>

            <button data-bs-slide-to="2" data-bs-target="#banners"></button>

            <button data-bs-slide-to="3" data-bs-target="#banners"></button>

        </div>

    </div>

</body>

</html>
```

Note: If your carousel items have a light background then you can set dark controls.

a) Bootstrap 5 previous versions use “carousel-dark” class

b) Bootstrap 5x latest versions use “data-bs-theme=dark” attribute

```
<div class=“carousel” data-bs-theme=“dark”>
```

5. Collapse: - It is used to toggle show / hide. - Dynamically you can show / hide content in page.

classes:

.collapse

.show

attributes:

data-bs-target

data-bs-toggle

Syntax:

```
<div class=“collapse show”  id=“”>
<button data-bs-target=“#id” data-bs-toggle=“collapse”>
```

Ex_102_Collapse.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
```

```
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h2>FAQ</h2>

    <button data-bs-target="#q1" data-bs-toggle="collapse" class="btn btn-dark w-
100">What is Netflix?</button>

    <div class="mt-2 collapse" id="q1">

        Lorem ipsum

        Lorem ipsum

     </div>

</body>

</html>
```

6. Accordion: - It is a collection of items to show or hide dynamically. - It uses “Mutex” mechanism. - Mutex is mutual exclusion. - It can collapse all items in collection while showing any specific.

classes

.accordion

.accordion-item

.accordion-header

.accordion-body

.accordion-button

.accordion-collapse

attributes

data-bs-target

data-bs-toggle

data-bs-parent: It is defined for collapse to setup Mutex. It refers to accordion ID.

Syntax:

```
<div class="accordion">
    <div class="accordion-item">
        <div class="accordion-header">
            .. accordion button ..
        </div>
        <div class="accordion-collapse collapse">
            <div class="accordion-body">
            </div>
        </div>
    </div>
</div>
```

Ex_103_Accordion.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
```

```
    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h2 class="text-center">Frequently Asked Questions</h2>

    <div class="accordion" id="faqs">

        <div class="accordion-item">

            <div class="accordion-header">

                <button data-bs-target="#q1" data-bs-toggle="collapse" class="accordion-button">What is Netflix?</button>

            </div>

            <div class="accordion-collapse collapse" id="q1" data-bs-parent="#faqs">

                <div class="accordion-body">

                    <p> Lorem ipsum dolor, sit amet consectetur adipisicing elit. Magni adipisci vero quaerat, ea corrupti nulla laborum eveniet? Error odio expedita repellendus sequi facere eum, ex rem ut, nulla totam harum! Lorem, ipsum dolor sit amet consectetur adipisicing elit. Impedit fugit, aspernatur nam similique earum ea laborum, porro deleniti iusto inventore molestias sit laudantium voluptate esse ratione ut quia voluptatum exercitationem.</p>

                </div>

            </div>

        </div>



        <div class="accordion-item">

            <div class="accordion-header">
```

```
            <button data-bs-target="#q2" data-bs-toggle="collapse" class="accordion-button">How much does Netflix cost?</button>

        </div>

        <div class="accordion-collapse collapse" data-bs-parent="#faqs" id="q2">

            <div class="accordion-body">

                <p> Lorem ipsum dolor, sit amet consectetur adipisicing elit. Magni adipisci vero quaerat, ea corrupti nulla laborum eveniet? Error odio expedita repellendus sequi facere eum, ex rem ut, nulla totam harum! Lorem, ipsum dolor sit amet consectetur adipisicing elit. Impedit fugit, aspernatur nam similique earum ea laborum, porro deleniti iusto inventore molestias sit laudantium voluptate esse ratione ut quia voluptatum exercitationem.</p>

            </div>

        </div>

      </div>

    </div>

</body>

</html>
```

Bootstrap Nav, Tabs, Navbar, etc

Day-42

03-12-2025

Bootstrap Components

1. Alert 2. Modal 3. Offcanvas 4. Carousel 5. Collapse
6. Accordion

7. Dropdown: - It can have a collection of options with complex content. - HTML <select> & <option> are RC type and will not allow formats.

classes:

.dropdown

.dropup

.dropend

.dropstart

.dropdown-toggle

.dropdown-menu

.dropdown-item

.dropdown-item-text

.dropdown-divider

attribute

data-bs-toggle

Syntax:

```
<div class="dropdown">
    <button class="dropdown-toggle"> </button>
    <ul class="dropdown-menu">
        <li class="dropdown-item">
            <span class="dropdown-item-text"> Text </span>
        </li>
    </ul>
</div>
```

Ex_104_dropdown.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

        <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h3>Dropdown</h3>

    <div class="dropend">

        <button data-bs-toggle="dropdown" class="btn dropdown-toggle btn-dark">
<span class="bi bi-person-circle"></span> Login </button>

        <ul class="dropdown-menu" data-bs-theme="dark">

            <li class="dropdown-item"> <span class="dropdown-item-text"> <a> <span
class="bi bi-person-fill"></span> Profile</a> </span> </li>

            <li class="dropdown-item"> <span> <a class="dropdown-item-
text"  href="../public/amazon-cards.html" target="body-frame"> <span class="bi bi-
heart-fill"></span> Wish List</a> </span> </li>

            <li class="dropdown-item"> <span class="dropdown-item-text"> <a> <span
class="bi bi-gift-fill"></span> Gift </a> </span> </li>

            <li class="dropdown-divider"></li>

            <li class="dropdown-item"> <span class="dropdown-item-text"> <a> <span
class="bi bi-question-circle"></span> Help </a> </span> </li>

        </ul>

    </div>
```

```
    <iframe name="body-frame" class="w-100 mt-2" height="500"></iframe>

</body>

</html>
```

8. Nav & Tabs: - It is used to create a tab based navigation. - It allows to display multiple pages content in one page.

classes:

.nav, .nav-tabs, .nav-pills, .nav-menu, .nav-item, .nav-link, .tab-content, .tab-pane

attributes: data-bs-toggle

Syntax:

```
    <ul class="nav nav-tabs">

        <li class="nav-item">  <a class="nav-link"  href="#tab-id"> </a> </li>

    </ul>

    <div class="tab-content">

        <div class="tab-pane" id="tab_id"> </div>

    </div>
```

Ex_105_nav_tab.html:

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
```

```
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <h3>HDFC Bank</h3>

    <ul class="nav nav-tabs">

        <li class="nav-item"><a href="#accounts" data-bs-toggle="tab" class="nav-link
active">Accounts</a></li>

        <li class="nav-item"><a href="#cards" data-bs-toggle="tab" class="nav-
link">Cards</a></li>

        <li class="nav-item"><a href="#deposits" data-bs-toggle="tab" class="nav-
link">Deposits</a></li>

    </ul>

    <div class="tab-content mt-3">

        <div class="tab-pane active" id="accounts">

            <img src="../public/images/accounts.png"  width="400" height="300" >

        </div>

         <div class="tab-pane" id="cards">

            <img src="../public/images/cards.png"  width="400" height="300" >

        </div>

         <div class="tab-pane" id="deposits">

            <img src="../public/images/deposits.png"  width="400" height="300" >

        </div>

    </div>
```

```
</body>

</html>
```

9. Navbar: - It configures a responsive navbar. - Responsive is defined with media queries, which fit according to device.

classes:

.navbar, .navbar-brand, .navbar-toggler, .navbar-toggler-icon, .navbar-nav, .navbar-collapse, .nav-menu, .nav-item, .nav-link, .navbar-expand-lg | sm | md, .navbar-dark, .navbar-light

attributes: data-bs-target, data-bs-toggle

Syntax:

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">

    <div class="container-fluid">

        <span class="navbar-brand"> </span>

        <button class="navbar-toggler">

            <span class="navbar-toggler-icon"> </span>

        </button>

        <div class="navbar-collapse collapse">

            <ul class="navbar-nav">

                <li class="nav-item"> <a class="nav-link"> </a> </li>

            </ul>

        </div>

    </div>

</nav>
```

Ex_106_nav_bar.html:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

      <link rel="stylesheet" href="../node_modules/bootstrap-icons/font/bootstrap-
icons.css">

    <link rel="stylesheet" href="../node_modules/bootstrap/dist/css/bootstrap.css">

    <script src="../node_modules/bootstrap/dist/js/bootstrap.bundle.js"></script>

</head>

<body class="container-fluid">

    <nav class="navbar navbar-expand-lg navbar-dark bg-dark">

        <div class="container-fluid">

            <button data-bs-toggle="collapse" data-bs-target="#menu" class="navbar-
toggler">

                <span class="navbar-toggler-icon"></span>

            </button>

            <span class="navbar-brand">Amazon</span>

            <div class="navbar-collapse collapse" id="menu"">

             <ul class="navbar-nav">

                <li class="nav-item"> <a class="nav-link"> Home </a> </li>

                <li class="nav-item"> <a class="nav-link"> Shop </a> </li>

                <li class="nav-item"> <a class="nav-link"> Deals </a> </li>

                <li class="nav-item"> <a class="nav-link"> Contact </a> </li>
```

```
            <li class="nav-item">
                <div class="input-group">
                    <input type="text" placeholder="Search Amazon.in" class="form-
control">
                    <button class="btn btn-warning bi bi-search"></button>
                </div>
            </li>
        </ul>
        </div>


      </div>
   </nav>
</body>
</html>
```

Tailwind

Day-43

04-12-2025

Tailwind CSS

- Tailwind CSS is a utility framework.

- It provides various classes similar to bootstrap, but has multiple properties.

Setup Tailwind for project:

1. Open terminal in your project location and run the following command

```
> npm install tailwindcss @tailwindcss/cli    --save
```

2. Go to “src/styles/input.css” [add **input.css** file]

3. Add following line into input.css

```
@import  “tailwindcss”;
```

4. Generate an output file by using the following command in terminal

```
> npx @tailwindcss/cli -i ./src/styles/input.css -o ./src/styles/output.css --watch
```

5. Link “output.css” to your HTML page

```
<link rel=“stylesheet” href=“../src/styles/output.css”>
```

Ex: home.html

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Document</title>

    <link rel="stylesheet" href="../src/styles/output.css">

</head>

<body class="p-4">

    <div class="border border-2 p-2 border-orange-600 shadow-lg shadow-orange-
100">

        <h1>Tailwind CSS</h1>

        <p>Utility framework for rapidly building interactive UI.</p>

    </div>

</body>

</html>
```

Note: Go to "https://tailwindcss.com/docs" to explore all utility classes.

[Margin, Padding, Border, Font, Grid, Flex etc..]